



UnionPay Integrated Circuit Card Specifications

— Auxiliary Specifications

Part II Debit/Credit Application - Member Implementation Guide for Issuer



Version 2017

THIS PAGE IS INTENTIONALLY LEFT BLANK.

Table of Contents

SUMMARY OF REVISIONS.....	1
1 APPLICATION SCOPE.....	2
2 REFERENCES.....	3
3 TERMS AND DEFINITIONS.....	4
4 SYMBOLS AND ABBREVIATIONS.....	7
5 CARD SELECTION AND VERIFICATION.....	13
5.1 CARD PRODUCT SELECTION	13
5.1.1 Card Types	13
5.1.2 Financial Application Specifications.....	13
5.1.3 Industrial Application Specifications.....	13
5.1.4 Product Testing and Certification Requirements	14
5.1.5 Card Operating System Requirements.....	14
5.1.6 Communication Interfaces and Communication Protocols	15
5.1.7 Card Technical Requirements	16
5.1.8 Special Requirements.....	17
5.2 CARD VENDOR SELECTION	17
5.2.1 Qualification Requirements.....	17
5.2.2 Supplier's Capacity.....	17
5.2.3 Project Experience.....	18
5.2.4 Logistics Requirements.....	18
5.2.5 After-Sales Services Requirements	18
5.2.6 Purchasing Quantity.....	18
5.2.7 Integration Services.....	19
6 CARD ISSUANCE.....	20
6.1 OVERVIEW.....	20
6.1.1 Card Types.....	20
6.1.2 Application Requirements.....	21
6.1.3 Key Management	21
6.1.4 Equipment Upgrade.....	21

6.1.5	<i>Learning Curve</i>	22
6.2	ISSUER PRE-PERSONALIZATION ACTIVITIES.....	22
6.2.1	<i>Card Design</i>	22
6.2.2	<i>Personalization Input File</i>	22
6.2.3	<i>Personalization Key</i>	23
6.2.4	<i>Implementation</i>	24
6.3	CARD MANUFACTURER ACTIVITIES	25
6.3.1	<i>Prerequisite</i>	25
6.3.2	<i>Operations Performed</i>	25
6.4	PERSONALIZATION ACTIVITIES.....	26
6.4.1	<i>Prerequisite</i>	26
6.4.2	<i>Operations Performed</i>	27
6.5	POST-PERSONALIZATION ACTIVITIES	28
7	PERSONALIZATION DATA REQUIREMENTS.....	29
7.1	PERSONALIZATION DATA TEMPLATES.....	29
7.2	PERSONALIZATION DATA REQUIREMENTS FOR STANDARD DEBIT/CREDIT	31
7.2.1	<i>Application Selection</i>	31
7.2.2	<i>Application Initialization</i>	38
7.2.3	<i>Offline Data Authentication</i>	39
7.2.4	<i>Processing Restriction</i>	47
7.2.5	<i>Cardholder Verification</i>	50
7.2.6	<i>Terminal Risk Management</i>	58
7.2.7	<i>Card Action Analysis</i>	60
7.2.8	<i>Online Processing</i>	64
7.2.9	<i>Transaction Completion</i>	67
7.2.10	<i>Issuer Script Processing</i>	69
7.3	PERSONALIZATION DATA REQUIREMENTS FOR CHIP/MAGNETIC-STRYPE	73
7.3.1	<i>Relationship between IC Card Application and Magnetic-stripe</i>	73
7.3.2	<i>IC Card Verification Number</i>	74
7.4	PERSONALIZATION DATA REQUIREMENTS FOR ELECTRONIC CASH	75

7.4.1	Data Elements Involved.....	76
7.4.2	Risk Management	78
7.4.3	Implementation.....	79
7.5	PERSONALIZATION DATA REQUIREMENTS FOR CONTACTLESS IC CARD PAYMENT	79
7.5.1	Summary of General Requirements	80
7.5.2	Quick Debit/Credit Payment Application	80
7.5.3	Other Data Requirements	94
7.6	EXTENDED APPLICATION FOR CONTACTLESS IC CARD MICROPAYMENT	94
7.6.1	Data Elements Involved.....	95
7.6.2	Risk Management	96
7.6.3	Implementation.....	96
7.7	OTHER CONSIDERATIONS	97
8	ISSUER HOST SYSTEM MODIFICATIONS.....	98
8.1	NEW AUTHORIZATION DATA.....	98
8.1.1	Upgrading Message Fields.....	98
8.1.2	New Message Fields for IC Card	98
8.1.3	Electronic Cash Transaction	104
8.1.4	qUICS Transaction	104
8.2	ONLINE CARD AUTHENTICATION	105
8.2.1	Summary.....	105
8.2.2	Prerequisite.....	105
8.2.3	Implementation.....	105
8.3	ONLINE ISSUER AUTHENTICATION.....	106
8.3.1	Summary.....	106
8.3.2	Prerequisite.....	107
8.4	FUTURE CRYPTOGRAM SUPPORT	107
8.5	DATA RECORDING/LOGGING	107
8.6	ISSUER SCRIPT PROCESSING	108
8.6.1	Summary.....	108
8.6.2	Prerequisite.....	109

8.6.3	Implementation.....	110
8.6.4	Chip-Initiated Transactions.....	112
8.6.5	Fallback Transactions	113
8.7	HSM (ENCRYPTION SERVICE) PROCESSING	114
8.8	OTHER CONSIDERATIONS	115
9	ISSUER KEY MANAGEMENT SYSTEM	116
9.1	ISSUER CA MANAGEMENT	116
9.1.1	Issuer Certificate Request.....	116
9.1.2	IC Card Static Data Signature.....	116
9.1.3	Issuance of IC Card Certification.....	117
9.1.4	Implementation.....	117
9.2	SYMMETRIC KEY MANAGEMENT	118
9.2.1	Transaction Master Key Management.....	118
9.2.2	IC Card Master Key Management.....	118
9.2.3	Other Transmit Keys.....	118
9.2.4	Implementation.....	118
10	ISSUER DATA PREPARATION AND PERSONALIZATION SYSTEM.....	120
10.1	IC CARD SYSTEM DATA CLASSIFICATION	120
10.2	IC CARD DGI DATA CLASSIFICATION.....	120
10.3	INTERACTIONS WITH IC CARD KEY MANAGEMENT SYSTEM.....	121
10.4	PERSONALIZATION PROCESSING.....	121
10.5	IMPLEMENTATION	122
11	ISSUER BACK-OFFICE ACCOUNT MANAGEMENT AND BUSINESS SECURITY	124
11.1	ACTUAL BANK CASE FOR ACCOUNT AND TRANSACTION IMPLEMENTATION	124
11.1.1	Upgrading of Account and Transaction Structure	124
11.1.2	Description of featured transaction processing of the IC card.....	125
11.2	STATEMENTS	126
11.2.1	UPI Statement.....	126
11.2.2	Customer Statement.....	126

11.3	CUSTOMER SERVICE.....	127
11.4	CARD MANAGEMENT SYSTEM	128
11.5	CARD REPLACEMENT	128
11.6	DISPUTE RESOLUTION	128
11.7	SETTLEMENT AND RECONCILIATION.....	129
11.8	REPORTS.....	129
11.8.1	Chip Transaction Statistics	129
11.8.2	Fallback Transaction.....	130
11.8.3	Enhancing Reporting Functions.....	130
11.8.4	Transaction Research Tool	130
11.9	INTERNAL STAFF TRAINING.....	131
11.10	IMPLEMENTATION	131
12	ISSUER HOST AUTHENTICATION.....	133
12.1	AUTHENTICATION ENVIRONMENT	133
12.2	AUTHENTICATION PROCESS.....	133
APPENDIX A	(INFORMATIVE APPENDIX) RECOMMENDATIONS ON PROJECT IMPLEMENTATION	134
A.1	KEY SUCCESS FACTORS.....	134
A.2	PROJECT ORGANIZATION	135
A.3	PROJECT MANAGER	136
A.4	PROJECT TEAM.....	136
A.5	IMPLEMENTATION PLAN	137
A.6	PROJECT TASK LIST	138
APPENDIX B	(INFORMATIVE APPENDIX) REFERENCED PROCESS FLOW OF IC CARD ISSUANCE FOR UPI MEMBERS.....	147
B.1	BUSINESS QUALIFICATIONS FOR ISSUING MEMBER INSTITUTIONS	147
B.2	BASIC IC CARD ISSUANCE PROCESS FLOW FOR MEMBER INSTITUTIONS	147
APPENDIX C	(INFORMATIVE APPENDIX) KEY QUICK REFERENCE TABLE	149
C.1	PERSONALIZATION KEYS.....	149
C.2	ONLINE CARD AND ISSUER AUTHENTICATION KEYS	150

C.3	PUBLIC KEY	151
C.4	ISSUER SCRIPT PROCESSING KEYS	152
APPENDIX D (NORMATIVE APPENDIX) UPI IC CARD NETWORK ENTRY		
PROCESS 154		
D.1	NETWORK ENTRY TESTING PROCESS	154
D.2	SPECIAL INSTRUCTIONS.....	154

Summary of Revisions

The change listed below is associated with the current version.

Change Description	Revised/ Deleted/ Added	Location
Deleted “credit/debit application 0101” in Table 2 UPI Proprietary Application Identifier Extension (PIXs)	Deleted	7.2.1.1.1

1 Application Scope

This book applies to all UPI Participants.

2 References

The terms and conditions of the following documents quoted in this guide have become the terms and conditions of this guide. The modification list (excluding corrected contents) or revised edition attached to the dated documents shall not apply to this Specification. However, Participants may study whether to apply the latest versions of such documents. The latest versions of non-dated documents shall apply to the guide.

EMV - Books 1 to 4 of the EMV 4.1 Integrated Circuit Card Specifications for Payment Systems

3 Terms and Definitions

The following terms and definitions are applied in this guide.

qUICS

Quick debit/credit application (qUICS) ensures quick transaction through contactless interface.

Financial IC card Debit/Credit Applications Root CA

Root certificate authority serves financial IC card security, hereinafter referred to as “Root CA”. The system to implement the root CA functions is the “Financial IC card Debit/Credit Applications Root CA System”, hereinafter referred to as “Root CA System”.

Application

Application protocols and relevant data sets which are used between cards and terminals.

Clearing

A process where the issuer processes the transaction data submitted by the acquirer. As to the issuer, clearing provides another chance to synchronize the back-office counter with the counter embedded in cards. As to the acquirer, clearing is to submit the transaction data to the issuer.

Command

A message sent from the terminal to IC cards to initiate one operation or request a response.

Cryptogram

Result of cryptographic operation.

Electronic Cash (EC)

Function supporting low-value payment based on debit/ credit application.

Electronic Cash Balance

A counter that shows the available offline spending amount on the card.

Electronic Cash Balance Limit

The maximum amount cardholder can spend offline.

Financial Transaction

The commodity or service exchanged among cardholder, merchant and acquirer based on the ways of collection and payment.

Function

A sequence combined by one or more commands, which complete the whole or part of transaction.

Integrated Circuit (IC)

Electronic components designed to perform processing and/or memory functions.

Integrated Circuit(s) Card (IC card)

A card into which one or more integrated circuits are inserted to perform processing and memory functions.

Interface Device

That part of the terminal into which the IC card is inserted, including mechanical and electrical parts.

Issuer Action Code

The actions selected by issuer to be taken based on the analysis of Terminal Verification Result.

Java Card

Smart card that can run Java Applet programs, supports Java card and GP specification, and can dynamically download applications after card issuance.

Key

A sequence of symbols that controls the operation of a cryptographic transformation.

Load

A process to increase electronic cash balance in the card. There are a variety of ways to load, such as transfer from main account, cash deposit or transfer from other accounts. Electronic cash balance cannot exceed electronic cash balance limit.

Magnetic-stripe

The stripe containing magnetically encoded information.

Native Card

COS self-developed by each COS developer; there is no uniform standards and interfaces, and does not support dynamic download after card issuance.

Path

Concatenation of file identifiers without delimitation.

Payment System Environment

The set of logical conditions established within the ICC when a payment system application conforming to UICS has been selected, or when a Directory Definition File (DDF) used for payment system application purposes has been selected.

Private Key

That key of an entity's asymmetric key pair that should only be used by that entity. In the case of a digital signature scheme, the private key defines the signature function.

Public Key

That key of an entity's asymmetric key pair that can be made public. In the case of a digital signature scheme, the public key defines the verification function.

Public Key Certificate

The public key information of an entity signed by the certification authority and thereby rendered unforgeable.

Response

A message returned by the ICC to the terminal after the processing of a command message received by the ICC.

Script

A Command or command sequence transmitted to terminal by issuer for the purpose of being sent serially to the ICC as commands.

Terminal

The device used in conjunction with the ICC at the point of transaction to perform financial transactions. It includes interface device, and also includes some other components and interfaces such as host communications.

Terminal Action Code

Terminal action code (default, decline and online) reflects the action selected by acquirer based on Terminal Verification Result.

Transaction Log

Detailed records of recent transactions, from which the history of cards' transactions can be learned.

Unload

Process of reducing electronic cash balance to zero. The result of unload is to transfer all remainder amount in electronic cash account to cardholder's main account or return the equivalent amount in cash. Unload is used when cardholder or Issuer cancels electronic cash function.

4 Symbols and Abbreviations

The following abbreviations and symbols are only applicable to this specification:

- AAC Application Authentication Cryptogram
- AAR Application Authorization Referral
- AC Application Cryptogram
- ADA Application Default Action
- ADF Application Definition File
- AEF Application Elementary File
- AFL Application File Locator
- AID Application Identifier
- AIP Application Interchange Profile
- APDU Application Protocol Data Unit
- ARPC Authorization Response Cryptogram
- ARQC Authorization Request Cryptogram
- ATC Application Transaction Counter
- ATM Automated Teller Machine

- ATS Answer To Select
- AUC Application Usage Control
- BER Basic Encoding Rules
- BIN Bank Identification Number
- CA Certificate Authority
- CAM Card Authentication Method
- CDA Combined DDA/AC Generation
- CDOL Card Risk Management Data Object List
- CID Cryptogram Information Data
- CLA Class Byte of the Command Message
- cn compress numeric
- C-TPDU Command TPDU
- CTTA Cumulative Total Transaction Amount
- CTTAL Cumulative Total Transaction Amount Limit
- CTTAUL Cumulative Total Transaction Amount Upper Limit
- CUPS China UnionPay System

- CVM Cardholder Verification Method
- CVR Card Verification Results
- CVN Card Verification Number
- dCVN Dynamic Card Verification Number
- DDA Dynamic Data Authentication
- DDF Directory Definition File
- DDOL Dynamic Data Authentication Data Object List
- DES Data Encryption Standard
- DF Dedicated File
- DIR Directory
- DOL Data Object List
- GPO GET PROCESSING OPTIONS
- EC Electronic Cash
- EF Elementary File
- EMV Europay MasterCard VISA
- FCI File Control Information

- fDDA Fast DDA
- FWI Frame Waiting Time Integer
- GP Global Platform
- IAC Issuer Action Code
- IC Integrated Circuit
- IC card Integrated Circuit Card
- iCVN Integrated Circuit Card Verification Number
- IDD Issuer Discretionary Data
- K_{DEK} Data Encryption Key
- K_{ENC} Encryption Key
- K_{MAC} Message Authentication Code Key
- KMC Issuer Master Key
- Lr Length of Response Data Field
- M Mandatory
- MAC Message Authentication Code
- MDK Master DEA Key

- MF Mater File
- n Numeric
- O Optional
- P1 Parameter 1
- P2 Parameter 2
- P3 Parameter 3
- PAN Primary Account Number
- PDOL Processing Options Data Object List
- PKI Public Key Infrastructure
- PIN Personal Identification Number
- PIX Proprietary Application Identifier Extension
- PPSE Proximity Payment Systems Environment
- PSE Payment System Environment
- qUICS Quick UICS
- RFU Reserved for Future Use
- RID Registered Application Provider Identifier

- R-TPDU Response TPDU
- SAD Signed Static Application Data
- SDA Static Data Authentication
- SFI Short File Identifier
- SW1 Status Word One
- SW2 Status Word Two
- TAC Terminal Action Code
- TC Transaction Certificate
- TDOL Transaction Certificate Data Object List
- TLV Tag, Length, Value
- TSI Transaction Status Information
- TVR Terminal Verification Results
- UDK Unique DEA Key
- Dedicated Dedicated Not defined in this specification or/and beyond the scope of this specification

5 Card Selection and Verification

5.1 Card Product Selection

Prior to meeting with vendors and assessing their available products, Issuers should develop requirements for their cards. This section provides the following suggestions to assist Issuers in establishing selection criteria for card vendors:

- Card types
- Financial application specifications
- Industrial application specifications
- Product testing and certification requirements
- Card operating system requirements
- Communication interfaces and communication protocols
- Card technical requirements
- Special requirements

5.1.1 Card Types

Issuers may choose to issue different types of card products in a combination of financial functions, including but not limited to:

- Standard debit IC card
- Standard credit IC card
- Quasi-credit IC card
- Standard debit/credit + electronic cash card
- Standard debit/credit + electronic cash card (including contactless low value payment extended application)
- Financial IC card with magnetic-stripe function (commonly known as Hybrid card)

At the same time, Issuers may choose to cooperate with industries to issue multi-application card products, such as financial social security card, all-access transportation card, all-access city card and citizen card, to meet the various needs of cardholders.

5.1.2 Financial Application Specifications

UnionPay IC cards shall be in accordance with UICS.

5.1.3 Industrial Application Specifications

Issuer shall decide whether to support other chip-based industrial applications. A multi-purpose card is currently an important driving force for the development of IC card. Using a multi-purpose card, Issuer can realize more benefits in a single chip card and provide better and more complete services to cardholders.

Card issuing systems vary considerably between regions, and Issuers shall satisfy corresponding industrial demands and requirements when partnering with regional industries. On the other hand, nationwide card issuing systems have their respective regulatory requirements. If Issuers issue industrial applications cards, they shall meet the requirements of corresponding industry standards and pass the corresponding industrial application testing.

5.1.4 Product Testing and Certification Requirements

Prior to issuing, financial IC card products shall pass a wide range of testing and certifications, including product testing, security testing and operational testing, and the testing report shall cover the entire validation period.

Product testing includes testing on:

- Magnetic-stripe
- UICS debit/credit (supporting low value payment)
- Contactless IC payment card
- Contactless low value payment extended applications
- Dual-currency low value payment IC card based on debit/credit applications
- Dual-currency electronic cash card based on UICS

Prior to issuance, each card function, based on the product characteristics chosen by Issuers, shall pass the testing by Bank Card Test Center. Card products that passed testing are publicized on the Bank Card Test Center's official website.

Security testing includes:

- Assessment on IC card security
- EAL4+ product testing

UPI recommends above-mentioned security tests as optional test items.

Operational testing includes:

- Testing on bank card's magnetic-stripe track data format
- Business data validation for personalized UICS credit card
- UPI network entry testing
- Testing on business functions specified by Issuers

5.1.5 Card Operating System Requirements

Card Operating System (COS) can be categorized into open and closed operating systems based on implementation. An open operating system is developed by complying with uniform standards and specifications, and its applications are separated from operating system for the ease of application development and deployment; a closed operating system

does not comply with uniform standards and specifications, and its applications are closely integrated into the operating system and application specifications must be standardized prior to card issuance. A common open operating system is Java Card, and a common closed operating system is Native Card. Issuers may select the appropriate card operating system according to actual needs and the characteristics of different card operating systems.

The advantages and disadvantages of open and closed operating systems are given in table below:

Table 1 Comparison between Open and Closed Operating Systems

Open Operating System	Closed Operating System
Compliance to general standards and specifications, and thus facilitating application development	Implemented privately with no uniform standards and specifications
Support card re-issuing	Does not support card re-issuing; all desired applications are required to be preset in the card
Use high-level language with slower performance	Use low-level language with faster performance
Require larger chip program space	Require smaller chip program space
Higher cost	Lower cost

For multi-application cards, open operating systems generally set up firewall between applications with international standards to ensure the independence and security of applications; while closed operating systems use underlying file system permissions to achieve isolations between applications. Please consult the card vendors for specific implementation mechanisms.

Operating system is an important part of the card software, and affects the cost and issuance strategy. Issuer should determine the type of card operating system according to product characteristics.

5.1.6 Communication Interfaces and Communication Protocols

Card vendors may ask Issuer if there are preference requirements in data transfer protocol. IC card specification supports two types of contactless data transfer protocols, in which these protocols specify the data transmission between the card and the terminal. The two types are T = 0 (protocol by character) and T = 1 (protocol by data block). Card acceptance

is not affected by which type of protocols is in use, because all acceptance devices must simultaneously support both protocols.

For contactless cards, relevant domestic and foreign specifications require contactless receiving terminals in the financial sector should support both Type A and Type B; contactless financial IC card should at least support either Type A or Type B.

For UPI IC card, Issuer may combine different data transfer protocols for product design, including contact, contactless and dual interface (which includes both contact and non-contact) cards. It is currently recommended using contact cards for financial social security cards and dual interface IC cards for fast payment market.

5.1.7 Card Technical Requirements

5.1.7.1 Memory

Cards issued by Issuers shall have sufficient memory to process all the card applications. As chip vendor will continue to optimize the UICS application code, recommended memory sizes may not be sufficient. Issuers can jointly determine appropriate memory allocation with vendors by providing specific requirements. In the market, common memory sizes are 16K, 32K and 72K.

Note: The two most important memory components for chip cards are Read-Only Memory (ROM) and Electronically Erasable Programmable Read-Only Memory (EEPROM), and EEPROM is more expensive than ROM. To help manage these costs, UPI works with chip vendors to ensure that the bulk of the UICS application is placed in ROM and to reserve EEPROM for Issuer's personalization and cardholder information.

5.1.7.2 Transaction Speed

Performance of various card applications will impact the total transaction time. Issuers should consider describing their transaction speed requirements to chip vendors. If Issuers are issuing cards for quick payment, then they will have greater card transaction speed requirements, and card transaction speed will be one of the factors for card procurement. Based on the interaction between the card and the terminal, qUICS transaction time shall not exceed 500 milliseconds and this time does not include the time required for static or dynamic signature verification using online authentication or qUICS terminal offline data authentication.

5.1.7.3 Crypto-Processor

If Issuer plans to use DDA and offline cryptogram PIN, the chip card is required to include crypto-processor. UPI strongly recommends that if Issuers select to use these functions, they shall use cards with crypto-processors, because built-in crypto-processor will considerably reduce transaction processing time for DDA and offline cryptogram PIN.

Note: UPI has no specifications on offline cryptogram PIN. Please refer to UnionPay Integrated Circuit Card Specification – Basic Specification Part 4: Debit/Credit Application Security Specification for details.

5.1.7.4 Indicators for Card Hardware

Technical indicators for card hardware generally include: chip type, chip architecture, storage capacity, operating voltage, operating temperature, EEPROM endurance, data retention time, physical and electrical requirements of the card, transfer rate and etc. Issuer should consider describing requirements for each indicator to chip card manufacturers.

5.1.8 Special Requirements

It is important to describe special requirements to potential vendors as special requirements of Issuer may require specialized development, and this will affect the costs and timelines. For example, some regional exclusive cards might require specialized regional algorithms, and Issuers should propose special card and chip requirements, such as procuring special dedicated chips.

5.2 Card Vendor Selection

While meeting with vendors and assessing their available products, Issuers should assess Vendor's qualifications and abilities in order to make decisions on subsequent formal cooperation. Assessment on qualifications and abilities includes:

- Qualification requirements
- Supplier's capacity
- Project experience
- Logistics requirements
- After-sales service requirements
- Purchasing quantity
- Integration services

5.2.1 Qualification Requirements

Basic qualifications for card vendors include:

- Pass and obtain Qualification Certificate for UPI Card Product Manufacturer
- Pass and obtain ISO 9001 quality system certification
- Pass and obtain ISO 14001 environmental management system certification
- Card vendors' products are required to pass product testing by BCTC (Bank Card Test Center)

Issuers may specify additional qualifications for suppliers.

5.2.2 Supplier's Capacity

Card vendor should provide a complete product supply program according to Issuer's overall development plan and market expansion needs for all kinds of cards. Card vendor should also

ensure the supply of cards to Issuer and supply in accordance with Issuer's requirements in chip, application, packaging and others.

5.2.3 Project Experience

The expertise of card vendors in financial and non-financial IC cards is largely from previous project experience. Issuers should consider project experience accumulated when choosing card vendors, in order to shorten product cycle in subsequent cooperation and to reduce project investment. Card vendors should provide a sound project implementation plan to ensure the successful completion of the project, and the project plan should take into account of integrity, rationality, progressiveness.

5.2.4 Logistics Requirements

The logistics of cards is in accordance with UPI's logistics requirements for bank cards, signs strict security and confidentiality agreements for bank card products transportation with freight forwarders, and purchases insurances for the cards. Issuers and card vendors should negotiate logistics and packaging costs, and should define a clear accountability mechanism accordingly regarding the losses, damages and other accidents during transportation of goods.

For the method of delivering finished cards to Issuers, it is recommended to use transit escort or air freight. Issuers should perform rigorous safety qualification validation to freight forwarders, and sign strict security and confidentiality agreements for bank card product transportation with freight forwarders, to ensure the delivery of cards to specified locations in a safely and timely manner.

If finished cards are delivered directly to cardholders, it is recommended that Issuers or card vendors to use registered mail at local post offices or other express mail services (EMS). Any finished cards that are not timely mailed or scrap cards should be stored in a dedicated warehouse.

For card refund processing, it is recommended post offices to mail cards and detailed electronic list together to Issuers after receiving card returns.

5.2.5 After-Sales Services Requirements

Issuers should require card vendors to contact and communicate with Issuers through establishing a regular or irregular visit system, and to track the quality of cards. When there are customer complaints or quality problems, card vendors should respond and resolve the problems promptly for ensuring seamless cardholder services.

Issuers may also require card vendors to promptly provide training, technical support and other professional services upon request.

5.2.6 Purchasing Quantity

Issuers should determine the quantity to be purchased from suppliers. Quantity is one of the key factors affecting price, which is usually a ladder pricing system based on the amount of procurement.

5.2.7 Integration Services

Issuers should outline the services that card vendors are required to provide, such as card production and personalization. These integration services will affect the overall pricing. Given IC card's characteristics of multi-purpose usages and progressive application architectural designs, card vendors must have strong technical support capabilities for continuing to provide development, production, personalized and integrated technical support according to application expansion of IC cards.

6 Card Issuance

This chapter elaborates activities required to issue UPI IC cards. It assumes that this card has already passed the card test by Bank Card Test Center and provides higher level information on related vendor activities as well as detailed information on issuer activities:

- Issuer pre-personalization activities

Issuer conducts card design, establishes personalization input files (include all data to be personalized into the card), generates and transmits personalization key to related party. For information on establishing personalization input files, please refer to chapter 7.

- Card manufacturer activities

Card manufacturer places card surface designs and non-encoded magnetic-stripes into cards, embeds chips and initializes applications.

Note: “Card Manufacturer” here generally refers to vendors involved in the chip manufacturing process, including the chip operating system; embedding chip module, plastic printing and initializing applications in the chip. It could be more than one vendor.

- Personalization activities

The personalization embosses or prints data onto the card, encodes the magnetic-stripe, and loads data into the chip.

- Post-personalization activities

Cards are checked to ensure personalization data is correct prior to sending the card to cardholder.

Issuer should regard pre-personalization, card making, card personalization and post-personalization activities as one overall process. It can be adjusted between activities according to the differences in products and personalization treatment (such as self-service personalization and outsource personalization), but it is not recommended to simplify the process.

This chapter is an overview of card issuance and personalization processing.

6.1 Overview

Card personalization has become much more complex with card’s migration to chips-based. This section summarizes the changes regarding to this process for helping Issuers to understand the scope of card personalization.

6.1.1 Card Types

UPI IC cards include standard cards (including contact, contactless and dual interface cards), Hybrid cards, special-shaped cards, mobile payment cards and other card types. For

different types of cards, vendors should develop and test cards to ensure that they meet UPI standards. Complex and innovative products may affect time and cost.

6.1.2 Application Requirements

UPI applications shall be uploaded to chip's ROM. Vendors should develop and test these applications to ensure that they meet UPI standards, and these works require time and resources.

Based on the supported UPI functions, UPI applications will require many new data elements in the chip. Data currently used to emboss and to encode magnetic stripes is still required. The processing and risk management of these new data has made personalization more complex.

6.1.3 Key Management

Based on functions Issuers plan to implement, UPI relies on public key algorithm and DES algorithm to perform key management. In addition, chip personalization must use DES algorithm to process new security requirement. These new activities require risk management methods and new key management control and program. As key management requires special technology, if Issuer has no sufficient internal technical reserves, it is suggested to contact UPI representatives for assistance.

For DES algorithm used in IC cards, it is recommended to adopt double-length key algorithm (3-DES).

6.1.4 Equipment Upgrade

Chip card personalization requires Issuers to introduce new tools, upgrade current personalization devices, integrate new tools into the current system and provide staff training. The new tools include:

- Personalization Preparation Tool

It assists Issuer to create personalization files, which will either be provided to personalization department or loaded to personalization device.

- Personalization Device

It is used to personalize cards and must be upgraded for chip card personalization. The device needs to add one read and write chip module (contact or contactless) to write all chip-based application data onto the chip.

- Personalization Verification Tool

Quality assurance test should be performed for personalized chip cards before issuing cards to cardholders. This tool is required to ensure chip data has been correctly loaded onto the card during personalization. For more information please refer to “Post-personalization” in section 6.5

6.1.5 Learning Curve

As magnetic stripe card technology has been applied in the payment field for more than 20 years, there had been plenty of time to perfect the procedures and train all party participants. However for chip cards, time and experience is required to build professional and smooth processing procedures. UPI suggests that Issuer should allow sufficient time to train personnel involved in project implementation. Contact UPI representatives for any training requests.

6.2 Issuer Pre-personalization Activities

Issuer mainly accomplishes the following activities during pre-personalization:

- Complete card design and provide it to card manufacturer;
- Set business strategy for personalization;
- Generate and distribute personalization key;
- Create personalization input files.

The following sections explain these activities.

6.2.1 Card Design

Issuers create their card designs and send them to card manufacturers so that designs can be applied to the card. Designs may be required modifications to accommodate the chip.

Card designs for different card types (such as special-shaped cards) will be different. Personalized card products will also affect the packaging design for components such as card chip and antenna module.

Reference: For requirements on chip placement and other modifications on chip cards, please refer to *UnionPay Integrated Circuit Card Specifications* .

Note: All new or modified card designs require UPI's approval to ensure chip cards are complied with the requirements such as locations of chip and signature panel.

6.2.2 Personalization Input File

Issuers shall create personalization input file before personalization. This file includes all the data required by the card:

- Coding data for magnetic-stripe;
- Emboss/printing data;
- UPI application data that are required to be placed in the chip;
- Other chip data. This may be any optional chip data required for applications supported on the card, such as loyalty point or stored value data.

Note: The current requirements on the embossing/printing and encoding of designated UPI cards are not affected by this Specification.

UPI suggests that Issuer purchases professional personalization preparation tool to help creating personalization input file. This tool can help:

- Collecting data from various sources, for example: retrieving account numbers and cardholder names from customer database;
- Generating chip data required on the card;
- Generating symmetric and asymmetric keys through the connectivity to HSM (Hardware Security Module). All sensitive data in personalization input file must be encrypted by Key Exchange Key (KEK).

Note: for details on KEK, please refer to UnionPay Integrated Circuit Card Specifications – Auxiliary Specifications Part 5: Debit/Credit Card Personalization Guide.

6.2.3 Personalization Key

DES key is required to perform the following security functions during personalization:

- Securing debit/credit application;
Used to secure debit/credit application between the time it is initialized and when it is sent to be personalized.
- Securing the data in personalization input file ;
Used between Issuer and personalization device to safely transmit sensitive data in the personalization input file.

6.2.3.1 Securing Debit/Credit Application

After the card manufacturer completes card settings, card applications must be blocked by a set of initial personalization DES keys to ensure the card remains secured from initialization to personalization. Without this protection, it may be possible for stolen cards to be personalized by an unauthorized party and be used inappropriately. Apart from blocking, these keys also verify personalization data loaded onto the card during personalization to ensure it is completed, undamaged and not manipulated. The initial personalized keys contain the following three keys:

Message Authentication Code Key (K_{MAC}) is used to calculate and verify the MAC value within personalization instruction for ensuring the legitimacy of the card and Issuer.

Encryption Key (K_{ENC}) is used to generate encrypted messages for IC cards and the encrypted message to verify Issuer's host. If the security level of personalization message requires encryption for the data field in personalization instruction, K_{ENC} is also used to decrypt the data blocks of this command.

Distributed Encryption Key (K_{DEK}) is used to encrypt all sensitive data transmitted to the card by personalization devices. The card first decrypts sensitive data using this key, and then loads the plaintext data onto the card.

These three keys are derived from a master DES key called Issuer Master Key (KMC). The initial KMC may be decided by negotiation between Issuers and card manufacturers, or solely generated by Issuers and passing it to card manufacturers and personalization departments. This key can be generated by Issuer using key management system, and it can also be generated by UPI or a third party representing Issuers.

6.2.3.2 Securing Personalization Input File

As personalization input file contains confidential information, sensitive data must be encrypted before being transmitted among various parties.

To accomplish this, Issuer must generate a Key Exchange Key (KEK) and share it with personalization devices. Issuer can generate this key by using personalization preparation tool or authorizing UPI or a third party to generate this key.

All sensitive data in the personalization input file to be transmitted by Issuer must be encrypted by KEK. These data include:

- Cardholder's offline PIN (if supporting offline PIN);
- ICC private key (if implementing offline enciphered PIN or DDA);
- Unique key (if supporting online card and Issuer authentication or issuer script processing).

The personalization device first decrypts personalization file data using KEK, then re-encrypts the file data using K_{DEK} , and loads the data onto the card.

6.2.4 Implementation

Pre-personalization requires the following implementation activities:

- Strategy
 - ✓ Create card design;
 - ✓ Complete business decision for personalization data preparation;
 - ✓ Decide the number of KMC to be generated; (UPI recommends generating at least one KMC for each card vendor)
 - ✓ Decide the number of KEK to be generated; (UPI recommends generating at least one KEK for each personalization provider)
- Operations
 - ✓ Complete card design and pass it to card manufacturer after being approved by UPI;

- ✓ Generate appropriate number of KMC and KEK;
- ✓ Create personalization input file and ensure all sensitive data are encrypted by KEK;
- ✓ Safely transmit KMC to card manufacturer and personalization provider;
- ✓ Safely transmit KEK to personalization provider.
- Risk Management
 - ✓ Issuer shall decide where the keys are generated according to whether personalization is outsourced to professional providers or implemented by Issuer itself.

6.3 Card Manufacturer Activities

This section introduces the activities that must be performed by card manufacturers.

6.3.1 Prerequisite

Before card production, card manufacturers must:

- Pass and obtain the certification of qualification to manufacture UPI cards;
Onsite inspection and card tests will be conducted by UPI qualification and certification office.
- Pass and obtain functionality certification by Bank Card Test Center;
- Receive Issuer's card design;
- Receive KMC from Issuer and load into Hardware Security Module (HSM), this key is used to generate three keys:
 - ✓ K_{MAC} — used to verify personalization data loaded onto card during personalization, and to ensure the data is complete and unmodified.
 - ✓ K_{ENC} — used to generate IC card encrypted messages and the encrypted message for host verification.
 - ✓ K_{DEK} — used to encrypt confidential data written onto the card during personalization process.

KMC is unique to each issuer. However, K_{MAC} , K_{ENC} and K_{DEK} are unique to each card.

6.3.2 Operations Performed

During chip card manufacturing, card manufacturers typically execute the following steps:

Step 1: Place card design to white cards.

Step 2: Embed chip onto plastic card base.

Step 3: Initialize debit/credit application, for example: set counter in the card at 0 and set basic data elements.

Step 4: Determine internal file structure of the card, create files and decide which records are stored in which data elements.

Step 5: Obtain a unique K_{MAC} using KMC and load into the card. This key is used to calculate and verify the MAC value within personalization instruction, thereby ensuring the security of applications from the time between initialization and personalization. It is also used to ensure that data is not manipulated during personalization process.

Step 6: Obtain a unique K_{ENC} using KMC and load into the card. This key is used to generate IC card encrypted messages and the encrypted message for verifying Issuer's host. It may also be used to encrypt and decrypt data blocks of personalization instruction.

Step 7: Obtain K_{DEK} using KMC and load it to the card. This key is used to encrypt/decrypt sensitive data before loading them onto the card during personalization.

Step 8: Safely transfer cards to Issuer, or directly transfer cards to personalization provider under Issuer's permission.

6.4 Personalization Activities

Issuers may process personalization in-house or outsource it to third-party personalization providers who passed certification and qualification requirements in accordance with Section 5.2.1. This section provides an overview of the personalization process.

6.4.1 Prerequisite

Prior to personalization begins, personalization system must be upgraded to support chip card personalization. If Issuer is using a third-party personalization service provider, the provider must pass and obtain the certification of qualification for personalization of UPI cards. However, if Issuer performs personalization in-house (or within the same company group), UPI's certification is not necessary.

HSM in personalization system should include following DES keys:

- KMC— personalization system uses KMC to generate following keys:
 - ✓ K_{MAC} — used to calculate and verify the MAC value within personalization instruction, and verify personalization data loaded onto the card during personalization, to ensure the data is complete and not manipulated.
 - ✓ K_{ENC} — used to generate IC card encrypted messages and the encrypted message for verifying Issuer's host. If the security level of personalized encrypted message requires encryption in the data blocks of personalization instruction, it may also be used to encrypt and decrypt these data blocks.

- ✓ K_{DEK} — used to encrypt confidential data written to the card during personalization.
- KEK — personalization system uses KEK to decrypt sensitive data in personalization input file.

6.4.2 Operations Performed

Below are the personalization steps taken by card personalization providers:

Step 1: Receive card from Issuer or card manufacturer. The card should already contain a card design, a chip with initialized applications, K_{MAC} , K_{ENC} and K_{DEK} are stored on the card and a non-encoded magnetic-stripe.

Step 2: Receive personalization input file from Issuer. This input file should contain all the card data, including embossing data (if applicable), printing data (if applicable), magnetic-stripe encoding data, chip's IC card data and other chip application data, such as loyalty point program or stored value. All sensitive data in the file should be encrypted by KEK.

Step 3: Upload the chip card and personalization input file to personalization system.

Step 4: Use KMC, stored in HSM, to derive K_{MAC} , K_{ENC} and K_{DEK} for each card.

Step 5: Use K_{MAC} and K_{ENC} to establish a secured personalization channel between personalization system and the card.

Step 6: If the security level of personalization command requires the use of MAC for the data within the command, K_{MAC} is used to calculate the MAC value within personalization instruction. If the security level of personalization instruction requires encryption on the data blocks within the instruction, K_{ENC} is used to encrypt these data blocks of the instruction.

Step 7: Use K_{DEK} to re-encrypt sensitive data.

Step 8: Load data into the chip and each data is placed in appropriate file. The chip uses K_{MAC} to verify personalization data has not been manipulated, K_{ENC} to decrypt data blocks with possible instructions and K_{DEK} to decrypt sensitive data, and then safely stores the data into the chip in plaintexts.

Chip structure designed by manufacturer can safely store sensitive data, and can ensure that unauthorized party cannot access it even though the data is in plaintexts. Card manufacturer (or chip manufacturer) has the responsibility to establish this security mechanism in the chip, and Issuer is not required to take any actions.

Step 9: Use another KMC to generate another set of K_{MAC} , K_{ENC} and K_{DEK} to replace the existing set within the card.

This is to prevent further personalization by unauthorized party, but it does not preclude Issuer's authorized post-issuance personalization data changes through "issuer script processing".

Note: For technical details, please refer to

UnionPay Integrated Circuit Card Specifications – Auxiliary Specifications Part 5: Debit/Credit Card Personalization Guide.

6.5 Post-personalization Activities

After card personalization, UPI recommends Issuer or its service provider to check the validity of personalized cards prior to distribution to cardholders.

Recommended post-personalization activities to be performed include:

- Determine whether personalization quality assurance is necessary;

UPI strongly recommends Issuer to perform quality assurance process prior to issuance for ensuring the validity of personalized cards. This check is very necessary to avoid cardholders encountering acceptance problems due to personalization error.

- Describe the requirements on personalization quality assurance process;
- Determine the implementation method: build an in-house personalization quality assurance tool, or purchase an existing tool from vendors, or have personalization providers complete this task;

For most Issuers, purchasing a tool is more feasible than building one in-house as there are many international or local providers specialized in card quality assurance tool. In the case of planning to purchase tools, Issuer should assess available products and select the most appropriate tools. UPI's testing tool package also provides a similar tool.

- Install and testing tools, ensure the validity of card personalization;

UPI recommends Issuer to perform extensive and detailed test on the first batch of personalized cards and use the test result of this first batch as basis to sample test subsequent batches.

7 Personalization Data Requirements

This chapter focuses on strategic and operational activities of financial IC card personalization and aims to help Issuers in creating personalization input file by providing in-depth technical information.

Issuers should be aware that the success of its product is driven by decisions as a whole and not by a particular one; therefore, good decision-making is very important. UPI, card vendors or personalization providers can provide assistance in many business and technical activities. For other activities required by Issuers, Issuers can assess what would be supported by vendors and what would be implemented in-house.

7.1 Personalization Data Templates

This section describes personalization data templates, which are defined in UnionPay Integrated Circuit Card Specification – Auxiliary Specifications Part 1 Personalization Template Based on Debit/Credit Application. There are a total of 12 templates, and they cover debit/credit applications over various combinations of contactless and contact small-value payment applications.

UPI is responsible for the interpretation, maintenance and update releases of personalization data templates. UPI should make timely updates to personalization data templates according to the development of card business.

UPI personalization data templates define all the data and key objects required during personalization process for UPI IC cards satisfying certain application functionality. The templates also define the principles and specific requirements of obtaining the values for these data objects, which serve as reference standards to Issuers for certain UPI IC card issuances.

UPI personalization data templates are mainly to:

- Help member institutions in choosing the correct card parameter values;
- Provide a reference template as IC card personalization guidance to member institutions;
- Ensure that products of member institutions are in accordance with UPI specifications.

Issuers may directly select to use one of the templates with matching application functionality, generate and organize data according to the template's defined data set and create personalization input file. Issuers may use a template and revise small amounts of data assignments according to their own business logic, thus creating their own dedicated template to serve as their personalization standards.

Data templates can be divided into the following three categories based on their defined card application functionality:

- Personalization template for defined standard debit/credit application data

Template 1 debit card: DDA, online PIN + signature, bypass PIN allowed;

Template 2 debit card: DDA, online PIN, bypass PIN not allowed;

Template 3 credit card: DDA, online PIN, bypass PIN allowed;

Template 4 credit card: DDA, signature, online control;

Template 5 credit card: DDA, online PIN, bypass PIN not allowed;

Template 6 credit card: DDA, offline PIN, bypass PIN allowed;

Template 7 quasi-credit card: DDA, online PIN, bypass PIN allowed;

Template 9 signature debit card (outside of Mainland China only): DDA, Online PIN + signature, bypass PIN allowed

Template 10 debit card and certain pre-paid card (U.S. only): DDA, online PIN + signature, bypass PIN allowed

- Personalization template for defined pure electronic cash application data

Template 11 pure electronic cash card;

- Personalization template for debit/credit application data supporting contact/contactless electronic cash

Template 12 common E-cash cards, version 1;

Template 13 common E- cash cards, version 2;

Template 14 dual-currency E-cash cards;

Template 15 E-cash cards of extended applications;

Template 16 E-cash cards that support enhanced functionality;

Template 17 online only cards with no E-cash functions.

If Issuers select standard debit cards, they may choose to use template 2; if they select standard credit cards, they may choose to use template 3; if they select pure electronic cash cards, they may choose to use template 11; if they select standard debit + electronic cash + contactless IC card payment, they may choose to use template 12; if they select standard credit + electronic cash + contactless IC card payment and require cardholders to input online PIN for authentication, they may choose to use template 13.

For details and specific definition on data templates, please refer to UnionPay Integrated Circuit Card Specification – Auxiliary Specifications Part I Personalization Template Based on Debit/Credit Application.

7.2 Personalization Data Requirements for Standard Debit/Credit

7.2.1 Application Selection

When a standard debit/credit card is presented to a terminal, the terminal determines which applications are supported by both the card and terminal by comparing the Application Identifier (AID) list stored in the card with the AID list maintained by the terminal. Terminal creates an application list supported by both terminal and card using directory selection method or AID list selection method. Directory selection method is mandatory for terminal and optional for card; AID list selection method is mandatory for both card and terminal.

If the card and the terminal both support directory selection method, terminal reads payment application list in the card through a file called payment system environment (PSE). Terminal compares applications listed in PSE with the applications it supports and establishes the candidate list – the terminal internal list that is supported by both card and terminal. If the card does not have PSE, the terminal reverts to use AID list selection method.

With AID list selection method, the terminal reads terminal application list in sequence and ask the card one by one if it also supports the application in its list. Terminal adds applications supported by both to the candidate list.

An application is then selected from the candidate list to conduct the transaction. One of the following methods can be adopted based on terminal capability:

- Cardholder selects application
 - ✓ Terminal supports cardholder confirmation

Terminal displays applications supported by both ends in a prioritized sequence defined by Issuer for cardholders to confirm.
 - ✓ Terminal supports cardholder selection

Terminal displays all applications supported by both ends in a prioritized sequence defined by Issuer for cardholders to select from.
- Terminal automatically selects application

Terminal automatically selects the application with the highest priority by using application priority indicator.

Terminal shall support the application selection methods defined in *UnionPay Integrated Circuit Card Specification – Basic Specification Part 3 Debit/Credit Application Terminal Specification*. Issuers may investigate all kind of terminals on the market to determine terminal's abilities in supporting application selection. Investigation should focus on display type, storage capability, terminal integrity and merchant location.

Issuers must set up card applications and support application selection. This section describes related decision, business and technical activities for implementing application selection:

- Application selection data;
- Cardholder confirmation;

For each application in the card, Issuers must decide whether application can be selected without cardholder's confirmation, or terminal is allowed to automatically select application with the highest priority.

- Directory selection method and AID list selection method.

Card must support AID list selection method. Directory selection method is optional for card and Issuers must decide whether card should support this function.

7.2.1.1 Application Selection Data

Application selection data is part of personalization data. Issuers should ensure that appropriate data are set up for application selection, and these data include:

- Application identifier (AID)
- Application label
- Application preferred name
- Issuer code index table (if application preferred name exist)
- Application priority indicator
- Preferred language

7.2.1.1.1 Application Identifier

Application identifier (AID) is a string of numbers representing application type and it is used in application selection. Terminal compares the AID list stored in the card with the AID list it supports and builds a candidate list – a terminal internal list supported by both the card and terminal. Personalization must assign each card application an AID.

AID consists of two parts:

- Registered Application Provider Identifier (RID)

This component identifies the payment scheme. UPI RID is A000000333.

- Proprietary Application Identifier Extension (PIX)

This component represents application. UPI PIXs are listed in the table below.

Table 2 UPI Proprietary Application Identifier Extension (PIXs)

Application	PIX
Debit application	010101
Credit application	010102
Quasi-credit application	010103
Electronic cash application	010106

If there are more than one application using the same AID in a card (for example, two credit applications), card AID must attach a suffix to show the difference. The suffix notify the terminal that there are more than one application in the card using the same AID, ensuring that all the applications in the candidate list will be assigned with an unique AID.

Suffix is 2 numbers appended to the end of AID, for example, if Issuer supports two credit applications, the following format can be used to personalize these two application AIDs:

- A00000033301010201 – first credit application
- A00000033301010202 – second credit application

Note: this suffix does not determine application priority.

7.2.1.1.2 Application Naming

Terminal may display the names of applications to cardholders so that cardholders can select or confirm which application they want to use on a given transaction. This name is the medium for cardholders to identify card applications, and is consists of two data elements: application label and application preferred name. This section provides guidance in helping Issuers to clearly understand how UPI application supports and promotes application selection.

7.2.1.1.2.1 Application Label

Application label is a mandatory data element and must be stored in the card. This data element identifies application, and application selection process will display this data as application name to cardholders unless the card contains application preferred name.

Application label must contain the word “UICS” and clearly explain the payment functions, product or both in an effort to differentiate these card applications. The data can contain space and be correctly identified.

7.2.1.1.2.2 Application Preferred Name

Application preferred name is an optional data element and can be placed in the card. It is the preferred naming convention for applications. If the terminal is capable of displaying

applications, it will search and display application preferred name; the terminal should display application label if application preferred name does not exist.

If application preferred name exists, terminal checks another data element in the card – issuer code index table to determine whether it supports the character set (defined by ISO 8859) corresponding to the application preferred name in order to display its name correctly. If the terminal does not support character set defined in issuer code index table, terminal cannot properly display application preferred name, and in this case, terminal will display application label instead.

Application preferred name data should be the same with application label. However, UICS should not specify definite naming rules for application preferred name, Issuers may use this data field to place its brand name.

7.2.1.1.3 Application Priority Indicator

When card supports more than one payment application, application priority indicator must be assigned to each application to distinguish its priority. This priority indicator decides the display sequence of applications supported by both the terminal and the card during cardholder selection process. If terminal does not support cardholder selection, terminal will automatically select applications supported by both ends according to priority indicator.

Issuers can assign the same priority to multiple applications. However, if the terminal does not support cardholder application selection, the terminal will decide using the application mutually-supported by both card and terminal and this may confuse cardholders.

Note: If application priority indicator is used, chip payment application with mapping in magnetic-stripe must be set as the highest priority. If non-UPI application is also present, UPI payment application is required to have higher priority than non-UPI applications.

7.2.1.1.4 Preferred Language

To provide cardholders with customized experience at the point of transaction, terminal is capable of displaying transaction information in cardholder's preferred language. Issuer can personalize the card through "preferred language" data element, and set one or multiple preferred language. The language setting supports up to four preferred languages. If more than one language is provided, priority must be assigned to each language.

Note: "preferred language" data element is encoded according to ISO 639, these codes adopts lowercase to represent language. Lowercase must be used to set preferred language in the card.

7.2.1.1.5 Implementation

Related card data elements for application selection require the following implementation activities:

- Policy

Since application preferred name, application priority indicator and preferred language are all optional, Issuers may decide whether to include them in the card. Since these data elements provide greater control over Application Selection and a more customized experience to cardholders, it is recommended to include them on the cards.

- Operations
 - ✓ Decide what applications reside in the card;
 - ✓ Determine how application name should be displayed to cardholders during application selection;
 - ✓ When multiple payment applications are supported, use application priority indicator to distinguish priority sequence among UPI applications; Magnetic-stripe applications that corresponded to the account embossed or printed on card surface should be given the highest priority;
 - ✓ If preferred language is supported, decide which languages to support in the card; if more than one language is supported, priority sequence of languages needs to be determined.
- Technology

Issuers should include the following data elements in the personalization file to ensure that these data will be personalized:

 - ✓ If two applications use the same AID in the card, an unique suffix should be added to the end of AID;
 - ✓ Application label;
 - ✓ Application preferred name and issuer code index table (if supported);
 - ✓ Application priority indicator (when more than one payment application is supported);
 - ✓ Preferred language (if supported, and determine priority if more than one language is provided).

7.2.1.2 Cardholder Confirmation

Issuers may request cardholders to confirm UPI applications. To implement this option, cardholders are requested to interact with terminal at the point of transaction, and select displayed application by pressing buttons on the terminal.

Issuers can regard cardholder confirmation as a service option selected by customer. It helps the customer to understand which application is used to conduct the transaction.

Note: if card only supports one payment account, then cardholder confirmation is not required.

7.2.1.2.1 Implementation

Activities related to cardholder confirmation for policy makers, operations department and technical staffs include:

- Policy
 - ✓ Determine whether to request cardholders to confirm selection for each payment application in the card.
 - ✓ Issuers can choose to set up applications that required cardholder's confirmation. If terminal does not support cardholder confirmation, then applications required cardholder's confirmation will not be supported. Then the terminal automatically selects the application with highest priority and does not request cardholder's confirmation. This type of terminals does not allow cardholders to select applications to execute transactions and cannot choose to use card applications that required cardholder's confirmation.
 - ✓ If Issuers have such market demands for cards requiring cardholder's confirmation and terminals supporting cardholder's application selection, please contact UPI representatives to understand Issuer's market demands.
 - ✓ If Issuers have only one payment application on the card, it is recommended that Issuers do not require cardholder's confirmation of the application since not all terminals support this functionality. For multi-application cards, it is recommended that the application with the highest priority (magnetic-stripe application with mapping) does not require cardholder's confirmation, so that to minimize acceptance problems when the terminal only supports cardholder's confirmation.
- Operations

If applicable, include cardholder's confirmation in requirements to card vendors.
- Technology

Add cardholder's confirmation parameters to personalization file and ensure the personalization of these parameters.

7.2.1.3 Directory Selection and AID List Selection Method

UPI has designed two different methods to implement application selection:

- Directory Selection Method

This method is optional for the card and mandatory for the terminal; terminal should attempt this method first by reading the payment system environment

(PSE) file from the card. For card supporting directory selection method, PSE is a high-level directory file containing at least the list of all payment applications in the card. Terminal compares this list with the list maintained by it and adds applications supported by both the card and terminal to the candidate list it created. Depending on whether the terminal supports cardholder's application selection or terminal's automatic application selection, application is then selected for transaction.

- **AID List Selection Method**

This method is mandatory for both the card and terminal. Using AID list selection method, terminal sends a command to the card for each application it supports, enquiring whether the card also supports this application; if the card response indicates the application in question is contained in the card, and then the terminal will add this application to the candidate list. Depending on whether the terminal supports cardholder's application selection or terminal's automatic application selection, application is then selected for transaction.

Since directory selection method is optional for card, it will switch to AID list selection method when card does not support directory selection method.

7.2.1.3.1 Implementation

Activities related to directory selection method and AID list selection method includes:

- **Policy**

Decide whether the card supports directory list method in addition to mandatory AID list selection method.

UPI believes it is sufficient to support AID list selection only, unless terminals in Issuer's market support many applications. Although there are many other factors that influence the total transaction time, directory selection method can reduce the duration of application selection when terminal supports many applications.

- **Operations**

Define requirements for application selection method and collaborate with vendors to ensure the card contains appropriate functions.

If Issuers plan to implement directory selection method, it is very important to make the requirements clear to vendors as cards from some vendors do not support this function currently.

- **Technology**

If Issuers plan to implement directory selection method, cards must contain a PSE; otherwise, it is not required to establish PSE in the card. For technical details on PSE, please refer to EMV and UCIS files.

7.2.2 Application Initialization

After application selection, the terminal must request card to read its application data. Depending on transaction characteristics, card may indicate different data or supporting functions. For example, different cardholder verification methods can be set for domestic and international transactions, and card can send different cardholder verification methods (CVM) list to the terminal. For information on CVM list, please refer to section 7.2.5.1 “Cardholder Verification Customization”

Issuers can perform a check that allows terminating the transaction based on the requirements related to domestic or international usage. For example, Issuers can restrict the application to domestic usage. In the event that the card is used outside of the country, the transaction is terminated and the terminal returns to the Application Selection phase to determine if another application can be used for the transaction.

Issuers should prepare Processing Options Data Object List (PDOL) based on application support. Terminal prepares initialization data according to PDOL return data. PDOL should generally be prepared in accordance with *UnionPay Integrated Circuit Card Specification – Auxiliary Specifications Part 1 Personalization Template Based on Debit/Credit Application*.

During transaction process, in addition to PDOL contains the data of authorized amount, the subsequent card behavioral analysis will also include authorized amount as inputs. If the terminal requests online but the two authorized amounts are inconsistent, then the card will also go online and process with standard debit/credit process flow; but if the terminal requests offline approval, the transaction will be rejected.

7.2.2.1 Implementation

Activities related to application initialization include:

- Policy
 - Decide different process rules (if applicable, for example, different affects to transactions based on terminal type, business environment, cross-regional use and etc.)
 - Decide whether to restrict the card for domestic or international usage.
- Operations
 - Communicate with vendors for Issuer’s special processing requirements. Clearly state what requirements are important as some vendors may not support this function.
- Technology
 - If Issuers would like the card to send different data elements based on transaction characteristics or terminate transactions based on domestic or overseas usage, the card must contain PDOL. PDOL contains data object tags

that are sent to the card by the terminal, and any terminal data requires processing is indicated in PDOL, for example: to implement domestic and overseas check, PDOL must contain terminal country code tag.

7.2.3 Offline Data Authentication

Data in the card can be verified by public key technology at the point of transaction. This verification is called offline data authentication. Offline data authentication protects card data against being manipulated and helps to find counterfeit cards.

There are two types of offline data authentications: static data authentication (SDA) and dynamic data authentication (DDA). DDA is further subdivides into standard DDA and combined DDA/generate application cryptogram (CDA). SDA can ensure data selected by Issuers will not be manipulated after personalization. Compared to SDA, DDA can additionally protect against copying of chip data in offline environment as the card generates a unique cryptogram for the terminal to perform verification. Since DDA requires more encryption calculations in the card, it is necessary for the card to be equipped with crypto-processor. CDA combines dynamic signature generation and card application cryptogram generation to ensure the returned application cryptogram during card behavioral analysis comes from a valid card.

It is optional for card to support SDA, DDA and CDA. If card supports DDA or CDA, then it must also support SDA; therefore SDA processing can be performed on terminals that do not support DDA or CDA.

If the card does not support any offline data authentication method, all transactions will be sent for online authentication requests.

This section includes the following topics to help Issuers in conducting offline data authentication:

- Overall policy making
- SDA implementation and technical activities
- DDA implementation and technical activities

7.2.3.1 Overall Decision Making

This section describes policy making requirement to support offline data authentication:

- Data to be validated
Determine the data is to be validated in SDA or DDA.
- Actions for offline data authentication at point of transaction
Decide what actions should undertake according to offline data authentication result at point of transaction (approve offline, decline offline, send online).
- Public key strategy

Formulate a series of strategies on public key mechanism to implement offline data authentication.

Note: For the sake of Risks, SDA is not recommended according to the request of the people's bank of China

7.2.3.1.1 Data to be validated

SDA validates a set of data elements to ensure that they are not manipulated since card issuance. Issuers should determine which data elements are required to be validated. UPI recommends that if the card also supports DDA, add AIP to the static authentication data set.

If using DDA, the card may involve terminal data in signature generation. This data adds variability to signature and provides protection against data copying. The card requires terminal data must include an unpredictable number, and of course, the card may also request additional data from the terminal, such as transaction amount.

Standard DDA designates terminal data elements that are involved in signature generation through DDOL while CDA does so through CDOL and PDOL.

DDA also validates static data in the card, and it verifies the same set of static data as SDA. UPI requires application transaction counter to participate in dynamic signature generation in order to increase the variability of signature.

In accordance with the requirements by Bank Card Test Center, even if the card supports DDA and CDA, terminals must verify SDA, and the card must also support SDA.

SDA static signature data lists vary between standard debit/credit and contactless electronic cash. Standard debit/credit should include 5F34 in accordance with UPI specification, while contactless electronic cash transactions do not include 5F34, in order to avoid duplication in return data.

For cards supporting contactless electronic cash transactions, the last data in return AFL data should have the shortest length, in order to minimize the risk of card rejection.

Issuers can also include other data element to meet risk management requirements.

7.2.3.1.2 Offline Data Authentication Risk Management Check

A series of risk management checks can be performed during terminal action analysis of offline data authentication results. Issuers need to decide what actions should be taken on their own benefits and protections when risk management conditions have been triggered. These actions include approve offline, decline offline and send online. Issuers set these check points through issuer action code (IAC) and personalize it into the card.

If Issuers decide to send online, Issuers still need to decide what actions should be taken if they are unable to send online: approve offline or decline offline.

The following table gives recommended actions to these policies by UPI:

Table 3 Offline Data Authentication Risk Management Check

Condition	Description	Recommended Action
Offline data authentication was not performed	During offline data authentication, SDA or DDA was not performed. This will occur when terminal does not support offline data authentication.	Send online. If online is not available, decline offline.
Offline SDA failure	Terminal performs SDA and it fails.	Send online. If online is not available, decline offline.
Offline DDA failure	Terminal performs DDA and it fails.	Send online. If online is not available, decline offline.
IC card data loss	Terminal checks card data required by offline data authentication and finds part or all of data was lost. This may happen in the following situation: card personalized incorrectly, fraud perpetrator deleted part or all card data related to offline data authentication.	Send online. If online is not available, decline offline.
CDA failure	Terminal performs CDA and it fails.	Decline offline.

7.2.3.1.3 Public Key Strategy Decisions

This section focuses on the overall public key management decisions in business, policy and risk management. These decisions apply to Issuers implementing offline data authentication and offline enciphered PIN.

Participants who conduct verification using public key technology must generate one or more RSA issuer public key pairs. Public key pair includes a public key and a private key, and they are mathematically related to each other. Message encrypted (or signed) by one key can be decrypted (or verified) by the other key.

Issuers send their public keys to UPI certificate authority (root CA) to generate issuer public key certificate. Certificate authority is trusted administration that issues and revokes certificates, and voluntarily pledges the consistency between its issued certificates and their corresponding keys. An issuer certificate (also called issuer public key certificate) is comprised of Issuer's public key signed by UPI private key.

Note: An independent public key is required for each UPI application utilizing public key technology.

Before applying to the certificate authority for public key certificate, Issuers need to make a series of key management decisions that include:

- Number of issuer public keys to be generated;
- Length of the key to be generated;
- Expiry date of each key;
- Exponent of each key;
- When receiving several public key certificates from certificate authority, Issuers shall choose the most appropriate certificate to load onto the card.

The Issuer's risk management staff must analyze the trade-off between cost and benefit and decide the best risk management model. UPI's recommendations in this section assist Issuers in their decision making. In addition to these recommendations, UPI suggests Issuers to follow new development in public key technology and consult public key experts.

7.2.3.1.3.1 Number of Issuer Public Key to be Generated

Issuers must decide the number of issuer public keys to be generated. In general, the more the public keys, the greater the security; however every public key requires a management fee. Issuers should consider: safety storage of all keys, requirements of installing monitoring devices, management of public key expiry data and filing of expired keys.

Based on risk management system, Issuers should have the flexibilities in deciding whether to use one key for all cards over a particular serial of cards, or to use multiple keys for mitigating risks. UPI does not limit the number of issuer public/private key pairs used by Issuers as it does not impact validation process of an IC card at terminals. Based on its business requirements, Issuers can select one or more issuer public/private key pairs for card issuance under consideration of some factors such as IC card product type and location of cardholders. However, for security concerns, UPI requires Issuers to apply for issuer public key certificate with their bank identification number (BIN) and use different issuer public/private key pairs for different BIN numbers in card issuance. This method can better control security and business risk when issuer signature private keys are leaked.

7.2.3.1.3.2 Issuer Public Key Length

Issuers shall decide the length of public keys to be generated. This section describes recommendations related to the decision of issuer public key length.

Note: Public key length is generally represented in bits, for example: 1024 bit, and it must be multiples of 8.

In general, the longer the public key, the greater the security. However, longer public key requires more transaction time while shorter public key has less transaction time with less security than longer public key.

If Issuers plan to support DDA, UPI recommends that Issuers assess public key processing performance of their products jointly with potential card vendors, and obtain accurate data. Then, Issuers can propose transaction time requirements.

Issuer public key must be signed by root CA private key. Root CA can use as many CA private keys as possible to sign issuer public key, given the following two conditions are satisfied:

- Issuer public key expiry date is earlier than or the same as root CA public key expiry date;
- Issuer public key length is shorter than or equal to root CA public key length.

The only requirement for issuer public key length is: the length of public key generated by Issuers is shorter than or equal to the longest root CA public key.

Currently, the longest root CA public key has 1984 bits. UPI will assess market security risk and related factors periodically and introduce new public key pairs in a planned manner. Please contact UPI representatives for latest information.

7.2.3.1.3.3 Issuer Public Key Expiry Date

This section provides recommendation on how to define proper public key expiry date.

Please refer to UPI official website for the current expiry dates of different root CA public key pairs. Issuer public key expiry date should be earlier or the same as root CA public key expiry date.

For example, if the expiry date of a public key submitted to root CA by Issuer is 2015, root CA will only use 1408 bits or longer private key to perform signature (assume this issuer public key length is less than or equal to 1408 bits).

In addition, issuer public key certificate should not expire earlier than the card. It is recommended that personalization device be set so that it is able to verify the validity of the relationship between root CA, issuer key and card expiry date.

7.2.3.1.3.4 Exponent of Issuer Public Key

Each issuer public key has an exponent, and it is used during RSA calculation associated with offline data authentication to restore issuer public key from public key certificate. Issuers can use one of the two following exponents:

- 3
- $2^{16}+1$

There is no significant difference in security level by using exponent of 3 or $2^{16}+1$; however, calculation time for the latter exponent is significantly longer than the former. Hence, UPI recommends using exponent of 3. Using other exponent will not enhance transaction security.

7.2.3.1.3.5 Issuer Public Key Certificate

This section helps Issuers to decide which certificate to place in the card when receiving multiple certificates from root CA.

As described previously, root CA uses all root CA private keys with length greater or equal to issuer public key to sign issuer public key certificate (At this time, issuer public key expiry date should be earlier than root CA public key pair expiry date). Issuers will only receive multiple certificates from CA when the issuer public key length is shorter than or equal to 2 (or more) root CA public key pairs and its expiry date is earlier than the expiry date of these public key pairs.

Root CA also provides CA public key exponent when sending issuer public key certificate. It is used to identify the root CA public key pair that signed this issuer public key certificate. “CA public key exponent” data element is required to be personalized onto the card.

7.2.3.1.4 Summary of Decisions

Decisions related to offline data authentication include:

- Decision on data to be validated using SDA or DDA.
- Decision on preferred actions to be taken (approve offline, decline offline, send online) when exercising risk management checks associated with offline data authentication results at the point of transaction.
- Determine public key management strategy, refer to the table below for details:
Table 4 Public Key Management Strategy

Public Key Management Decision	Recommended Strategy
Determine the number of public key to be generated.	UPI recommends: Issuers generate an issuer public key for each BIN.
Determine issuer public key length. The length of this key should be multiples of 8 and shorter than or equal to the longest root CA public key.	UPI recommends: issuer public key length to be longer than or equal to the shortest valid root CA public key.
Determine issuer public key expiry date.	Issuer public key expiry date must be later than or equal to UPI application expiry date and earlier than or equal to latest valid CA public key expiry date.
Determine issuer public key exponent.	Issuers can only select one exponent from 3 and $2^{16}+1$.

Public Key Management Decision	Recommended Strategy
Determine which certificate to be placed in the card if issuer receives more than one issuer public certificate from root CA.	The issuer should make decision based on risk management and customer service system. UPI recommends using longest certificate.

Note: These policies are also applicable to offline enciphered PIN.

Issuers shall submit public key certificate application form and issuer public key input file that comply with root CA requirements to root CA when applying for public key certificate. If the format of issuer public key input file is incorrect, root CA will not be able to issue certificate, and thus Issuers are required to re-submit files, whereby delaying the card issuance project.

UPI suggests Issuers should submit public key certificate application to root CA in early stages of the project to avoid delay.

In the case DDA is supported, it is required to generate an IC card public key pair using RSA technology. Each UPI application requires generating a public key pair.

Note: Length of IC card public key pairs must be shorter or equal to issuer public key length.

7.2.3.2 SDA Operation and Technical Activities

SDA related activities include:

Table 5 SDA Activities Performed

Steps	Responsible Party	Activities
1	Operations	Discuss public key and certificate generation matters with UPI representatives. This is an important step to assist Issuer in clarifying the requirements and timelines arrangement, and thus minimizing duration of delays.
2	Technical	Decisions on number, length, exponent value and expiry date of public keys; generate issuer public/private key pair using RSA technology.

Steps	Responsible Party	Activities
3	Operations	Issuers shall submit public key certificate application form and issuer public key input file that comply with root CA requirements to root CA for applying public key certificate. If the format of issuer public key input file is incorrect, root CA will not be able to issue certificate, and thus Issuers are required to re-submit files, whereby delaying card issuance project.
4	UPI	Root CA uses suitable root CA private key to sign issuer public key and generate issuer public key certificate. Root CA can use as many CA private keys as possible to sign issuer public key give the two following conditions are satisfied: Issuer public key expiry date is earlier than or the same as root CA public key expiry date; Issuer public key length is shorter than or equal to root CA public key length.
5	UPI	UPI returns issuer public key certificate to Issuers. Upon receiving multiple certificates, Issuers should decide which certificate is to be placed onto the card based on risk management and customer service system.
6	Technical	Develop a procedure to generate the Signed Static Application Data (SAD) for each UPI application. SAD is a hash result of the data obtained by issuer private key after validating by SDA. This procedure should be part of pre-personalization activities. The hash is created using the Secure Hash Algorithm 1 (SHA-1). SHA-1 is a public algorithm and standardized in FIPS 180-1.
7	Technical	Add issuer public key certificate, issuer public key exponent, issuer public key remainder (if required), CA public key exponent and SAD to personalization input file together with other personalization data. Issuers shall carefully manage this information, ensuring the correct certificate and SAD is placed in the correct card. If incorrect information is installed in the card, offline data authentication will fail and potentially impact acceptance process.

7.2.3.3 DDA Operation and Technical Activities

DDA related activities include:

Table 6 DDA Activities Performed

Steps	Responsible Party	Activities
1-6		Refer to “Table 5 SDA Activities Performed”
7	Technical	Generate an IC card public key pair using RSA technology; each UPI application requires one public key pair. Note: Length of IC card public key pair must be less than or equal to issuer public key length.
8	Technical	Generate an IC card public key certificate for each UPI application. Issuers generate chip card public key certificates by signing IC card public keys using issuer private key. This operation is part of pre-personalization activities.
9	Technical	Add issuer public key certificate, issuer public key exponent, issuer public key remainder (if required), CA public key exponent, SAD, IC card private key, IC card public key certificate, IC card public key exponent, IC card public key remainder (if required) and DDOL to personalization input file together with other personalization data. This information should be carefully managed to ensure that the correct data is placed in the card. Because IC card private key is sensitive data, it must be encrypted and stored in the personalization input file. It should be encrypted by KEK together with other sensitive data.

7.2.4 Processing Restriction

Terminal determines whether a transaction is allowed to continue through checking UPI application data with processing restrictions. Contents to be checked include application effective date, application expiry date, application version and other control restriction conditions defined by other issuers. Issuers can use Application Use Control (AUC) to restrict the card to domestic or international usage, or limit its usage in cash, shopping or service.

Processing restriction should implement the following checks:

- Application version check (mandatory)
- Application usage control check (optional)

- Application effective date check (optional)
- Application expiry date check (mandatory)

Following explanations are focused on optional checks.

7.2.4.1 Application Usage Control Check

Application usage control check is similar to service encoding check implemented in magnetic-stripe transaction. Chip data element “Application Use Control” (AUC) specifies where (domestic or international) and what type of transactions the card can be used. Terminal checks AUC to determine whether the transaction can continue.

Issuers shall decide the following conditions and personalize them into the card:

- Domestic cash transaction effective
- International cash transaction effective
- Domestic product effective
- International product effective
- Domestic service effective
- International service effective
- ATM effective
- Terminals other than ATM effective
- Allow domestic cash back
- Allow international cash back

Issuers should continue to use “service code” as the basis to usage control, and ensuring this data element is consistent between information in magnetic-stripe and chip.

7.2.4.2 Application Effective Date Check

Terminal compares application effective date in the card with current date. If this check is supported, “application effective date” data element must be personalized to the card. This date must be the first day of month in which the applications are loaded onto the card.

In case of “application not effective yet”, Issuers shall set IAC to decide: to decline offline or send online; if online is not available, whether to decline offline.

7.2.4.3 Processing Restriction Risk Management Check

A series of risk management checks are implemented during terminal action analysis based on processing restriction results. For each check, Issuers shall decide what actions should be taken for their benefits when risk management conditions have been triggered. These actions include approve offline, decline offline and send online. Issuers set these check points in issuer action code (IAC) and are personalized onto the card.

If decision is made to send online, Issuer shall decide what action to take if online is not available: approve offline or decline offline.

The table below provides UPI recommendations to these policies:

Table 7 Processing Restriction Risk Management Check

Condition	Description	Recommended Action
IC card and terminal have different version.	Terminal checks if card application and terminal application are different in version.	Ignore this condition, continue to process transaction.
Application expired	Terminal checks if card application expiry date is earlier than terminal current date.	Send online: if online is not available, decline offline.
Application not effective yet	Terminal checks if card application effective date is later than terminal current date.	Send online: if online is not available, decline offline.
Requested service is not allowed	Terminal checks card AUC and determines if current transaction supports card usage.	Decline offline

7.2.4.4 Implementation

Activities to implement processing restriction include:

- Policy
 - ✓ Determine whether to support AUC check;
 - ✓ Determine whether to support application effective date check;
 - ✓ Determine on preferred actions to be taken (approve offline, decline offline, send online) when performing risk management checks based on offline data authentication results at the point of transaction.
- Operation
 - Determine card AUC parameters.
- Technology
 - Generate data required by processing restriction and add it to personalization input file. This data includes application version, application effective date, application expiry date, AUC and etc.

7.2.5 Cardholder Verification

UPI allows Issuers to customize cardholder verification requirements according to the nature of transactions through the use of Cardholder Verification Method (CVM) list in the card. In addition to current CVM of signature and online PIN, UPI has introduced offline plaintext PIN verification method.

Following activities should be implemented for cardholder verification:

- Cardholder verification customization
Decide which CVMs are supported in the card.
- Terminal action analysis of cardholder verification result
Based on the outcome of cardholder verification as well as card and terminal parameters, terminal decides what actions to take (approve offline, decline offline or send online).
- Offline PIN
Develop specific policy associated with offline PIN, such as the number of PIN tries are allowed.

7.2.5.1 Cardholder Verification Customization

Issuers shall determine which cardholder verification methods the card will support. Issuers can set up the card to support multiple CVMs, as shown in the table below:

Table 8 Cardholder Verification Methods

CVM	Description
Signature	This method operates in the same manner as in the magnetic-stripe environment whereas a cardholder sign on sales slip and merchant compares the signature with the one on the card.
Offline plaintext PIN	This is a new method in which the terminal prompts the cardholder for a PIN and transmits the cardholder-entered PIN to the card in plaintext. The card then compares the cardholder-entered PIN to the offline PIN stored in the chip.
Offline plaintext PIN and signature	A combination of offline plaintext PIN and signature in which both methods are used to validate cardholders.

CVM	Description
Online PIN	This method operates in the same manner as in the magnetic-stripe environment whereas the cardholder-entered PIN is encrypted by the terminal using DES key technology and sent online to Issuers for verification.
No CVM required	This method operates in the same manner as in the magnetic-stripe environment where authorization does not depend on cardholder verification. Certain merchant environment allows the omission of cardholder verification, for example selected cardholder mobile terminal.
CVM failure	This method allows Issuers to select conditions under which CVM processing will default to a failure.
ID Verification	In this method, terminal indicates cardholder to show ID, and it displays ID type and number stored in the card to cashier for comparison.

Once the types of CVMs to support are decided, Issuers shall:

- Prioritize them to create a CVM list

The terminal will use this list during cardholder verification to execute all selected CVM in a priority sequence.

- Decide under what conditions each CVM is applicable, for example:

- ✓ Always required
- ✓ If terminal supports this particular CVM
- ✓ Based on transaction amount

This transaction amount requirement has the same currency as card application currency. For example, Issuers may set up to use signature for transaction amount less than 20 USD Dollar and use offline plaintext PIN for transaction amount greater than 20 USD Dollar .

- If current CVM fails, decide whether to apply the next CVM.

Below is a simple example of CVM list. An issuer would like to perform cardholder verification in the following order.

- Online PIN for all ATM transactions and cash back transactions.
- Offline PIN for POS transactions if terminal supports.

- Signature for POS transactions if terminal does not support offline PIN,
- CVM is not required if terminal does not support offline PIN or signature.

Table 9 CVM List Example

CVM Priority	Condition	Notes
Amount X	0	This CVM list does not check transaction amount
Amount Y	0	This CVM list does not check transaction amount
CVM entry 1		ATM transaction uses this CVM entry
CVM condition	01- if cash or cash back	
CVM type	000010b- encrypted verification online PIN	
CVM code	1b- cardholder verification fail if fails	
CVM entry 2		POS transaction uses this entry If terminal supports offline plaintext PIN check, execute this CVM
CVM condition	03- if terminal supports	
CVM type	000001b- offline plaintext PIN verification	
CVM code	1b- cardholder verification fail if fails	
CVM entry 3		If terminal does not support offline plaintext PIN check, execute this entry. If terminal supports signature collection, execute this CVM
CVM condition	03- if terminal supports	
CVM type	011110b- signature	

CVM Priority	Condition	Notes
CVM code	0b- execute next CVM if fails	
CVM entry 4		If terminal does not support offline plaintext PIN check and signature, execute this entry. CVM cannot fail
CVM condition	00- always	
CVM type	01111b- does not require CVM	
CVM code	1b- cardholder verification fail if fail	

7.2.5.1.1 Implementation

This section outlines the activities related to cardholder verification customization.

- Policy

Determine which CVMs Issuers are planned to support, in which signature verification method must be supported and “CVM is not required” method can only be placed at the end of CVM list. If Issuers would like their cards to be used on ATM that only supports online PIN and other unattended devices, they must add online PIN to the CVM list.
- Risk Management

Issuers should determine whether they support other CVMs in addition to current CVM based on commercial analysis and market impact. However, Issuers should seriously consider supporting offline plaintext PIN verification method to reduce the risk of stolen cards.
- Operation

Create CVM list, including determination of CVM priorities and conditions under which they apply.
- Technology

Include CVM list in personalization. If offline plaintext PIN is supported, offline PIN related data must be personalized into the card.

7.2.5.2 Cardholder Verification Risk Management Check

A series of risk management checks can be performed during terminal action analysis based on the outcomes of cardholder verification. For each check, Issuers shall decide what actions should be taken for their benefits when risk management conditions have been triggered. These actions include offline approval, decline offline or send online. Issuers set these check points through issuer action code (IAC) and personalized them into the card.

If Issuers decides to send online, Issuers shall also determine the action when online is not available, either approve offline or decline offline.

UPI's recommended actions for these policies are listed in the table below:

Table 10 Cardholder Verification Risk Management Check

Condition	Description	Recommended Action
Cardholder verification fails	Terminal performed all CVMs supported by the card and failed.	Decline offline.
Unknown CVM	Terminal is not able to identify a CVM in CVM list. This will occur if a new CVM is introduced but terminal has not been set up accordingly.	Ignore this condition and continue to transact.
Number of PIN try exceed limit	In current or previous transaction, number of PIN try by cardholder has exceeded the limit. It means offline PIN verification fails.	Send online: if online is not available, decline offline.
PIN is required, but PIN pad does not exist or does not work	Card and terminal support PIN, however PIN cannot be entered as PIN pad is not connected or does not work.	Send online: if online is not available, action taken by Issuers will depend on the balance of risk control and customer service.

Condition	Description	Recommended Action
PIN is required, PIN pad exist, but PIN is not entered	Card and terminal support PIN processing, however, cardholder did not enter PIN.	Send online: if online is not available, Issuers should balance between risk control and customer service to decide what actions to be taken. In early stages where UPI application is issued, cardholder may not remember his offline PIN; if Issuers select “decline offline”, this transaction will be decline offline even if the real cardholder is executing the transaction.
Enter online PIN	CVM for current transaction is online PIN and online PIN is entered.	Send online: if online is not available, decline offline.
IC card data loss	Terminal checks card data required to support cardholder verification and finds part or all data are missing. This may happen in the following situations: card personalized incorrectly, fraud perpetrator deleted part or all of card data related to offline data authentication.	Ignore this condition, continue to transact.

7.2.5.3 Offline PIN

This section outlines special considerations for Issuers supporting offline PIN.

- PIN generation
- PIN length
- PIN synchronization
- Post-issuance PIN change
- PIN try limit

7.2.5.3.1 PIN Generation

Issuers shall determine whether to generate PIN for cardholders or allow cardholders to select their own PIN before card issuance. Research shows that it is very easy for

cardholders to remember PIN selected by them while it is much more difficult to remember PIN automatically generated by computer.

7.2.5.3.2 PIN Length

Issuers shall determine whether PIN length is fixed or variable. According to ISO, PIN can be any numbers between 4 and 12 digits. Issuers often choose variable-length PINs when they are cardholder-selected and fixed length PINs when they are computer-generated.

Considering most of the ATMs currently do not support PINs with more than 6 digits, acceptance may be confined for cards with more than 6-digit PIN.

7.2.5.3.3 PIN Synchronization

If Issuers support offline PIN, UPI recommends offline PIN should be consistent with online PIN. Please note that if offline PIN value changes, Issuers shall develop procedures to concurrently update the online PIN value for magnetic-stripe and chip.

7.2.5.3.4 Post-Issuance PIN Change

Issuers shall decide whether post-issuance PIN change is allowed. This function is very important if Issuers do not allow cardholders to select their own PIN before card issuance. It is very possible that cardholders would like to set their own PIN.

If Issuers allow cardholders to change PIN after card issuance, it must be ensured that the offline PIN stored in the card is the same as the cardholder's online PIN. If the magnetic-stripe uses PIN Verification Value (PVV), it must correctly reflect the online PIN value. Magnetic-stripe-equivalent data stored in the chip must also update accordingly.

Note: most Issuers do not store PVV in magnetic-stripe and therefore the above problem does not exist.

In addition, once the number of PIN try exceeds the limit, offline PIN is automatically blocked (online PIN can still be used). Issuers should clarify the process flow to handle this type of situation, for example: to reset PIN try counter by post-issuance change function; this operation should be performed on secure devices controlled by Issuers.

7.2.5.3.5 PIN Try Limit

Issuers shall decide the maximum number of erroneous offline PIN entries by cardholders. Deciding the offline PIN try limit can be based on current online PIN try limit. Issuers can request the terminal to display a message (for example, "Last Chance") to notify cardholders there is only one PIN try attempt left. This enhances cardholder's customer experience, but also invites perpetrators to remove a stolen card before offline PIN is blocked.

In addition, Issuers may also set up in a way that the card automatically blocks applications when PIN try exceeds the limit. This can prevent fraudulent cardholder from using the card under all circumstances. However, UPI does not recommend doing this as chip blocking

mechanism by offline PIN is sufficient. In addition, Issuers has the ability to unblock in an appropriate manner.

UPI recommends that Issuers set the number of offline PIN try to 3. Please note that offline PIN try is independent of online PIN try. Hence, there are a total of 3 offline PIN tries and 3 online PIN tries.

7.2.5.3.6 Implementation

This section outlines policy, operation and technology required to support offline PIN:

- Policy
 - ✓ Decide PIN generation process;
 - ✓ Decide whether to allow post-issuance PIN change;
 - ✓ Decide the requirements of unblocking PIN;
 - ✓ Determine whether to set up a card option so that the terminal will display a message to warn cardholders when there is only one PIN try attempt left
 - ✓ Decide whether offline PIN to remain consistent with online PIN;
- Operation
 - ✓ Develop the procedures to generate cardholder's PIN or allow cardholders to select their PIN, depending upon which option Issuers have chosen;
 - ✓ Develop the procedure of post-issuance PIN change (if support);
 - ✓ Develop the procedures to unblock PIN;
 - ✓ Ensure synchronization between offline PIN and online PIN (if applicable);
 - ✓ Ensure when offline PIN is changed, online PIN and related data in chip and magnetic-stripe are changed accordingly (if applicable);
- Technology

Generate the necessary data related to offline PIN processing and include it in the personalization file. This data includes cardholder's PIN value (refer to as PIN) and PIN Retry Limit.
- Risk Management
 - ✓ Decide requirements for PIN length;
 - ✓ Decide the number of PIN try allowed before PIN is blocked;
 - ✓ Decide whether to block application automatically when PIN try exceeds limit.

7.2.6 Terminal Risk Management

Terminal must possess risk management functions but its check items are optional. Terminal can accomplish risk management by performing methods such as floor limit check, transaction velocity check, new card check, terminal exceptional file check, merchant mandatory online transaction and random selection online transaction through data provided by terminal and card.

These checks provide better control over transaction processing at the point of transaction. They help decide whether a transaction should be approve offline, decline offline or send online.

Checks related to the card include:

- Velocity check – is not recommended by UPI, but is not precluded.
- New card check – optional

Besides these optional checks, Issuers can support other special checks according to market demands.

Risk management allows the card to send online for authorization for transactions with amount smaller than floor limit. Following factors should be considered when setting these parameters:

- Domestic authorization environment
- Combination of multiple factors – customer credit score and fraud risks
- Expected location and transaction volume for international transactions – the risk level of destination

Issuers should set parameters according to authorization requirements for each combination of these factors (also include locations of expected usage). On considering that some international or domestic acceptance may not support online authorization, Issuers shall carefully balance between risk control and convenient acceptances.

Fraud and credit risk management should be taken to consideration when determining appropriate parameters. UPI risk management professionals can provide assistance.

7.2.6.1 Velocity Check

Velocity check allows Issuers to request send online after completing a preset number of consecutive offline transactions. Terminals that allow offline transaction must support terminal velocity check. Issuers may choose not to support terminal velocity check by not writing lower limit and upper limit of consecutive offline transactions (Tag ‘9F14’ and Tag ‘9F23’) data to the card during personalization.

If card provides lower and upper limit of consecutive offline transactions when reading application data, terminal will perform terminal velocity check. If none of these data appear in the card, terminal will not conduct this process.

Note: During card action analysis, the card can also execute similar velocity check. Card velocity check will not impact terminal verification result.

7.2.6.2 New Card Check

Terminal performing velocity check should also perform new card check. If last online ATC value was 0, terminal set “new card” in TVR to “1”. Terminal sets corresponding indicator in TVR for card used for the first time and requests transaction to be processed online. After the transaction is approved online according to issuer authentication result and card parameters, this register is reset.

Note: During card action analysis, the card may also perform similar new card check.

7.2.6.3 Terminal Risk Management Check

A series of risk management checks can be performed in terminal action analysis based on risk management process results. Issuers shall decide what actions to take when risk management conditions have been triggered, and these actions include approve offline, decline offline or send online. Issuers set these check points through issuer action code (IACs) and personalized them into the card.

If Issuers determine to send online, Issuers shall still need what action should be taken if online is not available, whether it is approve offline or decline offline.

The table below gives recommended actions to these policies by UPI.

Table 11 Terminal Risk Management Check

Condition	Description	Recommend Action
Card appears in exceptional file on terminal	Terminal checks whether primary account number (PAN) in card is listed in exceptional file.	Send online; if online is not available, decline offline.
Merchant requests online processing	On the terminal that supports online processing, merchant can set transactions to be processed online.	Send online; if online is not available, decline offline.
The amount of transaction exceeds floor limit	Terminal checks whether transaction amount exceeds merchant floor limit.	Send online; if online is not available, decline offline.
Transaction is randomly selected for online processing	Terminal supporting offline and online transaction will randomly select the transaction to be processed online.	Send online; if online is not available, approve offline.

Condition	Description	Recommend Action
Exceed consecutive offline transaction lower limit	Terminal checks whether the value of ATC minus last online ATC register is greater than lower limit of consecutive offline transactions.	Send online; if online is not available, approve offline.
Exceed consecutive offline transaction upper limit	Terminal checks whether the value of ATC minus last online ATC register is greater than upper limit of consecutive offline transactions.	Send online; if online is not available, decline offline.
New card	Terminal checks whether the card is being used for the first time.	Send online; if online is not available, approve offline.

7.2.6.4 Implementation

Activities required to perform terminal risk management include:

- Policy
 - ✓ Decide whether to support terminal velocity check;
 - ✓ Decide whether to support terminal new card check;
 - ✓ Decide preferred actions (approve offline, decline offline, send online) to be taken when performing risk management checks at the point of transaction.
- Operations

Determine card velocity parameters.

- Technology

Generate data required by terminal risk management and add to personalization input file. This data includes lower consecutive offline limit (LCOL, Tag '9F14'), upper consecutive offline limit (UCOL, Tag '9F23'), last online ATC register and etc.

7.2.7 Card Action Analysis

Card action analysis allows Issuers to perform velocity check and other risk managements in the card. This section describes proprietary UICS risk management features, including:

- Action of last transaction

- New card
- Transaction velocity counter

7.2.7.1 Card Risk Management

The table below summaries all card risk management checks in card action analysis, specifies whether these checks are compulsory or optional and describes check results.

Table 12 Card Risk Management Check

Risk Management Check	Conditions for Execution	Result (if conditions are satisfied)
Last transaction didn't complete online authorization	Conditionally compulsory — this check must be performed if the card supports issuer script command or issuer authentication.	Request online processing and set CVR indication bit.
Issuer authentication failed in last transaction (or issuer authentication was compulsory for last transaction but did not execute)	Conditionally compulsory — this check must be performed if the card supports issuer authentication.	Set CVR indication bit Check ADA whether it requests online processing
SDA failed in last transaction	Conditionally compulsory — this check must be performed if SDA is supported.	Set CVR indication bit
DDA failed in last transaction	Conditionally compulsory — This check must be performed if DDA is supported.	Set CVR indication bit

Risk Management Check	Conditions for Execution	Result (if conditions are satisfied)
Issuer script processing was used in last online transaction	Conditionally compulsory — this check must be performed if post-issuance is supported	The card stores the number of script commands in CVR. If script processing fails (use issuer script failure indication bit in the card), sets CVR indication bit. The parameter settings in ADA determine whether transaction is to be processed online.
Consecutive offline transaction lower limit velocity check	Optional — card performs this check if last online ATC register and consecutive offline transaction lower limit (Tag '9F58') exist.	If limit has been exceeded, the card requests online processing and sets CVR indication bit
Consecutive international offline transaction (based on currency) velocity check	Optional — card performs this check if application currency code, consecutive offline transaction counter (international – currency) and consecutive offline transaction limit (international – currency) exist.	If limit has been exceeded, the card requests online processing and sets CVR indication bit
Consecutive international offline transaction (based on country) velocity check	Optional — card performs this check if issuer country code, consecutive offline transaction counter (international – country) and consecutive offline transaction limit (international – country) exist.	If limit has been exceeded, the card requests online processing and sets CVR indication bit
Velocity check of accumulated offline transaction amount of designated currency	Optional — card performs this check if application currency code, accumulated offline transaction amount and accumulated offline transaction amount limit exist.	If limit has been exceeded, the card requests online processing and sets CVR indication bit

Risk Management Check	Conditions for Execution	Result (if conditions are satisfied)
Velocity check of accumulated offline transaction amount (dual currency)	Optional — card performs this check if primary currency code, secondary application currency code, currency conversion factor, accumulated offline transaction amount (dual currency) and accumulated offline transaction amount limit (dual currency) exist.	If limit has been exceeded, the card requests online processing and sets CVR indication bit. If the currency used in transaction is the secondary currency, currency conversion is required
New card check	Optional — card performs this check if last online ATC register and application default action (ADA) exist.	If the card never requested online, it can request online this time and sets CVR indication bit.
Offline PIN verification was not performed (PIN try exceeded the limit)	Optional — card performs this check if offline PIN is supported and data element application default action exist.	Set CVR indication bit if this transaction offline PIN verification was not executed and PIN try limit has already been exceeded. ADA sets transaction decline or send online in this situation

7.2.7.2 Implementation

Activities required to implement card risk management in card action analysis include:

- Policy
 - ✓ Determine whether conditions based on “last transaction ” for all checks are met;
 - ✓ Determine whether to support new card check;
 - ✓ Determine whether to support various card velocity check;
 - ✓ Determine preferred actions (approve offline, decline offline and send online) to be taken by card when performing card risk management checks at the point of transaction.
- Operations
 - ✓ Communicate with card vendors for data elements required in performing risk management functionality.

✓ Determine card velocity parameters.

- Technology

Generate data required by card risk management and add to personalization input file. This data includes consecutive offline transaction lower limit (Tag '9F58'), consecutive offline transaction limit (international – currency), consecutive offline transaction limit (international – country), accumulated offline transaction amount limit, accumulated offline transaction amount limit (dual currency) and etc.

7.2.8 Online Processing

During transaction processing, the card generates an authorization request cryptogram (ARQC) for online card authentication. This cryptogram is sent online to Issuer for validation and protects data integrity.

Issuers can also return an authorization response cryptogram (ARPC) in response message for online issuer authentication. Card verifies if the response comes from the valid issuer (or its agent), and whether data integrity has been violated.

These cryptograms are generated by application cryptogram (AC) key. AC unique derived key (UDK) is stored in the card while AC master DEA key (MDK) is stored in the issuer's host system. This section describes activities required to set the card to support online card authentication and issuer authentication; please refer to chapter 8 "Issuer Host System Modifications" for activities involving the host system.

7.2.8.1 Policy

This section describes policy related to online card authentication and issuer authentication:

- Number of master keys to be generated;
- To set the support of issuer authentication as mandatory or optional;
- Actions to be taken by card based on issuer authentication results.

7.2.8.1.1 Number of Master Key to be generated

Issuers shall determine the number of master keys to be generated. MDK is a double-length DES key. Similar to the decisions on the number of public keys to be generated, the more MDKs Issuers support, the greater the security of the entire transaction process. However, too many keys will cause a higher operational cost. UPI recommends Issuers to generate a minimum of one MDK for each BIN.

Each master key has a corresponding Derivation key index (DKI) to designate UDK is obtained by which master key in the card; DKI needs to be personalized to the card as UICS proprietary data, and its value cannot be 0.

7.2.8.1.2 Mandatory & Optional Issuer Authentication

Issuers can set whether it is mandatory or optional for cards to support issuer authentication. If issuer authentication is supported, it will only be performed when the issuer's (or its agent's) response message provides ARPC.

- **Mandatory**

If issuer authentication is mandatory, related counters and indicators can only be reset when issuer authentication has been successfully performed. In addition, the “if issuer authentication is mandatory but did not receive ARPC, decline transaction” bit of ADA should not be set.

- **Optional**

If issuer authentication is optional, related counters and indicators will be reset regardless of issuer authentication results.

The relationship between issuer authentication (mandatory & optional) in conjunction with card parameters and issuer script updates are shown in the table below:

Table 13 Counter¹ Reset for Approved Transaction

Issuer Authentication	Issuer Authentication Result		
	Successful	Failed	Not performed
Mandatory	Reset counter	Do not reset counter	Do not reset counter
Optional	Reset counter	Do not reset counter	Reset counter
Not supported	—	—	Reset counter ²

1— Related counters and indicators include:

- Velocity counter
- Issuer authentication failure indication bit
- Online authorization indication bit

2— If issuer authentication is not supported, related counters shall be reset by an approval response.

7.2.8.1.3 Issuer Authentication Risk Check

A series of risk management checks can be performed during transaction completion based on issuer authentication processing results. Issuers shall decide what actions to be taken when risk management conditions have been triggered.

UPI gives recommended actions to these policies in the table below:

Table 14 Issuer Authentication Risk Management Check

Condition	Recommended Action
Determine the actions the card should take if issuer authentication fails	Decline offline even if Issuers approved transaction. Must send the next transaction online.
Determine the actions the card should take if issuer authentication is mandatory, but ARPC is not received.	Do not decline and proceed with the transaction.

7.2.8.2 Decision Summary

This section outlines the decisions required to perform issuer authentication and recommendations.

Table 15 Issuer Authentication Decision and Recommendation

Decision	Recommendation
Decide the number of MDK to be generated	To generate at least one MDK for each BIN.
Decide the approach to support issuer authentication: mandatory or optional.	If issuer authentication is mandatory, card expects to receive ARPC for each online transaction; Card does not reset the counter if ARPC is not received.
Decide related actions to be taken based on issuer authentication results	Refer to table 14: Issuer Authentication Risk Management Check

7.2.8.3 Operation and Technical Activities

Related activities to set the card to support online card authentication and issuer authentication include:

Table 16 DES Key Management Activities for Online Card Authentication and Issuer Authentication

Step	Responsible Party	Activities
1	Technical	Generate suitable number of MDKs based on issuer decision.
2	Technical	Assign a Derivation Key Index (DKI) to each MDK. The DKI is an index that links the MDK to the UDK.
3	Technical	Develop a process to derive a UDK from the MDK for each application. This should be part of pre-personalization activities Note: UDK is double-length; it should be encrypted by double-length KEK for security purposes.
4	Technical	Add UDK and DKI to personalization file. UDK must be added to personalization input file after being encrypted by KEK.
5	Technical	During personalization, load UDK, DKI and other personalization data into the card. Loading the correct UDK and DKI are very important. Failure to execute either of these activities successfully may result in failure for online card authentication and issuer authentication.

7.2.9 Transaction Completion

During transaction completion, the card will perform other risk managements to decide approve or decline the transaction if it requests for online processing but terminal does not support online or online authorization does not complete.

7.2.9.1 Card Risk Management

The table below summaries all card risk management checks during transaction completion, indicate these checks are mandatory or optional, and describes check results.

Table 17 Card Risk Management Check

Risk Management Check	Performance Condition	Result (if conditions are met)
Consecutive offline transaction upper limit velocity check	Optional — card performs this check if last online ATC register and consecutive transaction upper limit (Tag‘9F59’) exist.	If limit has been exceeded, decline offline and set indication bit in CVR
New card check	Optional — card performs this check if last online ATC register exists.	Decide to approve offline or decline offline according to the setting of bit “For new card, decline transaction if online is available” Set indication bit in CVR
PIN try limit is exceeded	Optional	Recommend to set “decline transaction if PIN is blocked in previous transaction” bit in ADA to 1 if offline PIN is supported; Decline offline and set indication bit in CVR
Accumulated offline transaction amount (upper limit) velocity check	Optional — card performs this check if accumulated offline transaction amount and accumulated offline transaction amount upper limit exist.	If limit has been exceeded, decline offline and set indication bit in CVR
Accumulated offline transaction amount upper limit (dual currency) velocity check	Optional — card performs this check if accumulated offline transaction amount (dual currency) and accumulated offline transaction amount upper limit exist.	If limit has been exceeded, decline offline and set indication bit in CVR

7.2.9.2 Implementation

Required activities to perform risk management in transaction completion include:

- Policy
 - ✓ Decide whether to support new card check;

- ✓ Decide whether to support various card velocity checks;
- ✓ Decide preferred actions (approve offline or decline offline) to be taken when performing card risk management checks at the point of transaction.
- Operation
 - ✓ Communicate with card vendors for data elements to be supported according to functions performed.
 - ✓ Determine card velocity parameters.
- Technology

Produce required data for card risk management and add it to personalization input file. Data includes consecutive offline transaction upper limit (Tag '9F59'), accumulated offline transaction amount upper limit and etc.

7.2.10 Issuer Script Processing

Issuer script processing allows Issuers to modify account parameters residing on the card without re-issuing. These updates allow Issuers to enhance their risk management and customer service. For example, Issuers can block the account that has become delinquent and change velocity check counter to either provide better, less restricted usage for customers in good standing or tighter control over high risk customers.

7.2.10.1 Commands

The issuer needs to determine script commands to be supported. Following table provides definition and application background of these commands.

Note: Setting up your card to support a post-issuance application load is outside the scope of this document.

Table 18 Issuer Script Processing Commands

Command	Definition	Application Situation
Application block	This command allows Issuers to block a selected UPI application.	Issuers can use this command to shut down one application. Issuers may use this command when a card is reported lost/stolen or when an account becomes delinquent. This command helps to prevent fraudulent account being used in below-floor-limit and offline-only environments.

Command	Definition	Application Situation
Unblock application	This command allows Issuers to unblock a selected UICC application.	Issuers can use this command to unblock UPI application. It is used when a fraudulent account becomes reinstated. This command is only supported on special devices designated by Issuers, such as: issuer ATM or a terminal in bank branch.
Card block	This command allows Issuers to block the entire card.	Issuers can use this command to close all the chip applications in the card. Since the function of this command is equivalent to cutting the card in two, it should only be used in most severe situations. Once card is blocked, it cannot be unblocked; new card must be re-issued if account is reinstated. Issuers should consider using this command only when card is reported lost or stolen. If Issuer uses this command on a multi-application card for which the Issuer is not the sole application provider, Issuer shall develop the provisions for use of this command with other application providers. In addition, when Issuer initiates this command, Issuer shall have a mechanism to alert other providers that the card is blocked.
PIN change/unblock	This command allows Issuers to unblock offline PIN or unblock PIN plus perform PIN value change.	If PIN try exceeds the limit, card blocks offline PIN. PIN change/unblock command can be used in this situation for remedy: • If cardholders remember PIN and performed correct verification, unblock and keep original PIN value; • If cardholders forget PIN, unblock and change offline PIN value. If cardholders want a new PIN, this command can also be used to change offline PIN. This function is only supported by special devices designated by Issuers, such as issuer ATM or terminal in bank branch.

Command	Definition	Application Situation
Set data	This command allows Issuers to modify the values of some data objects, for example: data elements related to authorization control velocity checking.	Issuers can use this command to change velocity parameters after card issuance in order to better customize the cardholder risk control. For example: it can increase “consecutive offline limit” value from 3 to 5.
Modify records	This command allows Issuers to modify a record in the file.	This command may be used to modify PVN in magnetic-stripe data (“magnetic-stripe 2 equivalent data” and “magnetic-stripe 1 discretionary data”) in the chip.

Note: Issuers can also implement other commands in addition to this set of commands, as long as the commands are supported by the card and the host system.

7.2.10.2 Security

Secure messages must be used to transport issuer scripts to the card.

7.2.10.3 Number of Keys to Generate

The mechanism of secure messages uses two types of DES key:

- Message Authentication Code (MAC) Key

This key is used to secure issuer script processing. It should be used for all issuer script commands. It is used to confirm the data that has not been altered (in terms of integrity) and also can be used to confirm whether the commands sent by Issuers are valid (issuer authentication) at the same time; MAC MDK is stored in issuer host system, while MAC UDK derived from it and is stored on the card.

- Data Encryption (ENC) Key

This key is used to encrypt sensitive data in scripts, such as offline PIN etc. ENC MDK is stored in issuer host system, while ENC UDK derived from it and is stored on the card.

Issuers shall decide the number of MAC MDKs and ENC MDKs that are prepared to support. Similar to AC MDK, UPI recommends that Issuers support at least one MAC MDK and one ENC MDK for each BIN.

Note: There is no policy decision required in MAC UDK and ENC UDK generation, because they are application-level keys; Issuers must derive one MAC MDK and one ENC MDK for each UPI application.

7.2.10.4 Risk Management Check for Issuer Script Processing

There are two risk management checks for issuer script processing results:

Table 19 Risk Management Checks for Issuer Script Processing

Condition	Recommended Action
Script processing fails in last transaction.	Request online authorization.
Issuer script processing fails (after the last generated application cryptogram command)	Continue transaction, request online authorization in the next transaction.

7.2.10.5 Implementation

Issuers shall perform the following activities for issuer script processing:

- Policy
 - ✓ Decide which script commands the Issuers will support. Issuers should adequately communicate the requirements of these commands with card vendors as vendors may not support all the commands.
 - ✓ Decide whether to support other issuer script commands in the future.
- Operation
 - ✓ According to functionality that is implemented, communicate with card vendors on issuer script commands that currently needed to be supported.
 - ✓ Should Issuers prepare to support issuer script commands that will be supported in the future, Issuers should communicate with card vendors on the conditions of commands support for the relevant card.
- Technology
 - ✓ Implement issuer script commands that currently needed to support, as well as allocate some spaces for issuer script commands that will be supported in the future.

Related MAC and ENC key activities to support issuer script processing include:

Table 20 Issuer Script Processing DES Key Management Activities

Step	Responsible Party	Activities
1	Technical	Generate appropriate number of MAC MDKs and ENC MDKs.
2	Technical	Determine methods for associating MAC UDK/ENC UDK and MAC MDK/ENC MDK. Recommendation: Similar to AC key, use the same DKI to associate MAC and ENC keys; or use the BIN and expiration date of UPI application to register MAC and ENC keys.
3	Technical	Derive an UDK from MDK for each UPI application, and this is part of the pre-personalization activity: • Use MAC MDK to obtain MAC UDK; • Use ENC MDK to obtain ENC MDK.
4	Technical	Include MAC UDK, ENC UDK and their DKIs (if applicable) to personalization input file. MAC UDK and ENC UDK must be encrypted by KEK before being loaded to personalization input file.
5	Technical	During personalization, load MAC UDK, ENC UDK, DKI (if applicable) and other personalization data to the card. It is important to load the correct MAC UDK, ENC UDK and DKI; Operation errors may cause failure in online card authentication and issuer authentication.

7.3 Personalization Data Requirements for Chip/Magnetic-stripe

For adapting into different acceptance environments, Issuers usually accustom to retain physical magnetic-stripe on IC cards, and are called chip/magnetic-stripe composite cards (hereinafter referred to as Hybrid cards). This section mainly describes the relationship between magnetic-stripe and chip applications in this type of composite cards, and their personalization requirements.

7.3.1 Relationship between IC Card Application and Magnetic-stripe

For adapting into different acceptance environments, Issuers usually accustom to retain physical magnetic-stripe on IC cards. Issuers should set the priority for debit/credit application corresponding to magnetic-stripe information as the highest, and in which is called primary debit/credit application.

Issuers shall ensure the consistency between physical magnetic-stripe data and primary debit/credit application data, which includes but not limited to:

- Expiry date of primary debit/credit application in IC card is consistent with expiry date in the physical magnetic-stripe.
- If IC card has expiry date printed on card surface, expiry date of primary debit/credit application on chip and the one on physical magnetic-stripe are consistent with the expiry date on card surface.
- Cardholder's name in chip must be the same as the one printed on card.
- Service code on IC card's magnetic-stripe is set as "2XX" or "6XX"; information on second track, such as primary account number, expiry date and service code, should be reflected in relevant data elements in primary debit/credit application on the chip.
- For greater risk control, Issuers can set different iCVN stored in the chip from CVN in physical magnetic-stripe.

7.3.1.1 Implementation

Following activities shall be performed to implement UPI applications on a card with both chip and magnetic-stripe:

- Policy
 - ✓ Decide whether the card should retain physical magnetic-stripe.
- Operation
 - ✓ Determine how to ensure the consistency between common data on physical magnetic-stripe and chip.
 - ✓ Determine the processing logics for terminals supporting both chip and magnetic-stripe.
- Technology
 - ✓ Implement synchronized updates on common data in chip and physical magnetic-stripe, and determine terminal processing logics.

7.3.2 IC Card Verification Number

In addition to current card verification number (CVN) for physical magnetic-stripe, UPI recommends using alternative IC card verification number (iCVN) in IC card transactions; it can be used to protect against counterfeiting magnetic-stripe cards using magnetic-stripe data in the chip. CVN value in physical magnetic-stripe is different from iCVN value in the chip.

iCVN uses the same calculation method as CVN, apart from replacing the real service code with "999", and other calculation factors, such as DES key, primary account number and

expiry date, remain unchanged. If Issuers already support CVN generation and verification, supporting the new iCVN has minimal impact on host system. The following methods can be used to determine the use of CVN or iCVN:

- If PAN entry mode is 02, meaning the transaction is initiated by magnetic-stripe and Issuers shall perform traditional CVN verification.
- If PAN entry mode is 05, meaning the transaction is initiated by chip and Issuers shall replace service code with '999' to perform iCVN verification.

If iCVN is received for a transaction initiated by magnetic-stripe, CVN verification will fail and Issuers (or agents) shall decline this transaction. This check helps to reveal counterfeit cards, suspicious merchant and other issues.

7.3.2.1 Implementation

Activities related to magnetic-stripe data include:

- Policy
 - ✓ Decide the relationship between UPI application and magnetic-stripe.
 - ✓ Decide whether to support iCVN.
- Operation
 - ✓ Ensure magnetic-stripe data in personalized chip, encoded data in physical magnetic-stripe and data printed on card surface are consistent.
 - ✓ Communicate with vendors for CVN/iCVN processing requirements.
- Technology
 - ✓ Generate physical magnetic-stripe equivalent data and include it to personalization input file.
 - ✓ If iCVN is supported, replace service code with '999' and calculate iCVN; modify issuer host system to support iCVN verification.

7.4 Personalization Data Requirements for Electronic Cash

Electronic cash (EC) is an optional functionality. With this function, Issuers can recruit merchants mainly engaging in low-value transactions, thereby expanding IC card acceptance, providing cardholders with more applications and convenient services, and therefore increasing issuance volume for IC products. Since EC transaction is processed offline, there is no cost owing to online authorization. This benefit and faster transaction help to push merchants transitioning from traditionally accepting (low-value) cash transaction to accept card transaction.

The biggest difference between EC and electronic wallet is that EC is based on UPI debit/credit application and use its asymmetrical key system; while electronic wallet adopts

symmetrical three-level key system, in which its key transmission and update processes are more complex, with greater risk in PSAM card security management.

Note: Unless otherwise specified, UPI micropayment (EC) applications refer to in this specification are default to corresponding contact payment method.

7.4.1 Data Elements Involved

EC solution is formulated on the basis of UPI standards and its applications are almost the same as general UPI debit/credit card. However, for convenience of usage, some new data elements are added to EC solution and that do not appear in debit/credit. Refer to the following table for new data elements:

Table 21 Specialized EC Data Elements

Name of Data Elements	Requirement	Description
Electronic Cash Balance	Condition: if EC is supported	This data element stores the total balance which can be used for offline purchase. For each successful EC transaction, respective transaction amount is deducted from it. Once transaction amount exceeds the EC balance, transactions must be authorized online.
Electronic Cash Balance Limit	Condition: if EC is supported	It represents the maximum accumulated limit that can be spent offline by the cardholder, which is also the limit to recharge the card. Issuers can change this limit.
EC Issuer Authorization Code	Condition: if EC is supported	It is the code used to identify EC approval in the card. In offline transaction approval, this code is stored in authorization code of settlement message. Its format is "ECCxxx", in which xxx is the number defined by Issuers.

Name of Data Elements	Requirement	Description
EC Single Transaction Limit	Optional	It is the single EC transaction limit in the card and is used to control the risk of single EC transaction. It is written by Issuers during personalization and can be reset by Issuers.
EC Reset Threshold	Optional	It is the lower limit of balance to trigger the card to recharge automatically. Card requests online for automatic recharge when available offline balance in the card is below this threshold.

The utilization of original application data elements on IC card is also different when implementing EC function. While some data elements must be used, such as PDOL and application currency code, the values of some data elements, such as AFL and AIP, can be set differently from standard IC card and is used exclusively for EC transactions. Refer to the table below for data elements affected by EC:

Table 22 Data Elements Affected by EC

Name of Data Elements	Requirement	Description
PDOL	Condition: if EC is supported	Card must contain PDOL if it supports EC. It should at least request terminal to provide the following data elements: EC terminal supports indicator, authorization amount and transaction currency code.
Application currency code	Condition: if EC is supported	It is UPI proprietary data in accordance with ISO4217 encoding. EC requires application currency code to match with transaction currency code.

Name of Data Elements	Requirement	Description
AFL	Condition: if EC is supported	It indicates data storage location related to EC application (short file identifier and record number). The file record it refers to should include specialized EC data such as EC balance and EC issuer authorization code. It can also specify other records to use different data from non-EC transaction, such as CVM list.
AIP	Condition: See Description	It is the AIP used for EC transaction and it states the functionalities specified by EC card. Issuer can select functions different from non-EC transaction.
CVM list	Condition: See Description	If Issuers would like to have different CVM from that in non-EC transactions, Issuers shall personalize an independent CVM list.
IC card public key certificate	Condition: See Description	If Issuers support DDA in EC transaction and its signature data is different from non-EC transaction, Issuers shall personalize an independent IC card public key certificate.
IAC	Condition: See Description	If Issuers set different conditions for IAC-decline/online/default in EC transaction than in non-EC transaction, Issuers shall personalize an independent IAC.
Signed SAD	Condition: See Description	If Issuers supports SAD in EC transaction and its signature data is different from non-EC transaction, Issuers shall personalize an independent SAD.

7.4.2 Risk Management

Different from large-value transactions where risk control is continuously strengthened, Issuers can flexibly exercise a simplified risk control function for EC based on the requirements of micropayment feature and quick processing nature of EC. EC risk control

function can be relatively independent from standard process. For example, it can support different offline data authentication methods and different CVMs.

7.4.3 Implementation

The following activities shall be performed to implement EC function:

- Policy
 - ✓ Issuers decide whether to perform EC function;
 - ✓ Define the method to generate unique numbers for EC card's IAC – "ECCxxx". For example, it can be randomly generated or sequentially generated likes ATC;
 - ✓ Decide whether to use data element "EC single transaction limit" for EC transactions;
 - ✓ Decide whether to use data element "EC reset threshold" for EC transactions;
 - ✓ Decide risk management method to be adopted by EC transactions, such as offline data authentication method and CVM.
- Operation
 - ✓ Communicate with card vendors for EC requirements;
 - ✓ Define EC balance limit for cardholders in accordance with their risk records, account status and etc.;
 - ✓ Set EC single transaction limit and EC reset threshold value (if supported).
- Technology

Generate data required by EC functions and include it to personalization input file. This data includes EC balance limit, EC single transaction limit, EC reset threshold and etc.

7.5 Personalization Data Requirements for Contactless IC Card Payment

Development of new contactless technology has brought challenges and revolution to current financial transaction methods and technical environment. Considering usage behaviors of cardholders and preventing transaction disruption, it is critical to shorten the transaction time as much as possible while ensuring uninterrupted smooth transaction.

UPI specifications provide two types of contactless payment interfaces: quick debit/credit (qUICS) and contactless UICS. Based on debit/credit process, qUICS simplifies and compresses transaction process and instructions to achieve offline micropayment applications with contactless interface (contactless electronic cash). Contactless UICS is based on standard UICS transaction with contactless interface.

Given qUICS is based on standard procedure and uses current card system and operating regulations, qUICS reduces processing time between the terminal and the card by reducing the number of commands and responses. It also provides quick offline micropayment features, offline data authentication and online card authentication using existing cryptogram algorithm (version 01) or the new simplified algorithm (version 17). This section provides a brief description on the requirements of qUICS personalization.

7.5.1 Summary of General Requirements

Basic card requirements for contactless payment include the following:

- The card should at least support either Type A or Type B as defined in ISO 14443;
- If contact interface has been activated, the card should not respond to contactless interface;
- “Magnetic-stripe 2 equivalent data” is mandatory for qUICS;
- Card with offline function (qUICS) should support SDA or DDA (DDA is recommended);
- In order to meet the transaction time requirement with existing chips, UPI recommends adopting key generation method based on Chinese remainder theorem.

7.5.2 Quick Debit/Credit Payment Application

It is necessary to modify and optimize standard debit/credit application process so that it can meet the transaction speed requirements of contactless interface. This optimization includes:

- Reduce transaction time by compressing multiple interactive commands into as few number of commands as possible;
- Complete interaction between the card and terminal collectively. Terminal performs offline data authentication, terminal risk management and terminal action analysis after the card is out of card reader’s communication range. PIN operation is allowed before or after the card is out of reader’s communication range, which minimizes the time required for the card to remain within card reader’s communication range.

qUICS has two features:

- Online transaction using online card authentication;
- Offline transaction using offline data authentication.

7.5.2.1 Data Elements Involved

qUICS does not require all basic mandatory data to be included in the card, or do not require to read it if it is included on the card. In qUICS processing, all types of counters and indicators, as well as uninvolved variables in other files, will not be affected. Below are explanations on new data elements to qUICS card and data elements affected by qUICS:

Table 23 qUICS Proprietary Data Element

Name of Data Elements	Requirement	Description
Proximity payment system environment (PPSE)	Mandatory	It is a list of application identifier, application tag and application priority indicator supported by application which can be accessed by contactless interface. This list contains all the directory entries and is returned by card in FCI as response to SELECT PPSE ('2PAY.SYS.DDF01').
Available Offline Spending Limit	Optional	A calculation field used to allow reader/writer to print or display card offline transaction limit. Unless this tag is personalized to '1', card will not allow this tag to be included in records read by reader/writer or in response to GPO. Personalization of this data element does not affect it to be included in issuer discretionary data.
Card Additional Processes	Under the condition that card supports offline and micropayment option is not default EC or no card risk management options are supported.	Indicate card processing requirements and choices on parameters. Refer to Section 8.1.3.1 for detail.

Name of Data Elements	Requirement	Description
Card CVM Limit	Optional	If this appears, it means the card shall provide CVM when card and terminal currency type are matched and a contactless transaction exceeds this value. Current qUICS supports two CVMs: online PIN and signature.
Card Internal Indicators	Mandatory	Used to control internal process of qUICS card.
Card Transaction Qualifiers	Optional	Mainly used to indicate CVM required by the card to terminal.

Note: qUICS can provide offline quick micropayment features in which combining EC features in qUICS transaction process. Refer to “Table 21 – Specialized EC Data Elements” for EC data elements.

Table 24 qUICS Affected Data Element

Name of Data Elements	Requirement	Description
PDOL	Mandatory	qUICS card must set PDOL. PDOL will request different basic terminal data elements according to cryptogram version type (01 or 17) supported and whether offline is supported
Application Currency Code	Under the condition that card supports card supplementary processing	UICS proprietary data in accordance with ISO4217 encoding. qUICS card supplement process requires currency match check.

Name of Data Elements	Requirement	Description
AFL	Required	Indicates qUICS related data storage location (short file identifier and record number) for qUICS transactions. AFL will not be returned when card request online processing or decline; An AFL indicating data required by offline data authentication will be returned only in card approve offline.
AIP	Mandatory	AIP is used in qUICS transactions and it states capability of specified function supported by qUICS cards. Issuers may choose functions different from standard UICS functions.
Issuer Application Data	Required	If card supports to return available offline spending limit, it will return the corresponding value through issuer defined data in issuer application data; Issuers should use this to monitor offline capital risk.
IC Card Public Key Certificate	Refer to the description for condition	If qUICS application supports fDDA and its signature data is different from standard UICS application, independent IC card public key certificate need to be personalized.
SAD	Refer to the description for condition	If qUICS application supports SDA and its signature data is different from standard UICS application, independent SAD needs to be personalized.

7.5.2.2 Application Selections

Contactless application selection uses PPSE directory selection method. For contactless application selection, card personalization should pay attention to the following:

- Personalized PPSE uses the file name of “2PAY.SYS.DDF01” in all contactless cards.
- qUICS path is supported for single card application with UPI AID.

- If there are more than one application is personalized to FCI, then the application priority indicator should be personalized to all applications; application priority indicator Bits 8-5 should be set as “0000”.
- AID of contactless financial applications on the card should be returned in FCI by responding to SELECT PPSE command.
- Personalization of all UPI contactless applications should include PDOL, which should at least include data element “terminal transaction qualifier” (Tag ‘9F66’).

7.5.2.2.1 Implementation

Activities related to qUICS application selection include:

- Policy
 - ✓ Decide the number of contactless applications the issuer supports;

UPI recommends Issuers to list only one application in FCI if possible or minimize the number of applications.

- ✓ Decide which cryptogram version (01 or 17) to adopt.
- Operation
 - ✓ Communicate with card vendors for data elements the issuer needs to support based on functions issuer would like to perform;
 - ✓ If PPSE contains multiple contactless applications, Issuers shall prioritize them with application priority indicator;
 - ✓ Determine PDOL based on cryptogram version, card online/offline capability and requirements defined by Issuers.
- Technology
 - Generate data required by application selection and include it to personalization the input file. This data includes PPSE, application priority indicator, PDOL and etc.

7.5.2.3 Application Initialization

To further shorten the transaction time, qUICS has brought the following risk managements to an earlier stage of transaction – application initiation stage:

- Card risk management
- Online card authentication
- Offline data authentication
- Transaction certificate

In online transactions, only GET PROCESSING OPTIONS command is used. In offline transactions, READ RECORD command is also used to read related data of SDA or fDDA while signature (static or dynamic) verification is performed after the card leaves the induction area.

7.5.2.3.1 Card Risk Management

Risk management of qUICS cards is mainly controlled by personalizing a series of requirements through the card supplementary processing (Tag '9F68').

Table 25 Card Supplementary Processing

Condition	Description	Recommended Action
Support low-value check	This condition means that qUICS cards only use EC offline authorized limit (for short below: EC limit) for micropayment judgment. To support this option, qUICS transactions will only use EC limit without affecting CTTA limit.	Regardless of whether CTTA limit is set in standard application, EC limit is only used during qUICS transactions; limit control is simple but effective and is recommended.
Support low-value and CTTA check	By supporting this option, the usage of EC limit in qUICS transactions will change CTTA limit. In this case, EC limit forms part of CTTA limit (EC limit <= CTTA limit)	In this mode, two limits are related; simultaneous control is complex, therefore it is not recommended.
Support low-value or CTTA check	By supporting this option, qUICS considers using EC limit first and switch to CTTA limit if EC limit is insufficient. In this case, EC limit and CTTA limit are two independent offline limits.	The complexity of this mode is between the above two modes. Issuers should include factors such as card features and impact on host system into overall consideration for whether to adopt this process.
Support new card check	If this check is supported, card request online processing if last online ATC register is judged to be 0.	Recommend to support

Condition	Description	Recommended Action
Support PIN try limit check	If this check is supported, card request send online or decline offline (if terminal supports only offline) if PIN try counter exists and is equals to 0.	Recommend to support
Allow offline transaction when currency is not matched	To set whether qUICS offline transaction with unmatched currency is allowed.	Issuers should decide whether this is allowed based on card features and application scope.
Prioritize contact UICS online	In the case of requesting online, Issuers should decide whether to prioritize contact UICS and perform transactions according to the complete processes of standard UICS.	Issuers should decide whether it is allowed based on card features and risk management.
Return available offline spending limit	Set this option to request online return of qUICS card total offline authorized limit to better monitor capital risks.	Issuers should decide whether it is allowed based on card features and risk management.
Support prepayment	This is a condition of checks for prepaid cards. This type of cards do not distinguish primary account and EC amount in the back-office. Fund balance in the card need to be updated irrespective of whether transactions are completed online or offline.	Debit/credit products implementing EC functions do not need to consider this condition.
Not allow transaction with unmatched currency	To set whether qUICS transactions with unmatched currency is allowed.	Issuers should decide whether it is allowed based on card features and application scope.

Condition	Description	Recommended Action
In the case of new card and terminal only supports offline, transaction will be declined	With this condition, card declines offline if last online ATC register is judged to be 0.	Issuers should decide whether it is allowed based on card features and risk management.
Transaction with matched currency supports online PIN	To set whether to support CVM – online PIN for qUICS transactions with matched currency.	Issuers should decide whether to support transactions with matched currency based on (credit) card features, risk management or cardholder's selection.
Transaction with unmatched currency supports online PIN	To set whether to support CVM – online PIN for qUICS transactions with unmatched currency.	Issuers should decide whether to support transactions with unmatched currency based on (credit) card features, risk management or cardholder's selection.
Card requires CVM for transaction with unmatched currency	To set whether to request CVM for qUICS transactions with unmatched currency.	Issuers should decide whether it is allowed based on card features and risk management.
Support signature	To set whether to support CVM – signature for qUICS transactions.	Recommend to support

The table below summaries all card risk management checks of qUICS transactions during application initialization and states implementation conditions for these checks as well as descriptions of check results.

Table 26 qUICS Card Risk Management Check

Risk Management Steps	Check Item	implementation condition	Result (if conditions are met)
Set currency match tag	Currency match check	If “not allow transaction with unmatched currency” bit in card supplementary processing is 1, this check becomes effective.	If “match currency” bit in the card indicator is 0, and “not allow transaction with unmatched currency” bit in card supplementary processing is 1, decline offline
Terminal only supports offline	Offline check	If “In the case of new card and terminal only supports offline, transaction will be declined” bit in card supplementary processing is 1, this check becomes effective.	If last online ATC register is 0, and “In the case of new card and terminal only supports offline, transaction will be declined” bit in card supplementary processing is 1, decline offline
	PIN try limit check	If “support PIN try limit check” bit in card supplementary processing is 1, this check becomes effective.	If PIN try counter exists and equals to 0, decline offline and set indication bit in CVR
	Request CVM check	If “for transaction with unmatched currency, card require CVM” bit in card supplementary process is 1, this check becomes effective. “Support signature” bit in card supplementary processing is used in the check.	If card and terminal all support signature, the card shall continue to attempt offline processing; If either card or terminal does not support signature, the card shall terminate contactless transactions.
Terminal or card requires CVM	No CVM check	If data element “card CVM limit” exists; “card requires CVM for transaction with unmatched currency” bit in card supplementary processing is 0, this check becomes effective.	In the case where neither terminal nor card requires CVM, the card continues the next step “check online process request”.

Risk Management Steps	Check Item	implementation condition	Result (if conditions are met)
	Require CVM check	If data element “card CVM limit” exists or “for transaction with unmatched currency, card require CVM” bit in card supplementary processing is 0, this check becomes effective. “Transaction with matched currency supports online PIN”, “transaction with unmatched currency supports online PIN” and “support signature” bits in card supplementary processing are used in the check.	In the case where terminal or card requires CVM, if both support online PIN, card requests send online; if both support signature, the card continues the next step “check online process request”.
Check online process request	Terminal requests online check	Card data is not needed to initiate this check.	Card also requests online processing and proceed to “complete online transaction” step.
	Offline transaction with unmatched currency check	If “allow offline transaction with unmatched currency” bit in card supplementary processing is 0, this check becomes effective.	If currency does not match, card requests online processing.
	New card check	If “support new card check” bit in card supplementary processing is 1, this check becomes effective.	If last online ATC register is 0, card requests online processing and sets indication bit in CVR
	PIN try limit check	If “support PIN try limit check” bit in card supplementary processing is 1, this check becomes effective.	If PIN try counter exists and equals to 0, card requests online processing and sets indication bit in CVR

Risk Management Steps	Check Item	implementation condition	Result (if conditions are met)
Offline currency check	Offline currency check	“Support low-value check”, “support low-value and CTTA check” and “support low-value or CTTA check” bits in card supplementary processing are used in the check.	If currency does not match, proceed to “offline unmatched currency” step; If currency matches, but does not support any offline purchase check, decline transaction for terminal that only supports offline, request online processing for terminal that supports online.
Low-value check	Terminal supports online – EC single transaction limit check	If data element “EC single transaction limit” exists, this check becomes effective.	If authorized amount is greater than EC single transaction limit, card requests online processing and sets indication bit in CVR.
	Terminal supports online – EC reset threshold value check	If data element “EC reset threshold value” exists, this check becomes effective.	If authorized amount is greater than (EC balance minus EC reset threshold value), card requests online processing and sets indication bit in CVR.
	Terminal only supports offline – low-value check	Data elements “EC balance” and “EC single transaction limit” (if exist) are used in the check.	If authorized amount is greater than EC balance, or greater than EC single transaction limit (if exist), the card declines transaction and sets indication bit in CVR
Low-value and CTTA check	Terminal supports online - EC single transaction limit	If data element “EC single transaction limit” exists, this check becomes effective.	If authorized amount is greater than EC single transaction limit, card requests online processing and sets indication bit in CVR.

Risk Management Steps	Check Item	implementation condition	Result (if conditions are met)
	Terminal supports online – EC reset threshold value check	If data element “reset threshold value” exists, this check becomes effective.	If authorized amount is greater than (EC balance minus EC reset threshold value), card requests online processing and sets indication bit in CVR.
	Terminal supports online – CTTA check	Data elements CTTA and CTTAUL (if exist, otherwise use CTTAL) are used in the check.	If authorized amount is greater than (CTTAUL minus CTTA), card requests online processing and sets indication bit in CVR.
	Terminal supports online – micro and CTTA check	Data elements “EC balance”, “EC single transaction limit” (if exist), CTTA and CTTAUL (if exist, otherwise use CTTAL) are used in the check.	If authorized amount is greater than EC balance, or greater than EC single transaction limit (if exist), or greater than (CTTAUL – CTTA), the card declines transaction and sets indication bit in CVR.
Low-value or CTTA check	Terminal supports online — single transaction limit check	If data element “EC single transaction limit” exists, this check becomes effective.	If authorized amount is greater than EC single transaction limit, card requests online processing and sets indication bit in CVR.
	Terminal supports online — CTTA check	Data elements “EC balance”, CTTA and CTTAUL (if exist, otherwise use CTTAL) are used in the check.	If authorized amount is greater than EC balance, and is greater than (CTTAUL minus CTTA), card requests online processing and sets indication bit in CVR.

Risk Management Steps	Check Item	implementation condition	Result (if conditions are met)
	Terminal supports online — micro or CTTA check	Data elements “EC balance”, “EC single transaction limit” (if exist), CTTA and CTTAUL (if exist, otherwise use CTTAL) are used in the check.	If authorized amount is greater than EC single transaction limit (if exist), or (authorized amount is greater than EC balance, and is greater than (CTTAUL minus CTTA), the card declines transaction and sets indication bit in CVR.
Currency unmatched in offline transaction	International — currency velocity check	If consecutive offline transaction counter (international – currency) and consecutive offline transaction limit (international – currency) exist, this check becomes effective.	If limit is not exceeded, card approves offline; If limit is exceeded and terminal supports online, card requests online; otherwise card declines offline reject. Card sets indication bit in CVR.
Complete online transaction	prioritize select contact type check	If “support prioritize in selecting contact UICS online” bit in card supplementary processing is 1, this check becomes effective.	If terminal also support contact UICS, card requests terminating transaction.
	Low-value and CTTA prepayment check	If “support prepayment” bit in card supplementary processing is 1 and “support low-value and CTTA check” is 1, this check becomes effective.	If fund is insufficient, card declines transaction; If fund is sufficient, card requests online to update EC balance and CTTA.
	Low-value prepayment check	If “support prepayment” bit in card supplementary processing is 1 and “support low-value check” is 1, this check becomes effective.	If fund is insufficient, card declines transaction; If fund is sufficient, card requests online to update EC balance.

Note: Throughout the qUICS card risk management check, as long as the card requests online, “return available offline spending limit”, “support low-value check”, “support low-value and CTTA check” and “support low-value or CTTA check” bits of card supplementary processing will be used to judge whether the cards requires a calculation of available offline spending limit and its calculation method.

7.5.2.3.2 Implementation

Activities related to qUICS application initialization include:

- Policy
 - ✓ Decide whether Issuers should use available offline spending limit, card CVM limit and card transaction qualifier; Since these data elements give Issuers greater risk control over qUICS cards, UPI recommends Issuers to use these data elements in the cards.
 - ✓ Decide which offline data authentication method to support.

Contactless specification requires qUICS cards to support SDA or fDDA. UPI recommends supporting fDDA.
 - ✓ Decide which data items need to be signed

In general, qUICS card also simultaneously supports standard (contact) application, if qUICS signature data is different from standard application, then it is required to have two SADs or two IC card public key certificates, and this will further complicate the implementation.

UPI recommends: if card simultaneously supports qUICS and standard application, qUICS should use the same signature data as standard application.
- Operation
 - ✓ Communicate with card vendors for data elements requiring support according to implemented functionality;
 - ✓ Determine various risk parameter values for card supplementary processing;
 - ✓ Set related parameters for card CVM limit and card transaction qualifier (if supported).
- Technology

Generate data required for application initiation and add it to personalization input file. Data includes card supplementary processing, available offline spending limit, card CVM limit, card transaction qualifier and etc.

7.5.3 Other Data Requirements

1. During offline qUICS transactions, the card cannot return 5F34 during both GPO response and read record. The correct approach is that 5F34 shall be return only in GPO response. Issuers shall not use 5F34 as the “data for static signature” in personalized electronic cash. From September 27, 2011 onwards, 5F34 is no longer allowed 5F34 to appear repeatedly for card that passed testing.
2. It is recommended to set the bit 7 of byte 2 in 9F68 of the card supplementary attribute, “not allow unmatched currency transaction”, to 0 for electronic cash transactions. In this case during qUICS transactions, if transaction currency is unmatched, it is still able to complete transactions online in order to meet the needs of foreign markets. This action is optional and for Issuers to voluntarily to modify its value.
3. During qUICS transactions, 5F24 “application expiry date” shall not be placed in the last record that AFL indicated the terminal to read. Therefore, during personalization of electronic cash, 5F24 “application expiry date” shall not be placed in the last record of qUICS’s AFL that the terminal is going to read.
4. Based on standard debit/credit electronic cash payment and qUICS offline approve transactions, the card shall return 9F74 when reading record. The value of 9F74 is ECC001 and its format is 6-byte ASCII code. For standard debit/credit transactions, the card shall not return 9F74 when reading records.
5. If the bit 6 of byte 1 in 9F6C of the card transaction attribute, “if offline data authentication has failed and online is available for the reader, reader requests online”, = 1, please consider the following scenario:

Card offline approved and returned all the records; at this time, 9F79 value in the card has been deducted. But for some reasons the terminal determines fDDA fails and causing transaction processing online, resulting in Issuers deducting the amount again in which the transaction deducts the amount twice. To avoid this scenario, it is recommended to set bit 6 of byte 1 in 9F6C to 0.

6. For qUICS transactions, in order to ensure the success rate of card waving during offline transactions, please place minimal amount of data in the last record the terminal needs to read in AFL indicator. It is recommended that during personalization of electronic cash data, the last record of qUISC’s AFL should only contain 9F74 data, and its value is ECC001.

7.6 Extended Application for Contactless IC Card Micropayment

To meet the needs of sectional and time-based sectional charging and on the basis of standard qUICS transaction process, an “UPDATE CAPP DATA CACHE” command is added in the supplementary application data between GPO command processing and

READ RECORD command processing. Terminal can use data from READ RECORD to perform the next action logic.

7.6.1 Data Elements Involved

To satisfy the requirements for electronic cash extended application, the following data elements are added to electronic cash application:

Table 27 New Data Elements Added for Contactless IC Card Low-Value Extended Application

Name	Format Label Length	Conditions Requirement	Description	Retrieval	Value
CAPP transaction indicator bit	F: b8 T: DF60 L: 1	If card supports CAPP extended transactions, then must indicate this data in PDOL	Indicate the CAPP transaction types supported by terminal	N/A	0: indicate that terminal does not support extended application 1: indicate sectional chargeback transaction is supported 2: indicate offline pre-authentication is supported 3: indicate that it supports using offline pre-authentication to complete transactions
Sectional chargeback application indicator	F: b8 T: DF61 L: 1	If card only supports sectional chargeback application, then Issuers should personalize in BF0C	Is used to distinguish the functionality of extended application supported by the card	Returns by the issuer self-defined data BF0C in file control information (FCI)	0: indicate that card only supports sectional chargeback application 1: indicate card supports extended application

Name	Format Label Length	Conditions Requirement	Description	Retrieval	Value
Advance overdraft limit for electronic cash sectional chargeback	F: n12 T: DF62 L: 6 Format: cn	If card supports the functionality of sectional chargeback, then it supports Get Data and Put Data commands	Indicates the maximum overdraft limit during sectional chargeback transactions	Returns by Get Data command	
Overdrafted amount for electronic cash Sectional chargeback	F: n12 T: DF63 L: 6 Format: cn	If card supports the functionality of sectional chargeback, then it supports Get Data but does not support Put Data command	Indicates the current overdrafted amount in the card	Returns by Get Data command	

To satisfy the requirements for sectional and time-based sectional charging, card shall establish specific files for extended application in accordance with *UnionPay Integrated Circuit Card Specification – Product Specification Part 2 Extended Purchase Specification Based on Contactless Low-value Payment Application*.

7.6.2 Risk Management

Different from continued strengthening of risk controls for large-value transactions, Issuers can flexibly implement a simplified risk control functions to satisfy the low-value features and fast processing requirements for electronic cash. EC risk control function can be relatively independent from standard applications, for example, supporting different offline data authentication methods and different cardholder verification methods.

7.6.3 Implementation

- Policy
 - ✓ Decide whether the card should support contactless IC card micropayment extended applications.
 - ✓ Assess the requirements for new extended applications.

- Operation
 - ✓ Decide which extended applications the card is currently needed to support, and determine the relevant data.
- Technology
 - ✓ Generate necessary data for sectional and time-based sectional charging and add it to personalization input file. Data includes CAPP trading indicator bit, sectional chargeback application indicator, advance overdraft limit for electronic cash sectional chargeback, overdrafted amount for electronic cash sectional chargeback and etc.
 - ✓ Generate extended application-specific files required by sectional and time-based sectional charging.

7.7 Other Considerations

The network entry testing for NFC mobile phones, mobile phone accessories, SIM cards and SD card shall be written to issuer product information (9F63) of the card, and the data content and format shall be accordance with *UnionPay Integrated Circuit Card Specification – Basic Specification Part 2: Debit/Credit Table A.1*.

To allow cards issued by institutions can be more normally accepted by terminals in foreign countries, and to avoid decline transactions due to there are no common supporting authentication methods, it is recommended the CVM list shall include online PIN for debit cards and signature for credit cards in standard debit/credit application. When card supports both online PIN and signature authentication, the order in CVM list also reflects the differences in security policies: if online PIN is located before signature in CVM list, then signature is only required if online PIN is unsuccessful; otherwise, if signature is located before online PIN in CVM list, then online PIN authentication is only required when signature is unsuccessful. Since the order for different authentication methods in CVM list may lead to differences in security policies, Issuers should be noted when setting the related parameters.

IAC decline and the setting for CVM list should not decline transactions when cardholders bypass online PIN.

8 Issuer Host System Modifications

This chapter gives an overall description of issuer host system modifications required by IC card migration, and it helps Issuers in implementing IC card migration plan.

8.1 New Authorization Data

This section focuses on introducing new chip data required by authorization, financial and settlement types of transactions.

8.1.1 Upgrading Message Fields

Issuer host system shall upgrade field 22 and field 60, in order to recognize definitions of new values arising from IC card upgrade.

- POS entry mode (Field 22)

“POS entry mode” represents the input mode of cardholder’s data (such as PAN and PIN):

 - ✓ PAN entry mode 05 – IC card and card information is reliable;
 - ✓ PAN entry mode 07 – read contactless IC card with qUICS application;
 - ✓ PAN entry mode 95 – IC card but card information is not reliable;
 - ✓ PAN entry mode 98 – read contactless IC card with UICS debit/credit application.
- Terminal entry capability (Field 60.2.2)

This value is a decimal number code and indicates whether terminal can read IC card during IC card transaction:

 - ✓ New value 5 – readable contact IC card;
 - ✓ New value 6 – readable contactless IC card.

8.1.2 New Message Fields for IC Card

- IC card data field (field 55)

For online transaction message, most of the new data is stored in “IC card data field” (field 55); for settlement message, most of the new data is stored in “Section 2 – IC card featured information based on UICS debit/credit” of the file system.
- IC card serial number (field 23)

This is used to distinguish different cards with the same PAN.

Issuers shall also modify their back-office system, for example, cardholder statements, reports and etc. Please refer to “issuer back-office system modifications” for more information.

For IC card security upgrade, new IC card data elements are described in the table below:

Table 28 New UICS Data Elements

Field Name	Description	Transmission System (Field/Tag)	File System (section/displacement)
Card serial number	Used to distinguish different cards with the same PAN. Only used during IC card transactions.	23	Section 2, 19-21
Application cryptogram	Application cryptogram (TC, ARQC or AAC) generated by IC card	55—9F26	Section 2, 0-15
Cryptogram information data	Indicates the cryptogram type returned by card and operations that need to be performed by terminal.	55—9F27	Section 2, 198-199

Field Name	Description	Transmission System (Field/Tag)	File System (section/displacement)
Issuer application data	Special data need to be transmitted to Issuers during online transactions. 1st byte is the data length defined by IC card. Content Format: Length (07) (1 byte) DKI (1 byte) Cryptogram version number (1 byte) Card verification result (CVR) (4 bytes) Algorithm identifier (1 byte) If data is defined by Issuers, above data should be followed and appended by a byte containing the length of issuer-defined data and issuer-defined data that is 1-15 bytes long.	55—9F10	Section 2, 56-119
Unpredictable number	Contains a random number, used to generate application cryptogram to provide variability and uniqueness.	55—9F37	Section 2, 40-47
Application transaction counter	Records the number of transactions after personalization, maintained by application on the card.	55—9F36	Section 2, 120-123
Terminal verification result	Used to record a set of indication bits for all UICS function process results performed by the terminal. For example, offline data authentication results.	55—95	Section 2, 30-39

Field Name	Description	Transmission System (Field/Tag)	File System (section/displacement)
Transaction date	Local date of transaction authorization	55—9A	Section 2, 128-133
Transaction type	Financial transaction type represented by first 2 bits of processing code according to ISO 8583:1987 definition	55—9C	Section 2, 181-182
Authorized amount	Stores the current transaction amount	55—9F02	Section 2, 183-194
Transaction currency code	Transaction currency code according to ISO 4217 specification	55—5F2A	Section 2, 195-197
Application interaction feature	A list stating the card's capacity of supporting specified functions in this application.	55—82	Section 2, 124-127
Terminal country code	Terminal country code in accordance with ISO3166	55—9F1A	Section 2, 134-136
Other amounts	Second amount related to the transaction, representing cash back amount	55—9F03	Section 2, 200-211
Terminal capability	Represents data input on terminal, CVM support and security capability	55—9F33	Section 2, 24-29
CVM result	Represents the last CVM results.	55—9F34	Section 2, 212-217
Terminal type	Indicate the terminal environment, communication capacity and operation control	55—9F35	Section 2, 218-219

Field Name	Description	Transmission System (Field/Tag)	File System (section/displacement)
Interface device serial number	Unique and permanent serial number distributed to terminal IFD by the vendor.	55—9F1E	Section 2, 48-55
Dedicated file name	DF name in accordance with ISO7816-4 specification	55—84	Section 2, 220-251
Application version Number	Version number distributed to application by payment system	55—9F09	Section 2, 252-255
Transaction sequence counter	Counter maintained by terminal that increments by one for each transaction	55—9F41	Section 2, 256-263
Issuer authentication data	Data used for issuer authentication, received from Issuers and sent to the card through the terminal. In this document, issuer authentication data includes the following two parts: ARPC (8 bytes) Authorization response code (2 bytes)	55—91	—
Issuer script template 1	Template contains issuer specialized script data sent to the card before command to generate application cryptogram for the second time.	55—71	—
Issuer script template 2	Template contains issuer specialized script data sent to the card after command to generate application cryptogram for the second time.	55—72	—

Field Name	Description	Transmission System (Field/Tag)	File System (section/displacement)
Issuer script result	Records processing result of issuer script by card. This result should be included in the settlement message and the next online authorization.	55—DF31	Section 2, 137-178
Issuer product information	Records IC card issuer product information, such as information on citizen card, military card or social security card.	55 – 9F63	
IC card condition code	Represents whether IC card read/write capacity of the IC card terminal is available when magnetic-stripe information on IC card is used on IC card terminal. The value in this field can help determine whether the card or terminal is damaged and whether the transaction is initiated by counterfeit cards.	60.2.3	—
IC card verification reliability flag	Represents verification reliability of this card in IC card transactions. Acquirer will set this value when merchants or terminals have problems, or set by UnionPay System when both the acquirer and issuer cannot perform verification for this card.	60.2.7	—

Note: The table above contains new values or data sub-fields of original data field (such as field 22 and field 60) that are added to IC card.

Required activities to implement host system upgrade for realizing a complete IC card upgrade include:

Technicians should assess the impact on the host system, as well as the technical approach for the upgrade: in-house development, outsource to system vendor, or rely on third-party processors.

- In-house Development

Since the scope of upgrade from IC card implementation is extensive, it is necessary for Issuers to launch a formal project. Issuers should first build a team consisting of experts from different departments and assign a project manager to coordinate and manage this project. Like all formal projects, it is necessary to have project plan, periodic meetings, meeting minutes and delivery list; business requirement, technical requirement, design files, test and quality control process are also required.

- Host system Vendor

Most of Issuers request host system vendors to customize their software in an effort to meet their special requirements. UPI recommends Issuers should perform complete system tests after customization.

- Third Party Processor

Issuers shall also perform a complete functional test if they are using a system created by third party processor.

Issuers shall test its system after development to ensure processes are correct.

Issuers shall also perform integrated tests and IC card online tests required by UPI, to ensure the upgrade of issuer host system meets bank card online specification.

8.1.3 Electronic Cash Transaction

EC transactions under standard IC card application are all offline purchase transactions. Settlement messages uploaded should add at least the following new data:

- Electronic cash balance;
- Electronic cash issuer authorization code;
- TC and corresponding calculated data elements.

When card supporting EC functions requests online in standard IC card application, it should submit EC balance for Issuers to perform automatic load check, risk control and etc. This data is stored in IDD of original Tag '9F10' and has no additional impact on message transfer.

8.1.4 qUICS Transaction

Given qUICS already supports standard IC card application, there are not much system modifications required to support qUICS transactions. The modifications are mainly new data fields for existing data fields:

- POS entry mode (field 22)
PAN entry mode 07 – qUICS entry.
- Terminal Read Capability (field 60.2.2)

6 – Terminal has contactless read capability.

8.2 Online Card Authentication

This section describes host system modifications for Issuers to perform online card authentication.

8.2.1 Summary

When the card and terminal jointly decide to send a transaction online, chip will generate an ARQC. Chip will combine related card, terminal and transaction data with special algorithm, then generate ARQC using application cryptogram session key (AC SESK) and symmetric key algorithm. The cryptogram is unique to each transaction.

Note: AC SESK is generated by AC UDK stored in card secure zone.

Chip transmits ARQC and related data to the terminal and terminal transmits ARQC and original data used to generate this cryptogram to Acquirers, this original data includes offline risk management (processing restrictions, offline PIN, and offline data authentication) results. Acquirers place ARQC into the request message based on format and transfers to Issuers online; Issuers use HSM to verify this cryptogram and the verification results are used for authorization decision.

8.2.2 Prerequisite

Prerequisites for online card authentication include the following:

- Issuers must set card to support online card authentication.
- The issuer system must completed IC card upgrade.
- Acquirers accepting this type of transactions must completed IC card upgrade.

To prepare for system environment to support online card authentication, Issuers must:

- Load AC MDK and corresponding DKI to issuer host system.
- Support double-length keys.
- Upgrade issuer host system and support ARQC verification.

Note: ZCMK and MDK must be double-length.

8.2.3 Implementation

The table below lists technical activities related to online card authentication:

Table 29 Online Card Authentication Activities

Step	Responsible Party	Activities
1	Technical	After receiving a transaction request message, Issuers should send data required to verify ARQC to HSM, these data include: • ARQC; • data required to generate ARQC; • Data used to locate appropriate MDK and derived UDK (DKI, PAN and card serial number).
2	Technical	HSM uses DKI to locate MDK.
3	Technical	Finds MDK; HSM uses MDK together with PAN and card serial number to derive UDK.
4	Technical	HSM uses UDK together with application transaction sequence number to derive SESK.
5	Technical	HSM generates an ARQC using symmetric key algorithm to combine SESK and other data required to generate ARQC.
6	Technical	HSM compares ARQC generated by itself with ARQC generated by the card. If they match, online card authentication has succeeded; Otherwise, online card authentication has failed.
7	Technical	Host system records online card authentication result.
8	Technical	Host system uses online card authentication result for the online authorization decision making.

8.3 Online Issuer Authentication

This section describes host system modifications for the issuer to perform online issuer authentication.

8.3.1 Summary

To protect issuer's authorization response and ensure that this response comes from a valid issuer, Issuers can send a cryptogram for "online issuer authentication" to card, and this cryptogram is called ARPC. Data used to generate ARPC includes ARQC and issuer authorization response.

Issuers include ARPC in the response message, then terminal forwards it to the card, and card uses UDK stored in the card to verify ARPC. ARPC verification result will affect the final transaction dispositions, for example, if ARPC verification fails, the card may override the approval authorization response from issuers (or agent) to decline transaction. This can avoid transaction being approved by unauthorized party during authorization process.

The card will set and save online issuer authentication result. Issuers can access this indicator in clearing transaction or during next online authorization request.

This section describes activities for Issuers to perform online issuer authentication in their systems.

8.3.2 Prerequisite

Prerequisites for online issuer authentication include:

- Issuers must set card to support online issuer authentication.
- The issuer system must completed IC card upgrade.
- Acquirers accepting this type of transactions must completed IC card upgrade.

In preparation for system environment to support online issuer authentication, Issuers must:

- Load AC MDK and corresponding DKI to issuer host system.
- Support double-length keys.
- Upgrade issuer host system to support ARQC verification.

8.4 Future Cryptogram Support

With the needs of market development, new cryptogram version may be defined in the future to include different data elements to meet special needs. UPI recommends the issuer host system should be flexible enough to adapt to new cryptogram version. “Cryptogram version number” can be set as a process parameter in the host system, thus system infrastructure will easily adapt to the changes when a new version number is input.

Activities required by future cryptogram support include:

- Technology

By setting “cryptogram version number” as a logical parameter to process online card authentication and issuer authentication, system architecture can easily adapt to the introduction of new cryptogram.

8.5 Data Recording/Logging

UnionPay System has provided a large amount of new data that are supported in online messages and settlement files. Issuers shall determine the data most important to log based

on customer service, risk management, service assessment and transaction analysis. This section recommends Issuers to consider the following activities:

- Policy

Decide which data Issuers want to log. Issuers should continue to log the data that are required to be logged by current system; additionally, Issuers should log data and records related to online card authentication:

- ✓ Whether online card authentication has passed;
- ✓ Whether DKI is present in message;
- ✓ Whether MDK is found in HSM;
- ✓ Whether All data elements required by cryptogram version 01 exist;
- ✓ Whether there are any HSM parity errors.

Issuers should also log other chip data which benefits transaction analysis:

- ✓ Offline PIN result (in CVR);
- ✓ Offline authentication result (in TVR);
- ✓ Application transactions counter value.

- Technology

Modify systems to record new data.

8.6 Issuer Script Processing

This section describes host system modifications needed for Issuers to implement issuer script processing.

8.6.1 Summary

Issuer script processing allows Issuers to change information related to account or block all applications in the card after card issuance. Issuers place script commands in transaction response message and send it to the terminal, terminal forwards the commands to the card. The card performs commands after security requirements are met.

Secure message adopts two types of DES keys:

- Message Authentication Code (MAC) Key

This key is used to secure issuer script processing and must be applied to all issuer script commands. It is used to confirm data is not manipulated (integrity), and concurrently confirm whether the commands sent by Issuers are valid (issuer authentication); MAC MDK is stored in issuer host system's HSM, while MAC UDK is derived from it and is stored in the card.

- Data Encryption (ENC) Key

This key is used to encrypt sensitive data in the script, such as offline PIN and etc. ENC MDK is stored in issuer host system's HSM, while ENC UDK is derived from it and is stored in the card.

Commands supported by issuer script processing include:

- Block application
- Unblock application
- Block card
- PIN change/unblock
- Data configuration
- Change record

For details of each command, please refer to “Table 18 – Issuer Script Processing Commands”.

8.6.2 Prerequisite

Prerequisites for issuer script processing include:

- Issuer system must perform IC card upgrade.
- Acquirers accept this type of transactions must perform IC card upgrade, such that internal transactions by Issuers would send script processing commands.
- Configure card to support to receive and process commands and data of issuer scripts. Issuers should confirm this requirement with card vendors prior to procurement.
- Length of issuer script returned in transaction response message shall not exceed 128 bytes, because certain networks may not support this length, therefore UPI recommends that Issuers should try to avoid sending scripts with maximum length. It is also possible to send issuer script commands with issuer-controlled special devices, such as ATM and terminal in bank counter; according to EMV specifications, these commands can be extended to 261 bytes (if supported by issuer's network).

Issuer script processing must comply with *Unionpay Integrated Circuit Card Specification* and related operation regulations.

- Issuer script processing should not have negative impacts on POS transaction or ATM transaction.
- Only Issuers have the rights for post-issuance update.
- Issuer script can only be sent in transaction response message.

In preparing for system environment to support issuer script processing, the issuer must:

- Load MAC MDK and ENC MDK to issuer host system.
- Support double-length keys.
- Upgrade issuer host system to support issuer script processing.

8.6.3 Implementation

The table below lists technical activities related to implementation of issuer script processing:

Table 30 Issuer Script Processing Activities

Step	Responsible Party	Activities
1	Technical	To ensure that only Issuers can send script processing commands and any sensitive data in the script are secured, the issuer host system must perform the following key management operations: • Import MAC MDK to HSM, this key is used to secure issuer script processing; • Import ENC MDK to HSM, this key is used to encrypt sensitive data in the script, for example, if Issuers would like to use PIN change/unblock command to change cardholder PIN, Issuers must use this key to encrypt the new PIN; • Securely send these MDKs to the personalization system, these MDKs are used to derive UDKs and import them to the card.
2	Technical	Based on the analysis of issuer script commands, Issuers should determine command sets for implementation. A simple method is: only support block application command at the early project stage, and gradually support other commands as experience is accumulated for issuer script processing management. If Issuers support offline PIN, it shall also support PIN change/unblock command.
3	Technical	Issuer host system adds command files it supports.

Step	Responsible Party	Activities
4	Technical	Create back-office processing procedures: allow customer service and risk management in the master files, set to require issuer script processing symbols for specific chip card brand, card type or card segment setting. The following are examples for few commands: - APPLICATION BLOCK – Using existing procedures for flagging a card confiscation on the master file, adds logic to allow sending the APPLICATION BLOCK command in addition to a decline response. - DATA configuration – Through historical analysis of cardholder patterns, Issuers can indicate in the master file that specific cards or card ranges require velocity counter updates. UPI recommends that Issuers should not send “confiscate card” response to a multi-application card, instead, use either the CARD BLOCK or APPLICATION BLOCK command.
5	Technical	Search master file information during an online transaction processing, if master file indicates the need to initiate an issuer script processing, locate its issuer script command. This command is prepared in step 4.
6	Technical	According to the issuer script command, add specific data required by the command, such as new cardholder offline PIN and updated value of velocity limit.
7	Technical	If there is sensitive data in the script, perform the following operations to encrypt sensitive data; otherwise, proceed to step 9. - Locate ENC MDK in HSM; - Use ENC MDK to derive ENC UDK; - Use ENC UDK together with ATC to derive ENC SESK; - Use ENC SESK to encrypt sensitive data.

Step	Responsible Party	Activities
8	Technical	Generate a MAC value and attach to the end of issuer script: • Locate MAC MDK in HSM; • Use MAC MDK to derive MAC UDK; • Use MAC UDK together with ATC to derive MAC SESK; • Use MAC SESK to generate a MAC value for entire content of post-issuance data. Note: ensure card supports this MAC length (currently UICS specifies length of MAC to be 4).
9	Technical	Include issuer script to transaction response message (field 55, Tag 72).
10	Technical	After Issuers obtain issuer script processing result from settlement file or subsequent request messages, modify related data in the master file.

8.6.4 Chip-Initiated Transactions

Issuers should continue to use current criteria for authorization decision processing, such as the effort to ensure the account is in good standing and has available funds. Simultaneously, Issuers shall analyze all the new data added to the chip, especially CVR and TVR, to decide what information are very important to Issuers. UPI recommends that Issuers should include the data in the following table in authorization decision processing.

Table 31 Data Elements to Include in Authorization Decision Processing

New Data	Function
Online card authentication result	Protection against counterfeit cards and providing data integrity. Evidence of card-present transaction. Effective proof of offline processing results.
Offline PIN verification result	Avoid card being stolen Evidence of cardholder is present during transaction

New Data	Function
Offline data authentication result ¹	Protection against data alteration (SDA) or against alteration and counterfeit (DDA).
New card indicator ¹	Evidence of card is being used for the first time.
Merchant forced transaction online indicator ¹	Evidence that merchant thinks the transaction (or cardholder) is suspicious and mandatory requesting an online transaction.
PIN try limit indicator ¹	Evidence of card being stolen or the cardholder has forgotten PIN.
ATC	Since this counter provides a consecutive reference number for each transaction, if ATC value is repeated or separated with the previous one with a large span, it may indicate counterfeiting or other frauds.

1 – If Issuers have set card action to be decline offline for these conditions, issuer authorization system will not be able to obtain evidences for these events.

The activities required in issuer authorization decision processing using abovementioned data include:

- Policy
Decide criteria that Issuers will use for authorization decision. It is recommended to take the current criteria as the basis, and combine with data recommended by UPI for enhancement.
- Technology
Modify host system; including the usage of new chip data in authorization decision processing.

8.6.5 Fallback Transactions

Fallback transaction means that chip card performs non-chip transaction (magnetic-stripe transaction) on terminals with chip capability. In issuer host system, this situation can be identified through combination of the following data fields:

- Field 22 “POS entry mode” is not 05, 07, 95, 98 or 91 indicating a non-chip reading mode.
- Field 60.2.2 “terminal entry capability” is 5 or 6 indicating the terminal is able to read chip card.

- Field 60.2.3 “IC card condition code” is 2 indicating abnormality occurs when terminal attempted to read IC card.
- Field 55 “chip information field” is not presented.

Issuers shall modify the host system to check these values from magnetic-stripe transactions to determine whether these transactions are qualified as fallback transactions. Issuers shall decide whether they should treat fallback transactions differently from other magnetic-stripe transactions. It is an issuer decision after balancing customer service with risk control.

Activities required to process fallback transactions include:

- Policy
 - ✓ Issuers should decide how to treat fallback transactions based on market conditions. If chip migration is relatively mature in the market where issuer is positioned in, Issuers should incline to view fallback transactions as transactions with high fraud and credit risks. Issuers should decide when is the right time to approve or decline fallback transactions.
 - ✓ Issuers can decide not to decline fallback transactions when products are first launched or before new chip technology becomes stable in the market. When Issuers believe the market is sufficiently mature, Issuers should consider more severe handling for fallback transactions. Please note that factors causing fallback transactions are various, for example: card fraud, collusion with merchants, poor merchant acceptance procedures, chip inoperative and faults appear when terminal is reading chip can all cause fallback transactions.
 - ✓ UPI strongly recommends that Issuers should perform trend analysis on fallback transactions. This analysis can help Issuers to determine reasons causing fallback transactions. UPI, Issuers and other members involved can all use this information to trace real situation related to fallback transactions.
- Technology

Modify the host system to identify fallback transactions. For fallback transaction processing actions, develop fallback transaction trend analysis report.

8.7 HSM (Encryption Service) Processing

To support related cipher text authentications of IC card, HSM module shall be in accordance with *UnionPay Integrated Circuit Card Specification – Basic Specification Part 4: Debit/Credit Application Security Specification*, and support the following IC card cipher text processing requirements:

- ARQC authentication and ARQC computing services.
- IC card script MAC computing services.
- IC card script encryption computing services.

8.8 Other Considerations

In order to be supported by more foreign institutions, Issuers should mandatory verify field 35 for ATM transactions. For POS transactions, if Acquirers did not transmit field 35, Issuers would relax the field 35 verification and not decline the transaction.

9 Issuer Key Management System

This chapter describes the processing requirements for IC card security data during UICS upgrade. These IC card security data is generated securely through the IC card key management system and is part of the IC card personalization data.

9.1 Issuer CA Management

9.1.1 Issuer Certificate Request

Prior to UICS card issuance, Issuers shall apply to UPI for issuer certification in accordance with UnionPay Integrated Circuit Card Specifications – Auxiliary Specifications – Part 6 Financial IC Card Debit/Credit Applications Root CA Public Key Authentication Specification – Management Specification.

9.1.2 IC Card Static Data Signature

To ensure that important information on IC card has not been tampered, IC card key management system uses issuer private key for signature verification on related information, and writes them into IC card through personalization devices. UnionPay recommends the following data is used to generate static data signature:

Table 32 Application Data for Static Signature

Data Item	Condition
AIP	If DDA is supported
Application effective date	Recommended
Application expiration date	Recommended
Application PAN	Recommended
Application PAN serial number	Recommended, but filter this item during qUICS transactions
AUC	Recommended
CVM list	Recommended and there may be multiple lists in the card
Issuer action code – Default	Recommended

Data Item	Condition
Issuer action code – Decline	Recommended
Issuer action code – Online	Recommended
Issuer country code (“5F28”)	Recommended

9.1.3 Issuance of IC Card Certification

For each IC card, Issuers shall define the appropriate IC card public key length and public key exponent, and randomly generate IC card public and private keys. Issuers shall issue IC card certificates using issuer private keys in accordance with *UnionPay Integrated Circuit Card Specification – Basic Specification Part 4: Debit/Credit Application Security Specification*.

9.1.4 Implementation

Activities related to issuer CA management include:

- Policy
 - ✓ Decide the length of issuer public key to use.
 - ✓ Decide the length of IC card public key to use.
 - ✓ Decide the public key exponent to use.
- Operation
 - ✓ Choose appropriate issuer public key length and IC card public key length according to business needs and transaction speed requirements.
 - ✓ IC card public key length shall not be longer than issuer public key length, and issuer public key length should not be longer than root CA public key length.
 - ✓ UPI recommends setting issuer public key exponent to 3 in order to increase transaction speed.
- Technology
 - ✓ The processing speed of hardware encryption machine is relatively slow when generating RSA public and private key pair. Therefore, Issuers should rationalize the number of deployed hardware encryption machine or use key pool technology, in order to break the technical bottlenecks when hardware encryption machine processing RSA public and private key pairs.

9.2 Symmetric Key Management

9.2.1 Transaction Master Key Management

Issuers shall generate one or more groups of IC card transaction master keys: AC MDK/MAC MDK/ENC MDK and these keys are closely related to transaction authentication processing. When generating IC card system data, Issuers shall generate IC card transaction keys from IC card number and card serial number and write them into IC card system file.

9.2.2 IC Card Master Key Management

Issuers shall generate and manage their IC card master key, and this IC card master key can be used to add, delete and modify IC card application.

9.2.3 Other Transmit Keys

IC card key management system will interact with Issuer's other external systems to generate and manage various types of transmit keys, such as KEK keys, which are private keys used to encrypt IC card output, and IC card transaction keys.

9.2.4 Implementation

Activities related to symmetric key management include:

- Policy
 - ✓ Decide the length of transaction master key to use, and UPI recommends using double-length DES key.
 - ✓ Decide the number of group transaction master key to use.
 - ✓ For IC card master key, transit keys and other symmetric keys, Issuers shall develop a comprehensive key management practices to prevent key compromises.
- Operation
 - ✓ For symmetric key management, Issuers may use back-to-back management style and allow multiple people to own the same private key in sessions.
 - ✓ Issuers may generate only one unique set of IC transaction master keys or one set per each BIN, or use any other groupings to manage transaction master keys in order to satisfy various requirements Issuers may have for their own key management.
- Technology
 - ✓ IC card chip-level keys should be generated by real-time computation and written into IC card system file at once for card personalization. Otherwise, card-level keys should not be stored in any system.

10 Issuer Data Preparation and Personalization System

This chapter describes the Issuers involved complex IC card system data generation process during IC card upgrade. The system data generation process is integrated management of all different types of IC card data such as cardholder parameters, IC card limits, IC card controls and IC card security data (certificates and keys). In response to processing needs, Issuers generally need to create a new separate system, such as data preparation system, to support.

10.1 IC Card System Data Classification

Outputs of data preparation system are IC card system data and include all application data in an IC card. Classifying IC card data element in a reasonable manner can ensure the flexibility of IC card product parameter definition and the independence of each IC card product, thereby helping Issuers with systematic management on IC card data.

External Dynamic Data

This type of parameters is generally card-level dynamic data and includes: cardholder name, identification number and other personal information; upper limit (lower limit) of consecutive offline transaction, offline balance, upper limit of offline balance, upper limit of amount for single transaction and other amount limit type of parameters; card number, validity date, expiration date and other traditional magnetic-stripe card product information. Magnetic-stripe card parameters can be generated and managed in existing magnetic-stripe card system, and the data preparation system can treat them as externally import data.

IC Card Static Template Parameter

This type of parameters is generally product-level static data and includes CVM, IAC, issuer supplementary code, application label and other template data. Issuer may define the template according to different application scenarios and card types, and specifically can refer to the 14 templates defined in *UnionPay Integrated Circuit Card Specifications – Auxiliary Specifications Part 5: Debit/Credit Card Personalization Guide*. This type of parameters is used as internal parameters for the data preparation system and for template definition.

IC Card Security Data

This type of parameters is dynamically accessed from IC card key management system, and includes IC card symmetric key, IC card certificate information, issuer certificate and other information.

10.2 IC Card DGI Data Classification

Data Grouping Index (DGI) is a data grouping indicator for personalization data. The first byte of DGI denotes short file identifier (SFI) and the second byte denotes file record number. Issuers should assign IC card system data to different DGI packets in a reasonable manner, thereby making the internal file structure of UICS card will not be personalized

inconsistently when using different card vendors, in which is beneficial to Issuers with integrated management on IC card internal files.

For Issuers perform DGI group assignment planning on IC card data using data preparation system, the following principles are served as references:

Cardholder data is placed in the same DGI packet

Cardholder name, identification number and other cardholder information should be placed in the same DGI packet as possible, so that counter and terminal equipment can obtain cardholder information by reading only one IC card internal document record and they would also update all cardholder information simply with a single write command.

Return as few DGI packets as possible during qUICS transactions

Since qUICS is used in quick and offline payment environment, so qUICS transactions should return as few DGI packet data as possible. In general, it is possible for qUICS to simply return the following basic information: equivalent magnetic track information, issuer certificate information and IC card certificate information.

Set the last DGI packet to be returned as short as possible during qUICS transactions

The last DGI packet data to be returned shall be set as short data during qUICS transactions in order to enhance the transaction success rate. It is generally recommended that electronic cash issuer authorization code (9F74) is set as the only data element in the last DGI packet during qUICS transactions.

10.3 Interactions with IC Card Key Management System

IC card data preparation system interacts with IC card key management system to jointly undertake IC card security data generation.

- IC card data preparation system maintains issuer certificate serial number corresponding to IC card products, imports unsigned static data into key management system, and obtains issuer certificates, IC card certificates and IC card static signature data from key management system.
- IC card data preparation system imports card number, card serial number and key index corresponding to IC card products into key management system, and obtains IC card symmetric key from key management system.
- Sensitive data outputted from data preparation system (such as symmetric keys and private keys correspond to IC card) must be encrypted by IC card key management system before exporting securely.

10.4 Personalization Processing

Output data from data preparation system are IC card system files that contain all of IC card application data. Through card vendors or self-built IC card personalization center, Issuers use IC card system files to perform IC card personalization operation.

- Issuers may send IC card system files directly to card vendors in the case of manufacturing large quantities and conventional cards, and allow card vendors to be solely responsible for IC card personalization and printing and embossing of card surface.
- Issuers may also build their own personalization centers to meet the needs of small-volume and fast card production. Issuers should procure IC blank cards (in general, blank cards already contain IC card standard application) and IC card punch machines (or upgrade from existing magnetic-stripe card punch machines with chip module configuration), and develop personalization software system at the technical level that meets the requirements of building a personalization center. At the same time, Issuers shall equip with relevant security management system to ensure the security of personalization centers during operation and maintenance.

10.5 Implementation

Activities related to data preparation and personalization include:

- Policy
 - ✓ Issuers shall classify IC card system data in a reasonable manner, in order to flexibly meet the IC card system data processing requirements for a variety of IC card products.
 - ✓ Issuers shall assign IC card system data to different DGI packets in a reasonable manner.
 - ✓ Issuers shall decide using in-house personalization centers or card vendors to manufacture cards according to different business needs.
- Operation
 - ✓ Issuers may adopt the recommended strategy in this section to classify IC card system data and to assign groups for IC card DGI data.
 - ✓ Issuers may manufacture large quantity IC cards through card vendors; and manufacture small quantity and urgent IC cards through in-house personalization center.
- Technology
 - ✓ IC card data preparation system is an indispensable subsystem for IC card system card, and the system can greatly reduce the workload of existing card system migration and the difficulty in implementation.

11 Issuer Back-office Account Management and Business Security

This chapter mainly describes the technical modifications required by back-office to support IC card application:

- Actual bank case for account and transaction implementation
- Electronic cash
- Statements
- Customer service
- Card management system
- Card replacement
- Dispute resolution
- Settlement and reconciliation
- Reports
- Internal staff training

11.1 Actual Bank Case for Account and Transaction Implementation

11.1.1 Upgrading of Account and Transaction Structure

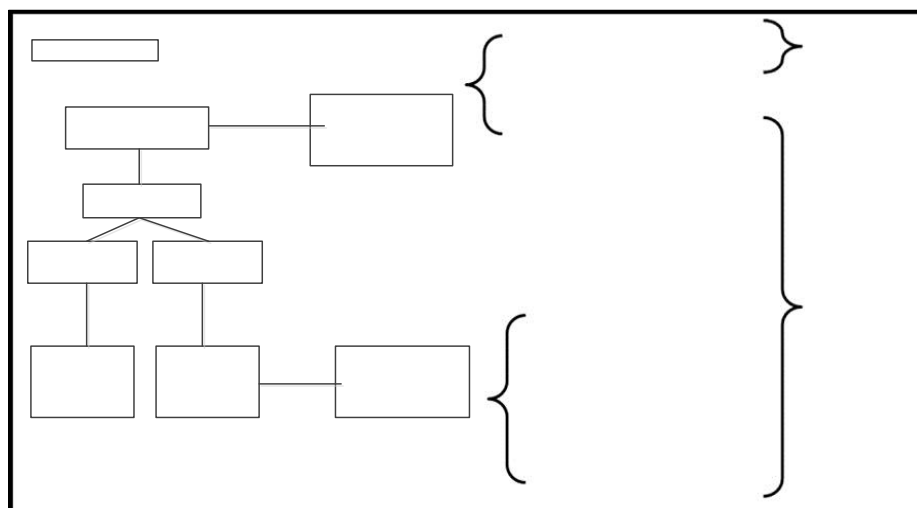


Figure 1 UICS Account and Transaction Structure

● Account Structure

The account structure of the original magnetic strip card (online account) remains unchanged while a card level offline account is established directly under each online account, and each IC card has an independent offline account limit to support the offline payment function of the IC card.

Although micro payment specification is rather flexible on the definition of account setting and limits application in issuer host end, issuer should avoid too complex account management and avoid cardholder confusion. It is recommended to manage account with following principle:

- ✓ Outward transfer the EC limit from the master account, or add an EC limit proportionally to the credit limit of the card holder.
- ✓ Identify the EC limit as a mapped account related to the master account and transfer cash from the current account to the mapped account, which is called EC account.
- ✓ The authorization and settlement of all online transactions are conducted based on the master account of the card holder. Neither the EC account amount nor the EC balance of the card is influenced.
- ✓ With regard to consumption transactions completed through EC balance offline authorization of the card, the subsequent settlement is based on the EC account.
- ✓ The EC account cannot be reported for loss or bear interest or support cash withdrawal (unload is applicable at the time of card cancellation).

● Transaction Structure

When an IC card holder initiates a traditional online transaction, such as consumption, cash withdrawal or transfer, with the medium of the IC card, relevant authorization processing module needs to be upgraded to analyze and verify the IC card data fields, while other processing logics (including account processing) are identical with the processing flow of the magnetic strip card.

With regard to featured transactions of the IC Card, such as load and unload, a new transaction interface needs to be added in the original system. In addition to the standard IC card field analysis and verification, the processing of a specific online account or offline account is also needed. Moreover, the card system also needs to support the card writing, such as updating the “EC balance upper limit” and other parameters of the IC card.

11.1.2 Description of featured transaction processing of the IC card

● Account description

- ✓ Debit/credit master account: online master account(i.e. online account of the original magnetic strip card).
- ✓ EC balance: offline balance recorded in the secure memory area of the IC card.
- ✓ EC account balance: offline balance recorded in the background system of the card. This account is unable to truly reflect the current offline balance of the IC card.

- Realization of load business of a specified account
 - ✓ Cash transfer through the background system of the card: the cash is transferred out from the online master account corresponding to the debit/credit card to a background offline account corresponding to the card number.
 - ✓ Updating of the EC balance of the IC card: the Issuer re-writes the EC balance of the IC card in form of load script in a secure manner. The script is protected by the key of the IC card to ensure security and reliability.
 - ✓ $EC\ balance = original\ EC\ balance + load\ amount.$
 - ✓ $EC\ account\ balance = original\ EC\ account\ balance + load\ amount.$
- Accounting of offline consumptions through background system
 - ✓ The Acquirer collects the offline consumption breakdown to form offline transaction documents, which will be sent to the Issuer through the UPI.
 - ✓ The Issuer analyzes the offline consumption record and verifies the transaction TC, as well as updates the EC account balance in the background system of the card.
 - ✓ $EC\ account\ balance = original\ EC\ account\ balance - offline\ transaction\ amount.$

11.2 Statements

Implementation of UPI application may affect two types of statements:

- Statements to Issuers by UPI may change;
- Customer statements may change.

11.2.1 UPI Statement

Issuers should contact UPI representatives to confirm whether statement changes will occur. For example, root CA processing may require charging new fees.

11.2.2 Customer Statement

When magnetic-stripe card is replaced by chip card and product service is not changed, prices generally will not change. The cost of issuing chip cards is assumed by Issuers in general; the use of chip cards can drastically reduce fraud loss and help offset some costs.

However, if products released by Issuers have incorporated new value-added services, this is an opportunity for price-resetting. Price adjustment can take the form of packaged or bundling account pricing or per transaction fees. Before prices are changed, Issuers should made corresponding modifications to the following account statement elements:

- New types of customer relationship and fees;
- New service types and fees;
- Application transaction counter;
- Explain new services and fees information in declaration section of account statement.
- All price changes should be assessed by legal department of Issuers and put into implementation after customer disclosures.

Note: UPI will not recommend particular pricing arrangements; product pricing is at the sole discretion of members.

11.3 Customer Service

Providing excellent customer services is a key factor to the success of implementing debit/credit chip cards. Handling customer enquiry quickly and accurately promotes the growth of IC card application. Achieving the necessary level of support will typically require modifications to customer service system. Information about chip card features and transactions will need to be added, along with updates to customer service manuals.

Customer service representatives need to know how to distinguish between chip cards and magnetic-stripe cards, and look up card offline parameters on operation screen. Information that should be displayed on customer service screen includes:

- Chip card indicator;
- Chip personalization information carried by issuer script processing commands;
- Transaction data according to standard IC card or magnetic-stripe processing;
- Application list that is stored in the card and its corresponding card number (account number);
- Data provided by card management system.

UPI recommends Issuers to use chip indicator instead of card number range to denote an IC card.

Customer service manuals should be updated to help customer service representatives to answer chip related questions. For example, questions related to a declined IC card transaction include:

- Is the card declined by chip or magnetic processing?
- Is the card declined as a result of wrong PIN?
- How many times trying to transact? How much is the transaction amount?
- Did ARQC authentication fails and caused the card is declined?

11.4 Card Management System

Upgrade the card management system to reflect new data and risk parameters in the card. By incorporating these parameters into the card management system, application of these chip data can be traced. According to these records, new card can be issued when card is stolen or expires. According to different functions supported, UPI recommends that Issuers include the following data:

- Card serial number;
- Authorization control parameters, especially velocity check parameters based on card type or card serial number range;
- CVM list setting;
- Cardholder offline PIN and PIN try limit;
- DKI (indicating MDK used by online card authentication, issuer authentication and issuer script processing).

In addition, all the issuer script updates throughout the card's life cycle should be updated in card management system simultaneously so Issuers can re-issue the card with correct setting.

11.5 Card Replacement

During IC card migration, Issuers should consider how to manage card replacement, especially for multi-application cards. The followings should be considered for card replacement process:

- If delivery time for chip cards is too long, should magnetic-stripe card be used for transition;
- Where chip card will be personalized and how is related information passed to personalization providers;
- How should card management system be modified to identify card replacement and trace the entire card replacement period;
- How should customer service representatives access card information during the card replacement process;
- How to replace all the applications in multi-application cards, including non-payment application, such as loyalty point program application.

11.6 Dispute Resolution

New IC card transaction data and process will affect processes on handling customer dispute resolution, chargeback, re-billing and arbitration. It is necessary to analyze potential influence by these changes on the host system and customer service systems. Dispute resolution allows Issuers to return transactions to Acquirers by chargeback.

Issuers shall decide how to capture, process, record and backup chip data concerning dispute resolution and implement corresponding modification.

UPI recommends maintaining this data for sufficiently long period of time to cope with possible dispute resolution, such as more than 180 days.

For a transaction, certain chip data in its request message provides evidence for the risk checks performed by card and terminal. Transaction response message provides evidence for performance process by Issuers.

11.7 Settlement and Reconciliation

Implementing IC card migration should not affect the settlement and reconciliation process. To facilitate trace and analysis, there should be independent statistics reports on IC card transactions. For automatic reconciliation system, chip transactions should also be included in the input files.

Offline authorized transaction has only one settlement transaction in the system, with no authorized transaction relating to it.

11.8 Reports

This section assists Issuers in modifying internal reporting system.

11.8.1 Chip Transaction Statistics

There should be independent chip transaction reports by separating chip and magnetic-stripe transactions, and number of transactions and transaction amounts should be tracked. These statistics data will be reflected in a series of reports, such as settlement, reconciliation, fraud, customer service and service charge report, helping Issuers to monitor the increasing usage rates and risk control results of UICS application.

The issuer can identify transactions initiated by the chip through following data fields:

Issuer host system shall upgrade field 22 and field 60, in order to recognize definitions of new values arising from IC card upgrade.

- POS entry mode (field 22)
“POS entry mode” represents the input mode of cardholder’s data (such as PAN and PIN):
 - ✓ PAN entry mode 05 – IC card and card information is reliable;
 - ✓ PAN entry mode 07 – read contactless IC card with qUICS application;
 - ✓ PAN entry mode 95 – IC card but card information is not reliable;
 - ✓ PAN entry mode 98 – read contactless IC card with UICS debit/credit application.
- Terminal entry capability (field 60.2.2)

This value is a decimal number code and indicates whether terminal can read IC card during IC card transaction:

- ✓ New value 5 – readable contact IC card;
- ✓ New value 6 – readable contactless IC card.

11.8.2 Fallback Transaction

Fallback transactions are magnetic-stripe transactions performed with chip cards at chip-capable terminals. UPI recommends that Issuers shall trace and report all fallback transactions to determine the reasons for occurrences. A high incidence of fallback transactions indicates there are problems with card or acceptance environment.

If fallback transactions occur frequently with a specific card number or card serial number range, the cards may contain faults, such as incorrect personalization data or chips cannot be used. In these situations, UPI recommends Issuers shall contact cardholders to have a face-to-face discussion on the matter; if possible, Issuers should request cardholders to bring the card to the bank or mail it over for further research.

If Issuers discover fallback transaction ratio is rather high for particular merchant or Acquirer, they should contact UPI representatives in a timely manner. UPI representatives are coordinating Issuers and Acquirers to jointly resolve this fallback problem.

11.8.3 Enhancing Reporting Functions

The data provided by IC card transactions can clearly reflect interactions between the card and terminal. When Issuers implement IC card applications, it is an opportunity to enhance the management of reporting functions using these data. The following reports should be considered to enhance:

- Fraud report, highlight the differences between magnetic-stripe and chip cards;
- Report to identify suspicious activities based on card parameter setting;
- Offline transaction and online transaction comparison statistics report;
- Online card authentication, offline data authentication, offline PIN verification result information report;
- Chip issuance volume and transaction volume statistics report.

11.8.4 Transaction Research Tool

Issuers should consider developing a tool to analyze IC card transactions. For example, a tool that can acquire UICS authorization information in real time or after transaction is completed and transmitted to a back-office database. By database inquiry, correctness of the processing, such as offline data authentication, offline PIN verification, online card authentication and issuer authentication can be confirmed. This database is also helpful to staffs responsible for customer services, dispute handling, settlement and reconciliation.

11.9 Internal Staff Training

IC card is a restructuring of current card products rather than a new product. It affects the current business in every way, including market development staff, customer service staff, business support staff, back-office process staff, system development teams, legal consultant and risk management staff.

When developing a training plan, Issuers should consider provide targeted professional knowledge training to staff in different positions, and the entire training should be segmented into multiple specific chapters. Chapters on interacting with cardholders should require extensive training as chip processing procedure is very complicated and has great differences from magnetic-stripe processing. UPI recommends organizing training plan as a continuous activity, because most of the staff may need several training courses to understand and master all the technical items. Issuers should also incorporate IC card application training into new staff training plan.

11.10 Implementation

This section collects activities that required for issuer back-office system, such as statements, card replacement, disputer resolution, reports and staff training:

- Policy
 - ✓ Analyze product pricing structure to decide whether to change fee standards;
 - ✓ Assess reporting requirements for new data in chip transaction.
- Operation
 - ✓ Assess influence of price changes to determine appropriate interchange rate;
 - ✓ Inform cardholders of new pricing structure (if applicable);
 - ✓ Establish and implement card replacement process;
 - ✓ Assess the influence of IC card on dispute resolution;
 - ✓ Decide which current reports need to change and which reports need to be added;
 - ✓ Establish and implement training plan which should involve all staffs.
- Technology
 - ✓ Modify system to support new interexchange rates and pricing structure (if applicable);
 - ✓ Enhance current reporting functions and design new IC card reports;

- ✓ Modify customer service and dispute resolution system, as well as modifications required to perform card replacement.

12 Issuer Host Authentication

This chapter describes issuer host authentication required to implement IC card migration. Host authentication is mandatory for Issuers.

This chapter does not involve other testing for IC card migration, such as internal system and back-office processing. Since IC card migration covers a broad scope, each constituent must perform a full testing. This procedure should also be included in the overall issuer's test plan.

UPI provides testing tools to assist Issuers in performing host authentication. Please contact UPI representatives for more information about testing tools.

12.1 Authentication Environment

Once internal testing for IC card modification is completed, Issuers should prepare for UPI host authentication. The first step of authentication process is to ensure all the required parts are in place:

- UPI testing tool;
- IC card authentication script;
- Personalization-ready testing chip card;
- UnionPay System connection.

Please contact UPI representatives to obtain authentication scripts and other authentication materials.

UPI recommends Issuers should use magnetic-stripe testing transaction to test connection with UnionPay System 1-2 weeks prior to the scheduled online testing. In this way if connection problem exists, it can be timely resolved.

12.2 Authentication Process

This section describes the host authentication process for Issuers. Issuers must perform a series of transactions according to the authentication testing script to certify issuer host system is able to send and receive new data fields in each message.

Please refer to Appendix D for Acquirer online testing process.

Note: UPI debit/credit issuer network entry test also includes “card authentication”. Each new type of card product or personalization data change may affect interoperability in UnionPay System, therefore they all require card authentication. For more information on card authentication, please consult UPI representatives.

Appendix A (Informative Appendix) Recommendations on Project Implementation

Activities and time period required to realize IC card migration depend on specific requirements by Issuers. In general, a complete plan to implement IC card migration usually takes 9 to 18 months. A full understanding of IC card product characteristics and benefits as well as how to promote issuer's commercial needs have significant influence on implementation of the entire project.

IC card migration plan covers a broad scope; it will affect bank staff, cardholders, product vendors, business processes and processing systems. It raises high requirements on the project management and entails the establishment of a multi-functional team with multi-tasking capabilities.

This appendix is used to help Issuers in IC card migration project planning and formulation of a detailed working plan, which includes:

- Key success factors
- Project organization
- Plan implementation
- Project tasks list

A.1 Key Success Factors

At the planning stage, Issuers shall determine project goals, scope and standards for success. They also need to determine project leaders, sponsors, organization structures and participating departments. Completing these steps in excellence lay the foundation for the smooth implementation of the project.

Clear goals can lead the project to the right direction and make participants focus on major issues. For example, if one of issuer's goals is to issue multi-application card with loyalty point program, it needs to ensure the structure, storage volume and parameter settings of the card selected are able to support this function; in which there is no need to focus too much on other irrelevant functions.

Issuers can decide on the methods to implement this plan in accordance with the project scope. Issuers should determine appropriate implementation methods in accordance with device, capital and market size.

Standards for success shall be defined during project initiation and the standards should be agreed by all participants and their key leaders. At each project stage, members can participate in problem solution and confirm the communication requirements based on these standards. Standards for success are also the basis for implementing quality assurance and user acceptance. When releasing the product or preparing to finish the project, these standards will be used to judge for the degree of success. Finally, these standards are used to summarize the project, measure whether the project has reached its full success.

Irrespective of project organization size, several key success factors should be considered:

- Leader and performer

An excellent leader is a must, who should be capable of managing different teams, sufficiently arouse enthusiasm of various parties to ensure project success. Supervisors with powerful execution capabilities are also very important to the project organization.

- Roles and responsibilities

The entire project organization involves multiple areas, therefore roles and responsibilities should be defined for each area before initiating the implementation processes. Many activities are mutually dependent or inherited, so therefore each team must clearly understand their roles in the entire project.

- Preparations

To successfully implement UICS migration plan, sufficient preparations at the early stage are critical. The development of many activities depends on the determination of business strategy.

- Decision and Management Committee

Since this project involves multiple business departments in the bank, arguments may occur when discussing the business needs. Therefore, set up a specialized decision-making and management committee is beneficial to collectively coordinate and rule tough issues, thereby avoiding negative influence on project schedule.

UPI recommends the management committee should exist throughout the life cycle. Members of management committee should include persons in charge in each related areas. While they are not required to participate in the project daily, they are only required to provide guidance to project implementation, perform arbitration for strategies that cannot be decided by project group, and provide funds and resources required by the project.

A.2 Project Organization

Most members will establish a special task team with specific responsibilities to manage everything from initial planning to product release. This team is formed by representatives from various departments affected by IC card migration in the bank. Each task team member provides professional opinions in member's own field and shoulders member's responsibilities. The various professionals required by the project should participate in the project as soon as possible; and of course, full input to the project is not required for all members. Below are descriptions of the functions, roles and responsibilities of a typical IC card migration plan task team.

A.3 Project Manager

Project manager is responsible for the entire operation, milestones and timelines of the project. Project manager also has to manage daily operations, coordinate issues between each team, follow up on project activities and tasks, maintain and distribute project documents, arrange meetings as well as management committee reporting and coordination. Project manager also plays the role of UPI contact person. To ensure project manager can focus on critical issue resolution, organization should consider assigning assistants to share part of project manager's responsibility.

A.4 Project Team

Project team members report to the project manager and project team is usually consists of members in the following area:

- **Card Product Manager**

Card product manager takes hold of migration plan structure from business perspective and decides which functions to be included and their business strategies. Card product manager is also responsible for communicating with and managing vendors, and may even be responsible for new key management activities.
- **Marketing**

Based on card services to be realized, marketing staff analyzes market prospects according to cardholders and merchants with relevant reports submitted and formulates marketing strategies.
- **Legal**

IC card migration plan involves signing new contracts or modifying existing contracts with vendors, and cardholder user clauses should also be modified. These tasks are handled by legal department.
- **Security and Risk Management**

Security and risk management staff is responsible for providing professional knowledge on risk control and information security, managing symmetric and asymmetric keys, monitoring correctness and security of the keys generated, and ensuring security of key transmissions.
- **System Development**

System developer is responsible for system modification related to IC card migration.
- **System and Network Administration**

System and network administrator is responsible for the installation, modification and maintenance in testing and production environment. This part of work should be timely arranged to avoid negative impacts to the project progress.

- Business Management

Business management staff is responsible for establishing business process involved in IC card migration, such as dispute resolution and customer service.

- Customer Service

Customer service processes cardholder inquiry, account maintenance, cardholder dispute and etc. Business processes need to be modified to suit IC card application.

- Training

Trainer is responsible for formulating the training plan, compiling training material and providing specific trainings to staff in various positions.

- Documentation

Document management staff is responsible for writing and editing business and technical documents.

- Quality Management

Quality management staff performs all tests required and ensures modified system operating correctly.

- User Acceptance

User acceptance staff performs tests to ensure that modified system meets the needs of end-user business.

A.5 Implementation Plan

Task team formulates implementation plans according to the project goals and this plan should cover all activities throughout the project life cycle. Although actual plan format and content will be different for each bank, the following should be included in general:

- Major milestones to be achieved;
- Major issues to be resolved;
- Sequence and timing of major events;
- Logical presentation of what are to be accomplished.

After the completion of the implementation plan, task team assesses each key function module, compiles a whole set of segmented tasks, assigns these tasks to each project member through the task plan and clarifies responsibilities and time requirements.

The following “Project Task List” can help Issuers in formulating IC card migration implementation plans.

A.6 Project Task List

Step	Task
1	Activities prior the project
1.1	Obtain approval from key managers, set up management task team
1.2	Determine management task team authority
1.3	Appoint management task team and start operating
1.4	Discuss the impacts of IC card migration with UPI representatives
1.5	Provide training courses to task team members on IC card-related knowledge
2	Decision making
2.1	Define business opportunities
2.1.1	Describe current card product business model – target audience, process and technology
2.1.2	Determine IC card application’s business model and organization value
2.1.3	Judge whether technology criteria are mature (internal & external)
2.1.4	Define business goals
2.1.5	Describe possible benefits
2.1.6	Clarify business needs
2.1.7	State business opportunities and risks
2.1.8	Describe market environment and external influence

Step	Task
2.1.9	State IC card business requirements
2.2	Seek approval
2.2.1	Report IC card migration plan to key managers
2.2.2	Recognize management task team recommendations
2.2.3	Apply for budgets and human resources
2.3	Establish preparatory committee
2.3.1	Select appropriate business team and technicians to participate the project team; third party vendor, outsource provider, UPI or external consultant can be added to the team if required
2.3.2	Determine preparatory committee authority
2.3.3	Obtain related documents on specification
2.3.4	Train working team members
3	Preparation (planning and initiation)
3.1	Analyze business solution
3.1.1	Describe operation and recommendation for business solution – target audience, process and technology
3.1.2	Define IC card security strategy
3.1.3	Describe implementation method – independently, customize, outsource or mixed methods
3.1.4	Analyze impacts on original business by IC card migration

Step	Task
3.1.5	Describe possible benefits – direct financial income, business process, customer recognition and organization innovation
3.1.6	Obtain IC card migration implementation plan template provided by UPI
3.2	Formulate project management plans
3.2.1	Determine implementation budgets – staff, technology, third-party vendor, devices, tools and other expenses
3.2.2	Define key success factors
3.2.3	Formulate a quantitative business plans
3.2.4	Describe how to manage, assess and conduct statistics on new benefits
3.2.5	Define internal-external communication and training strategy
3.2.6	Define check and testing strategy
3.2.7	Perform risk analysis
3.3	Project plan
3.3.1	Assess business model and business plan
3.3.2	Determine project goal and scope
3.3.3	Determine work flow
3.3.4	Determine required resources (including how to obtain resources)
3.3.5	Describe project management methods
3.3.6	Evaluate changes and the expected level of difficulty

Step	Task
3.3.7	Perform project risk evaluation and formulate risk management plans
3.3.8	Formulate communication and training plans
3.3.9	Formulate check and testing plans
3.3.10	Integrate the above-mentioned contents and formulate project management plans
3.4	Initiate project
3.4.1	Assign work flow resources
3.4.1.1	Formally establish project team
3.4.1.2	Train project team members
3.4.1.3	Perform project management methods
3.4.1.4	Perform plan controlling process
3.4.1.5	Formulate communication plans
3.4.1.6	Formulate plans for unit tests, integration tests, authentication tests, user acceptance tests and etc.
3.4.1.7	Formulate project stakeholder management methods
3.4.1.8	Perform risk and emergency management methods
3.4.2	Establish income model
3.4.3	Perform income management methods
3.4.4	Initiate each work flow

Step	Task
4	Issuer completes migration
4.1	Vendor selection
4.1.1	Establish and release RFI/RFP
4.1.2	Evaluate information returned
4.1.3	Select vendor
4.1.4	Negotiation and award bid
4.1.5	Manage vendor
4.1.6	Receive products delivered by the vendor
4.2	Construction of key management system
4.2.1	Construction of symmetric key system
4.2.2	Construction of asymmetric key system
4.3	Chip card selection and personalization
4.3.1	Card selection
4.3.1.1	Determine card function and security requirement
4.3.1.2	Determine requirement for multi-application platform
4.3.1.3	Evaluate card security performance
4.3.1.4	Analyze whether card specification meets each requirement
4.3.1.5	Card vendor to provide card quality management services

Step	Task
4.3.1.6	Confirm that card vendor has passed bankcard test center authentication
4.3.2	Card design and production
4.3.2.1	Card surface design
4.3.2.2	Determine card secure transmission mechanism
4.3.2.3	Generate card transmission key
4.3.2.4	Transmit key to the card manufacturer
4.3.2.5	Card packaging and printing
4.3.2.6	Obtain card for personalization
4.3.3	Generate and download of issuer certificate
4.3.3.1	Generate issuer public/private key pair
4.3.3.2	Create issuer public key file
4.3.3.3	Obtain root CA application form
4.3.3.4	Submit the form and public key file to UPI
4.3.3.5	Receive issuer public key certificate
4.3.4	Data preparation system installation and debugging
4.3.5	Card personalization
4.3.5.1	Produce personalization script according to data preparation system output
4.3.5.2	Connect the key management system and obtain card key

Step	Task
4.3.6	Select personalization data template
4.3.6.1	Correctly select IC card personalization data
4.3.6.2	Set up personalization system according to data template
4.3.7	Verify correctness of card data
4.3.7.1	Use IC card testing tools to perform card verification
4.3.7.2	Send card to UPI for UPI to perform card data verification
4.3.7.3	UPI feedback test report
4.3.8	Produce sample card
4.4	Host system modification
4.4.1	Transaction processing system modification
4.4.2	Business processing system modification
4.5	System network entry test authentication
4.5.1	Offline test
4.5.1.1	Use UPI simulator to test
4.5.1.2	Submit test log
4.5.1.3	UPI evaluates the log
4.5.2	Online test
4.5.2.1	Perform online test authentications

Step	Task
4.5.2.2	UPI issue test report
5	Internal test
5.1	Determine testing strategies
5.2	Formulate detailed test plans
5.3	Unit test, integration test and user acceptance test
5.3.1	Formulate acceptance standards
5.3.2	Prepare testing tools
5.3.3	Compile testing cases
5.3.4	Prepare testing data
5.3.5	Construct testing environment
5.3.6	Implement testing plans
5.3.7	Check and verify test results
5.3.8	Perform regression test
5.3.9	Pass acceptance
6	Product release
6.1	Determine product release strategy
6.2	Formulate product releasing plans
6.3	Formulate device delivery plans

Step	Task
6.4	Standard technical environment
6.4.1	Network connection
6.4.2	Environment test
6.4.3	System installation
6.5	Commissioning preparations
6.5.1	Plan and implement commissioning rehearsal
6.5.2	Import commissioning data
6.5.3	Operation trainings
6.5.4	Testing emergency system
6.6	Commissioning
6.6.1	Formulate commissioning scheme and forecast results
6.6.2	Prepare data verifications
6.6.3	Implement commissioning plans
6.6.4	Verify correctness of IC card application
6.6.5	Pass business department acceptance
6.6.6	Handover to daily operation departments

Appendix B (Informative Appendix) Referenced Process Flow of IC Card Issuance for UPI Members

This Appendix is prepared for the convenience of member institutions to understand the basic process of card issuance and facilitate more efficient development and implementation of the project. Member institutions may reference this Appendix and combine with actual situation to implement card issuance project.

B.1 Business Qualifications for Issuing Member Institutions

Member institutions who wish to apply to UPI for IC card issuance business shall have the following qualifications:

- For the issuance of IC cards with debit/credit applications, Issuers must have the appropriate UPI debit/credit card issuance business qualification (institutions/business network entry procedures are required for qualifications);

B.2 Basic IC Card Issuance Process Flow for Member Institutions

The following figure is the basic IC card issuance process flow for member institutions:

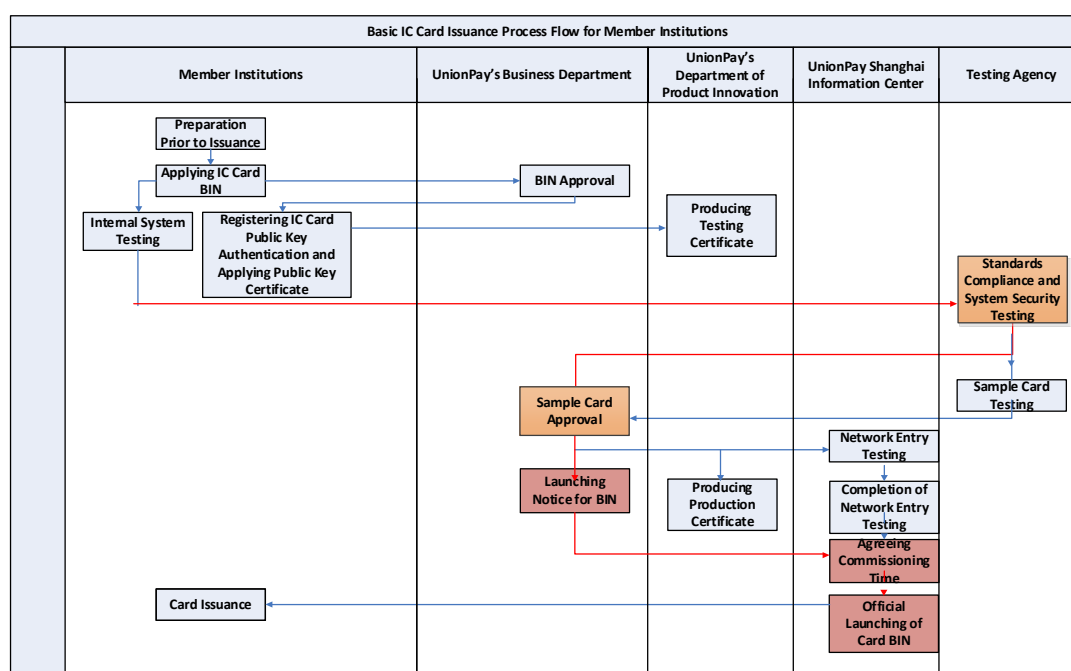


Figure 2: Basic IC Card Issuance Process Flow for Member Institutions

B.2.1 Preparation Prior to Issuance

Member institutions shall conduct a series of preparations prior to implementation of IC card issuance project, including consulting, training, researching, identifying needs, project planning, vendor bidding, system development and transformation and etc.

B.2.2 Applying IC Card BIN

Member institutions shall apply to UPI for card BIN during their initial application for issuance business. For member institutions that have issued magnetic-stripe cards, they may decide whether to apply for a new IC card BIN according to their circumstances.

B.2.3 Registering IC Card Public Key Authentication and Applying Public Key Certificate

After obtaining BIN that recognized by UPI, member institutions shall apply to UPI's Department of Product Innovation for IC card public key authentication registration, public key certificates and etc. UPI will first produce relevant test certificates.

B.2.4 Internal System Testing

After completing system upgrades related to card issuance project, member institution should perform internal system test to ensure the system can operate normally and stably in accordance with relevant requirements.

B.2.5 Sample Card Approval

After the sample cards submitted by member institutions passed the testing by the Bank Card Test Center, the Bank Card Test Center will submit test results to UPI's Business Management Department. Then, the department will review the cards for approval, any subsequent works on the cards should be performed after the approval and recording of sample cards.

B.2.6 Production Certificate and Network Entry Testing

After approving sample cards, UPI's Business Department will notify the UnionPay's Department of Product Innovation and UnionPay Shanghai Information Center. UnionPay's Department of Product Innovation will then produce a production certificate, while the Shanghai Information Center will perform network entry testing.

B.2.7 Launching Notice for BIN

After approving sample cards, UPI's Business Department will send UnionPay Shanghai Information Center a notice regarding the launch of card BINs.

B.2.8 Agreeing Commissioning Time

After UnionPay Shanghai Information Center received BIN launching notice from UnionPay's Business Department and under the premise of completing related production certificate and network entry testing, member institutions may negotiate with UnionPay Shanghai Information Center to agree upon commissioning time.

B.2.9 BIN Official Launch and Card Issuance

UnionPay Shanghai Information Center will officially launch card BIN applied by member institutions at the agreed commissioning time. From that on, member institutions can engage in card business.

Appendix C (Informative Appendix) Key Quick Reference Table

This appendix provides quick references for keys involved in the UPI application, and includes the following types of keys.

C.1 Personalization Keys

They are used to protect personalization process.

Table 33 Personalization Keys

Abbreviation	Name	Generation Level	Explanation
KMC	Issuer master key	Every card vendor	IC card vendor uses this KMC to generate card level keys (K_{ENC} , K_{MAC} , K_{DEK}) and write them into the card.
K_{ENC}	Data encryption key (card level)	Every application	Uses to create a communication key and this communication key can create cryptogram and encrypted confidential data with CBC mode.
K_{MAC}	MAC key (card level)	Every application	Uses to create a communication key and this communication key can create the C-MAC used during command processing.
K_{DEK}	Data encryption key (card level)	Every application	Uses to create a communication key and this communication key can encrypt DES key or other confidential data flexibly under ECB mode.
KEK_{ISS}	Issuer key exchange key	Every issuer	Protects offline PIN and other confidential data between issuer and data preparation device.

Abbreviation	Name	Generation Level	Explanation
DEK/TK	Data encryption key / transmission key	Every personalization device	Protects offline PIN and other confidential data between data preparation device and personalization device. Following special type of data transmission key may be used: PEK/TK – PIN encryption key and is used to protect PIN data. KEK/TK – key exchange key and is used to protect DES key.
MAC key		Every personalization device	Used to ensure the integrity of application data in personalization data files that are provided to personalization device.

C.2 Online Card and Issuer Authentication Keys

They are used for online card authentication and issuer authentication.

Table 34 Online Card and Issuer Authentication Keys

Abbreviation	Name	Generation Level	Explanation
AC MDK	Application cryptogram master key	Every BIN	Generates unique card key and is used for online verification between the card and issuer.
AC UDK	Application cryptogram unique key	Every application	Generates unique communication key – AC SUDK and is used for application cryptogram production and verification.

Abbreviation	Name	Generation Level	Explanation
DKI	Derivation key index	Every MDK	Uses to clarify which master key is used to obtain the unique key in the card.

C.3 Public Key

They are used for SDA, DDA and offline cryptogram PIN.

Table 35 Public Keys

Name	Purpose	Generation Level	Explanation
Certificate center public/private key pair	SDA, DDA and offline cryptogram PIN	Generated by root CA	Root CA is responsible for certificate center public key pair generation and management.
Issuer public/private key pair	SDA, DDA and offline enciphered PIN	Every BIN	Generated by Issuers. Public key is transmitted to root CA, which uses the key to create issuer public key certificate. Private key is stored in issuer HSM.
IC card public/private key pair	DDA and offline enciphered PIN	Every application	To support DDA, issuer is required to generate IC card public/private key pair for each IC card; private key is stored in secure storage area within the IC card; IC card public key is signed by the issuer's private key to create IC card public key certificate and is stored in the card.

Name	Purpose	Generation Level	Explanation
Signed static application data (SAD)	SDA and DDA	Every application	Used to verify the signature for card application data. For SDA, SAD signed by the issuer private key is stored in the card independently; for DDA, SAD is included in IC card public key certificate.
Issuer public key certificate	SDA, DDA and offline enciphered PIN	Every BIN	Certificate contains issuer public key signed by root CA private key.
IC card public key certificate	DDA and offline enciphered PIN	Every application	Contains IC card public key signed by the issuer private key, and is placed in the card during card personalization. The certificate contains SAD signature encrypted by issuer private key.

C.4 Issuer Script Processing Keys

They are used to protect issuer script processing data.

Table 36 Issuer Script Processing Keys

Abbreviation	Name	Generation Level	Explanation
MAC MDK	Secure MAC master key	Every BIN	Generates a unique key — MAC UDK and is used to generate communication key for message authentication required in post-issuance updates
MAC UDK	Secure MAC unique key	Every application	Generates a unique communication key — SUDK MAC and is used for generation and verification of secure message authentication code.

Abbreviation	Name	Generation Level	Explanation
ENC MDK	Secure message encryption master key	Every BIN	Generates a unique key — ENC UDK and is used to generate communication key to encrypt updated confidential data (offline PIN) after card issuance.
ENC UDK	Secure message encryption unique key	Every application	Generates a unique communication key — SUDK ENC and is used to encrypt/decrypt secure message.

Appendix D (Normative Appendix) UPI IC Card Network Entry Process

For issuing institutions to launch IC card functions, the process is similar with magnetic-stripe card issuance, wherein institutions shall pass IC card function network entry test (off-line testing and online testing) before applying for network entry applications.

D.1 Network Entry Testing Process

The process flow for issuer's IC card network entry test is as follows:

1. Offline testing phase: institutions apply for offline simulation tool and UPI issues authorization codes, dongles and other simulation tools. Institutions use offline simulation tools to perform offline test cases and generate offline test reports and transaction logs after offline tests are completed.
2. Online test application and acceptance phase: Online test application includes
1. Online test application forms and permission table; 2. UPI approval; 3. offline test reports and transaction logs.

Head office level institutions should send above materials to banktest@unionpay.com to apply for online test.

Local subsidiaries should send above materials to local UPI branch to apply for online testing. After receiving online test applications, Main Information Center will schedule online tests 4 days after reviewing and approving online test applications.

3. Executing online test phase:

Head office level institutions should complete all online test requirements with the coordination of testing personnel from Main Information Center. Alternately, testing within the jurisdiction of local subsidiaries should complete all online test requirements with the coordination of testing personnel from local UPI branches. Overseas acceptance testing should complete all online test requirements with the coordination of testing personnel from Shenzhen Information Center. All online tests should compile online test reports and are returned to institutions and UPI business department after passing audits.

4. Production and online stages: after passing relevant tests, institutions should negotiate and agree a commissioning time with the production team from Main Information Center. Institutions should complete online commissioning works according to agreed commissioning time.

D.2 Special Instructions

For the process flow of IC card network entry, there are several points need to be emphasized:

1. The offline simulation tool application form, online test application form and permission form and offline simulation tools are available for download at the official website of UPI.
2. Requirements for test cards: prepare two sets with each consists of at least two test cards, and the cards need to include all transaction functions to be tested. Head office level institutions should send the cards to Main Information Center and Shenzhen Information Center (refer to online test application form for mailing addresses). Local subsidiaries should send the cards to local UPI branches and Shenzhen Information Center.
3. The offline testing for institutions does not include offline electronic cash spending and refunds, clearing documents and testing on error handling. Online testing requires a complete test on all transaction functions supported by institutions, including offline electronic cash spending and refunds, clearing documents and testing on error handling.
4. For network entry test, offline and online test cases are basically consistent.
 - ✓ Offline test cases provide a more complete testing contents for institutions, to assess whether IC card applications and IC card online transaction messages are compliance to specifications and standards;
 - ✓ Online test cases are mainly testing whether card functions can transfer and process accurately on online interbank network.